

CULTIVATING NARRATIVE CHANGE IN COLLECTIVE SENSEMAKING

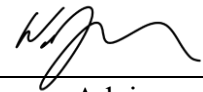
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AUTHORSHIP STATEMENT

I, XXXXXX XXXXX attest that the work in this thesis is substantially my own.

In accordance with the disciplinary norm of authorship in Design the following collaborations occurred in the thesis. Please consult Appendix S of the IIT Faculty Handbook for further information.

My primary advisor, Prof. Weslynn S. Ashton at the IIT Institute of Design, contributed to the design of the research project and provided guidance in the interpretation of results, as is the norm for a doctoral supervisor. She also acquired the financial support from the National Science Foundation for this research project. Prof. Laura Forlano, currently at the Northeastern University College of Arts, Media and Design, was my primary advisor for the first two years of this research. Prof. Forlano guided the theoretical and methodological framing of research and the interpretation of results. Prof. Carlos Teixeira and Assoc. Prof. Maria Villalobos Hernandez provided guidance for the framing of research, and refinement of contributions.

This dissertation is written in a multi-article format. The second chapter of this thesis, titled 'The Un-Common Good: Making Room for Radical Transition Imaginaries,' has been published as part of the EPIC Conference Proceedings in 2023, licensed under Creative Commons BY-NC-ND. The third chapter, 'Cultivating Narrative Resistance in Systemic Design,' has been presented at the RSD Symposium in October 2024 and accepted for publication as part of the RSD Proceedings in 2025. The fourth chapter, titled 'Beyond Hero's Tale: Future-making as Narrative Inquiry,' has been submitted to a journal. Prof. Ashton and Prof. Forlano are co-authors of all three publications that constitute Chapters 2, 3, and 4.

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ABSTRACT

The complexity of social and ecological crises and the need for collective action to tackle them brought forth collective inquiry as an essential capacity for societies to combine their knowledge and creativity, navigate complexity, and respond with adaptive solutions. However, current approaches to sensemaking often remain anchored in the same mindsets and worldviews that underpin these crises, constraining their ability to account for the fundamental shifts needed. This research addresses this gap by exploring how collective inquiry, when framed as a narrative practice, can open space for alternative perspectives and pathways in systems transformation. This study positions narratives as dynamic meaning systems that shape how groups interpret issues, determine relevance, and envision new possibilities. The dissertation is structured as three studies, situated in the context of food systems, each exploring ways to engage and mobilize these meaning systems across different contexts and scales of collective inquiry.

A central contribution of this research is the framework of narrative infrastructures—the social and material contexts through which narratives of change are constructed, circulated, and sustained within systems change efforts. This framework supports designers in navigating and shaping the spaces where narratives are articulated, contested, and maintained, to foster more critical, pluralistic, and transformative approaches to collective inquiry. Ultimately, this work enhances design's capacity to foster the shifts in mindsets through which societies envision their collective futures, using its narrative agency to disrupt harmful paradigms and open space for radical possibilities in the making.

CHAPTER 1

INTRODUCTION

1.1 Making Sense in Common

We stand at a critical juncture in history. Centuries of efforts to improve human life, driven by ideals of progress and prosperity, have strained the very systems upon which life depends. Ironically, the institutions built on these ideals seem to have let the masses down in service of a few. As humanity continues to exceed the planetary limits upon which our shared future depends, those who bear the greatest burdens are often the ones who contributed the least to the problem. During the period in which this work came into being, the world saw numerous global calls to action, urging nations to implement the measures they had pledged in response to mounting crises. Yet, despite these urgent appeals, coordinated action frustratingly stagnant. This raises a fundamental question: Why is it so difficult to act collectively, even when our shared future depends on it?

Tackling the urgent systemic shifts required for our collective well-being demands a shared capacity to make sense of their complex dynamics, envision alternative futures, and negotiate collaborative pathways forward. Over the past decades, a vast body of work has emerged across disciplines and domains, exploring how humanity might navigate these wicked problems—problems that are complex, interconnected, and resistant to traditional approaches.

Organizations and networks around the world are experimenting with new infrastructures for this kind of collective inquiry. These infrastructures bring together technology, participatory governance models, and frameworks for collaboration that

allow diverse groups to combine their knowledge and insights and mobilize resources in an iterative and adaptive way. International climate action networks, integrated data systems for sustainable urban planning, and community-led learning hubs are all examples of how groups are beginning to harness this collective capacity to address systemic challenges. This research considers collective inquiry as distributed sites of knowledge production on systems change, connected by networks, coalitions; whether this production of knowledge is explicitly codified or not.

Evolution of Design as a Collective Sensemaking Capacity. In this space of purpose-driven, action-oriented knowledge creation, design is increasingly recognized as a capacity for collective sensemaking. A growing repertoire of design methods and approaches has matured, enabling diverse groups to explore issues and potential solutions by contributing their knowledge and perspectives beyond disciplinary boundaries and organizational silos. Rather than prescribing specific solutions, these approaches focus on creating the social and material conditions that activate the collective intelligence of stakeholders, allowing them to explore pathways that evolve through ongoing dialogue, experimentation, adaptation, and negotiation. Design's approach to collective inquiry in systems change is grounded in the premise that combining stakeholders' unique vantage points—each offering a distinct lens—creates a more comprehensive understanding of present issues, desired future states, and shared pathways forward.

Stakeholder Groups Bring More Than Perspectives. However, these lenses are not pieces of a puzzle that, when combined, reveal a more complete picture of systems

change. Stakeholders bring more than discrete fragments of knowledge and perspective; their views are shaped by distinct visions for the future, guided by diverse values and mindsets. As Kania and Kramer (2012, p. 37) describe, this “kaleidoscope” of perspectives, each tinted by its own agenda, is often viewed as a barrier to collective impact.

This is part of the reason why alignment on high-level ideals, such as the Sustainable Development Goals, often fail to generate the collaborative action that systemic transitions require. While aiming to provide a simple, shared, action-oriented framework for systemic change, these goals overlook the divergent values and visions, as well as the deep-seated historic tensions that may exist among stakeholders. Yet, this very diversity—while complicating collective action—is the foundation of the democratic ideals that most societies uphold. It reflects the reality of our collective lives and shaping our shared futures, where multiple worldviews, values, and ways of being coexist, shaping how different groups see both challenges and possibilities. How can we, then, reframe this diversity as a shared resource to tackle transitions more inclusively and creatively, rather than treating it as an obstacle?

Systemic issues, while requiring some level of alignment, are also sites of inherent tension, sitting at the intersection of multiple, and often competing visions of what a desirable future looks like: socio-technical imaginaries (STIs). A socio-technical imaginary refers to a collectively held vision of a desirable future, shaped by shared social values, political ideologies, and technological aspirations (Jasanoff and Kim, 2009). These imaginaries are deeply embedded in institutions, public discourse, and cultural practices, guiding how societies envision their futures and pursue scientific,

technological, and policy pathways to achieve them. As such, they are a powerful dynamic in collective inquiry into our shared futures. Making sense of systems change is not merely generating plausible explanations and scenarios; it is a process of performing and negotiating imaginaries, framing issues, solidifying specific visions of the good life, and legitimizing certain pathways—while marginalizing others.

Without surfacing and engaging with diverse imaginaries, the process of collective inquiry risks reinforcing dominant mindsets and values that lie at the root of today's crises. This can marginalize alternative interpretations and conceal possibilities for transformative shifts. As Arora-Jonsson states (2023) SDGs may serve better for anchoring conversations between diverse groups with divergent views, rather than a universal and objective compass for coordinated action. This research views purpose alignment as an opportunity to surface diverse imaginaries and critically examine the framing of systems change.

This research is based on the premise that engaging with diverse socio-technical imaginaries is essential for fostering pluralistic and collective inquiry in systems change, reflecting the diversity of values and meanings at play - and potentially preventing superficial alignment that doesn't translate into action. But more importantly - it is critical to fostering the paradigm shifts that today's challenges require, to contend with the dominant framings of systems change that perpetuate capitalist and colonial paradigms of co-existence, envision radical possibilities grounded in other ways of being with each other and other species.

However, despite this need, we currently lack the tools and processes to engage with diverse imaginaries in sensemaking efforts. While approaches such as pluriversal

design (Escobar, 2020) point to the importance of embracing diversity, there remains a gap in the practical methodologies required to facilitate this engagement. This study explores how design can foster engagement with diverse imaginaries of transformative change in sensemaking processes—particularly in the framing and envisioning of systemic change. Through this engagement, it aims to deepen the understanding of transformative efforts, contend with their implications, and generate alternative possibilities for the future.

1.2. Approach: Sensemaking as Narrative Inquiry

A combinatorial understanding of collective intelligence, which emphasizes aggregating diverse inputs to solve complex problems, misses a fundamental mechanism of how we make sense of the world: telling stories. From ancient cave paintings to tweets, stories have always been the way we give meaning to our experiences—both lived and imagined—imbued with our worldviews, belief systems, and values. As organizations and communities engage in efforts for systems change, they produce diverse narratives that portray the approach they take and the role they claim in systems change. These narratives materialize in various forms—from policy papers and impact metrics that organize large-scale interventions to artworks and community spaces that offer glimpses of different worlds. They construct specific framings of systems change, locating issues, manifesting visions and advocating for specific pathways. As such, narratives are manifestations of socio-technical imaginaries: they shape our collective understanding of what is possible and desirable, often reinforcing certain visions while marginalizing others. These narratives are not static meaning systems; they are socio-

material performances that influence the trajectory of transformative change, capable of causing lock-ins to specific framing, as well we disrupting those.

When viewed through a narrative lens, collective inquiry in systems change is far from a neutral act of collecting, codifying, and interpreting information. Instead, it involves selectively capturing stories that shape how systems are understood, as well as constructing and disseminating narratives that interact with larger discourses. In this process, design plays a crucial role by creating the socio-material conditions—tools, spaces, and processes—through which specific stories are captured, and narratives of change come to being. By doing so, design influences how issues are framed, which possibilities for action are made visible (and which ones are rendered invisible), and ultimately, which futures are imagined and pursued.

As Inayatullah (1998) states, “the discourse we use to understand is complicit in our framing of the issue.” When sensemaking efforts are anchored in dominant narratives, our perception of the past, present, and future becomes narrowed, excluding ways of being and knowing that do not fit within the prevailing worldviews and systems of meaning. While narratives deeply influence how systemic issues are interpreted and which futures can be imagined, it often remains invisible, slipping into our vision like well-worn spectacles, subtly embedding itself in how we think, see, and act. This research aims to bring visibility to these lenses by uncovering how the design of socio-material conditions shapes the narrative construction of systems change. Furthermore, it explores how intentional design of these conditions can help groups diversify the narrative lenses through which they view systems change, to expand horizons for transformative action.

1.3. Aim of Research

This research aims to understand how narratives can be leveraged as a sensemaking mechanism to account for and engage with diverse imaginaries in collective inquiry on systems change. It considers collective inquiry as distributed and networked practices of sensemaking across various sites and scales where, where groups combine their perspectives and knowledges to frame issues and explore solution pathways - and in doing so, construct and perform imaginaries and form coalitions around these. I adopt a broad conception of sensemaking that encompasses activities ranging from scientific knowledge production to artistic expressions through which groups give meaning to shared experiences. This exploration is driven by the following question:

How might design engage narratives as a sensemaking mechanism to foster pluralistic, inclusive, and critical collective inquiry in systems change?

In order to answer this question, this research investigates socio-material conditions of collective inquiry, specifically asking:

- How do the social and material conditions of collective sensemaking influence activation or exclusion of socio-technical imaginaries?
- How can these conditions be intentionally designed to foster critical and generative engagement with diverse socio-technical imaginaries and the tensions between them?

This inquiry centers narratives as a sensemaking mechanism and considers their interaction with the broader discursive dynamics of collective inquiry, examining both the macro scale of organizational discourse, where various narratives compete and interact, and the micro scale of stories which represent the localized perspectives that

either feed into or get filtered out of sensemaking activities. It is important to define these categories in the context of collective inquiry where diverse collectives and organizations co-produce knowledge and narratives. These categories do not mark rigid boundaries but illustrate scales of sensemaking in the context of systems change.

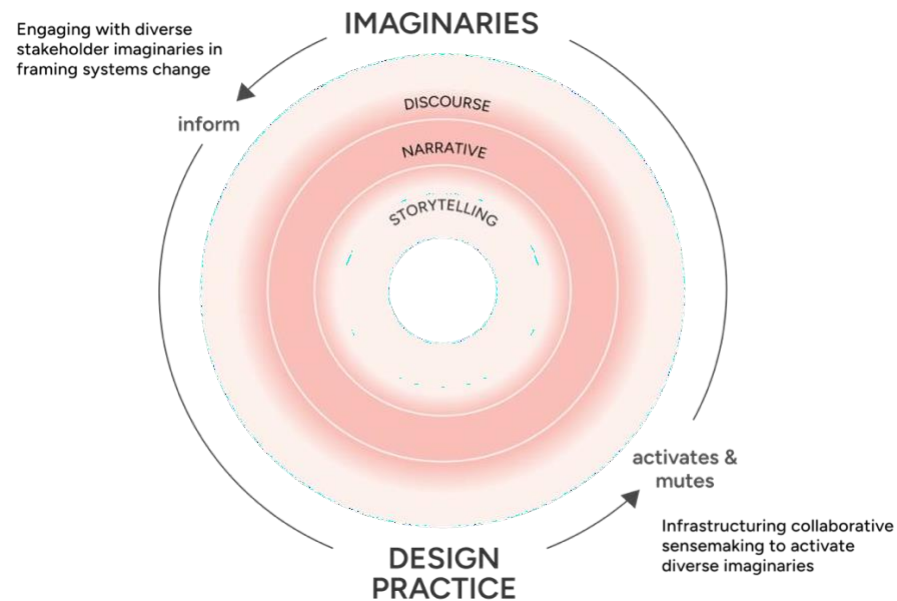


Figure 1.1. Intentional Engagement with Imaginaries in Design for Systems Change.

- Stories are defined as descriptions of phenomenon. These can represent past, present, or future events—whether lived, perceived, or imagined.
- Narratives are collectively stabilized yet dynamic meaning systems that weave together multiple stories, creating coherent interpretations of events or experiences shared by groups of people. Narratives link fragmented events into broader patterns that frame both past and future possibilities. They are performative, actively shaping how we understand and engage with systemic issues, capable of influencing existing power dynamics and worldviews.

- Discourse is a collective practice through which public understanding of matters are formed, where multiple narratives circulate and interact, organized by social and material contexts.

1.4. Literature Review

This research is informed by four areas of literature that help frame design's role in future-oriented collective inquiry for tackling complex social and ecological challenges. Figure 1.2 shows the preliminary review of the literature that grounded the initial framing of this research in 2020. With the evolving nature of insights, I refined the literature review into the following fields of theory and practice:

(1) Science and Technology Studies (STS) and Feminist STS scholarship provide the primary theoretical foundation for this research, approaching knowledge production as socially and materially situated practice, tied to public concerns, values, and interests. (2) *Object-orientation in Publics*, an approach largely shaped by STS, anchors my focus on the contexts that allow publics to form outside of institutionalized modes of public participation. (3) My vision for design's role in this broader context of collective inquiry is grounded in *Participatory Design's* ethos of creating the conditions for pluralistic and contested publics. (4) And lastly, I draw from Critical and Participatory Futures Practice for enacting these principles in future-oriented collective inquiry.

is research explores how design can inform the creation of infrastructures that can support the translation of knowledge generated across ongoing prototypes of socio-economic change to collective intelligence. The central diagram illustrates the recursive framework of this research, highlighting the 4 key uses that drive this research and the questions that drive their exploration. The diagram below proposes the relations between these key uses.



Literature Fields

- Design ● Innovation Studies
- Science and Technology Studies ● Future Studies
- Systems Studies

Negotiation in Action

How can design help create new forms of public matters as negotiations futures in action? What kind of infrastructures can support new sociations between ongoing prototypes of distributed collectives? What the value of distributed prototyping activities for systems change? What is a horizontal interpretation of scale mean for the agency of collectives items change?

- Infrastructuring** | ● Dantec & Disalvo, 2013
● Hilgren et al., 2011
● Yndigen et al., 2018
● Jiménez, 2014
● Ehn, 2008

- Agency** | ● Storni, 2015
● Yaneva, 2009
● Storni et al., 2015
● Kaghan and Bowker, 2001
● Latour, 2005

- Complex Systems** | ● Aguirre et al., 2014
● Sevaldson, 2013
● Naudé, 2012
● Wiltching et al., 2013
● Jonas, 2005

- Scale** | ● Hunt, 2014
● Westley and Antadze, 2010
● Pereira et al., 2015

Future Visions

How are the visions of plural futures embedded in the prototypes of change? Which design methods can help surface the future visions in the prototyping narratives? What would a non-linear understanding of time look like in prototyping narratives?

- Speculative Research** | ● Wilkie et al., 2017
● Rosner, 2018

- Time** | ● Mazé, 2016
● Lindström and Ståhl, 2015
● Haraway, 2013

- Design Futures** | ● Bleecker, 2010
● Burdick, 2019
● Slaughter, 2004

Collective Intelligence

What is the value and relevance of collective intelligence in processes of systems change? What kind of design objects and affordances can support translation of situated knowledges to collective intelligence without being reductionist? What is the value of collective intelligence for different collectives that prototype change in systems?

- Networks** | ● Pyka et al., 2009
● Aguirre et al., 2017
● Jiang and Thagard, 2014

- Open Knowledge** | ● Jiménez, 2014

- Trading Zone** | ● Sisondo, 2010
● Suchman, 2002
● Gallison, 1997
● Braidotti, 1999

Distributed Prototypes*

What is the value and relevance of situated knowledges distributed collectives that prototype change in systems? What are the ways in which design can codify the knowledge generated through prototypes? How can partial views of prototypes be translated into patched images of systems phenomena?

- Situated Knowledges** | ● Suchman, 2008
● Haraway, 2016
● Sisondo, 2010
● Woodhouse and Patton, 2004
● Haraway, 1991

- Design Epistemology** | ● Mattelmäki and Matthews, 2009
● Galey and Ruecker, 2010
● Redström, 2017

* Prototypes are defined as situated ongoing and distributed socio-material assemblies through which futures negotiated in action by different collectives.

Figure 1.2. Preliminary Literature Review Visualization

Science and Technology Studies (STS) and Feminist STS. Science and Technology Studies (STS) provides a critical perspective for understanding how large-scale transitions and their accompanying techno-scientific agendas are negotiated across various domains of society. At its core, STS rejects the notion that science and technology are purely objective and disinterested activities, dedicated solely to the pursuit of universal truths (Sismondo, 2010). Instead, the interdisciplinary field of STS posits that science and technology are inherently social activities, deeply embedded within and shaped by cultural norms, institutional structures, political agendas, and historical contexts.

A substantial scholarship in the field of STS is dedicated to understanding how knowledge production and technology development are socially organized and negotiated, while legitimizing specific transition pathways. A core concept for framing this is co-productionist approach to knowledge production, proposed by Jasanoff (2004; p. 2) stating that “the ways in which we know and represent the world (both nature and society) are inseparable from the ways in which we choose to live in it”. From co-production perspective, the development of national and institutional agendas of science and technology are interwoven with the production of sociotechnical imaginaries, shaping visions of common good and the organizations of social life needed (Jasanoff and Kim, 2009; Jasanoff and Kim, 2013). Sociotechnical imaginaries help explain how certain technological pathways are legitimized and pursued, while others are discarded or deemed undesirable.

The study of socio-technical imaginaries emerged primarily from analyses of national agendas in science and technology (Dawson and Buchanan, 2005; Chateau et al.,

2021). However, recent scholarship emphasizes the importance of local and trans-local networks as spaces where alternative imaginaries are actively produced and performed, challenging dominant framing of transitions. This approach emphasizes how communities express their hopes, concerns, and perspectives concerning transitions, instead of focusing solely on institutional and national narratives (Tidwell & Tidwell, 2008). Consideration of local imaginaries and their diverse manifestations helps uncover the messy reality of transition efforts as they relate to daily lives of people. Moreover, it challenges the norms of expert-driven knowledge production and seeks to engage diverse ways of knowing in making sense of transitions (Tidwell & Tidwell, 2008).

In order to understand what 'other knowledges' are excluded from dominant framings of transformative change, I will turn to Feminist Science and Technology studies. Feminist STS confronts the ideal of scientific knowledge as the objective unifying truth by questioning whose knowledge, worldviews, values, interests are accounted for in the scientific construction of reality, and at what expense. Feminist STS scholars advocate for alternative, plural epistemologies that are attuned to the interdependent relationships between humans, other species and entities and challenge ontological separations such as body/mind, nature/culture, human/machine (Haraway, 1988; Suchman, 2008; Boehnert, 2014; Forlano 2017). Proposed by Donna Haraway (1988), the concept of 'situated knowledges' defies the positivist understanding of reality and foregrounds partial, local, incomplete views of the world. These fragmented, subjective and situated views, through their connections and contradictions, can disrupt flattening of diverse realities and resist premature closure of debates concerning implications of science and technology on our collective lives (Haraway, 1988; Suchman,

2002). The partial views are not parts of a jigsaw which can align to present a complete vision of the whole, but they are always connected with imperfect seams and co-exist in tension.

My approach to collective inquiry into systemic shifts is informed by these critical perspectives, particularly the co-production of knowledge and imaginaries in sensemaking activities across various scales. The ability to shift values, worldviews, and mindsets—through which we envision shared futures—is inextricably tied to how we produce knowledge. Thus, engaging meaningfully with the transformative potential of local imaginaries requires designing conditions that embrace diverse epistemologies and resisting reductive formulations in favor of acknowledging complexities and contradictions.

This exploration centers narratives and storytelling as mechanisms for engaging situated knowledge in collaborative sensemaking activities, opening space for multiple, sometimes conflicting, ways of seeing and knowing. This approach grounds my exploration in the understanding that shifts in knowledge production are required for paradigm shifts in how systems change is conceived and acted upon.

Object-Orientation in Publics To lay the foundation for this research, it is essential to understand how social, ecological, and technical phenomena become 'matters' of public concern. This exploration is grounded John Dewey's pragmatism, which continues to inform how design engages with public issues today. Dewey (1946) describes the public as forming of coalitions through the associations between groups of people similarly affected by the consequences events. Importantly, Dewey argues that such dynamic and

reconfiguration of publics needs to be supported by forms of deliberation and organizing: “To form itself, the public has to break existing political forms. This is hard to do because these forms are themselves the regular means of instituting change” (Dewey, 1946, p. 31).

The decades that followed echoed Dewey's concerns in policy and urban planning fields, where researchers and government administrators devised ways to enable the contribution of citizens. Early examples of these are introduction planning cells in Germany and citizen juries in the US, and consensus conferences in Denmark, between 1970s and 1980s (Voß and Amelung, 2016). As citizen engagement started gaining prevalence and became procedural in shaping government agendas, “a new technocracy of political procedure” emerged, naturalizing standardized forms of public participation such as polling, surveys and citizen agendas, which can lead to “whitewashing policies” (Voß and Amelung, 2016; p. 16). When participation gets absorbed back to institutionalized structures of forming and negotiating public matters, creation of alternative forms of public remains a continued necessity, to challenge dominant framings of issues and structures they contribute to their stabilization.

Almost six decades later from Dewey's *The Public and Its Problems* (1946), Bruno Latour (2005) asks what a contemporary form of assembly might look like if parliament is not necessarily the best one. He introduces the idea of a Parliament of Things, where objects and issues themselves take a central role in political life. Latour's concept of Dingpolitik (or "thing-politics") shifts the focus from people-centered politics to object-oriented politics. In this framework, objects—such as environmental crises, technological innovations, or public health concerns, but also material environments of

participation—actively shape how publics form and the issues they address. Marres (2005) highlights that issues are not pre-existing entities but are constructed through public involvement and articulation that mobilize of associations to diverse matters of concern. Publics emerge when existing institutional structures fail to account for new or complex associations between issues and their impacts, thus flattening controversies. This approach foregrounds the pluralities of public formation, where multiple scales and spaces interact to shape how issues are framed. Marres (2012) emphasizes that material participation is not about instrumentalizing objects to solve problems. but about recognizing how object and material conditions can connect publics that may otherwise lack platforms to raise their concerns.

This plurality manifests in the contemporary networks of knowledge production around various environmental, social, technological matters where distributed and networked collectives form publics beyond institutional structures and procedures. Chilvers and Kearnes (2016) draw our attention to these emergent participatory collectives, which can redefine the objects of political concern. They draw on STS and expand on object-oriented politics to propose a relational co-productionist approach, which gives equal consideration to the agency of objects, procedural formats and human actions - each shaping and being shaped by participatory experiments of these emergent collectives. The implication is that public matters are not only discursively constructed but are also grounded in material commitments, capable of transforming practices and visions beyond the immediate sites of participation.

Participatory Design. My approach to design's role in this broader and evolving context of collective inquiry is informed by Participatory Design. This collaborative design approach emerged during the Scandinavian worker movements of 1970s with the goal of engaging the voice of workers in shaping their tools and workspaces, sparking a broader agenda of democratizing design (Björgvinsson, Ehn and Hillgren, 2010). Huyberts, Benesh and Geib (2017) note that in this early work, participatory design practitioners aimed at reconfiguring power relations and barriers of expertise for more democratic negotiation of issues. However, as the public sector transformed with technological developments and accompanying privatization, the once-tangible interfaces of institutions gave way way to more fragmented, networked forms of governance and participation (Huyberts et al., 2017).

The field of participatory design and its engagement with the publics has been transforming in response to this changing landscape where participation is not an activity delimited to a project site but a dispersed form of collective inquiry and collaboration (Ehn, 2008; Binder et al., 2008; Manzini 2015). DiSalvo (2011) posits collective articulation of issues as a design practice, which centers open-ended and pluralism in creating the socio-material conditions for bringing together people who hold shared concerns and commitments. However, the changing landscape of participation fostered by a networked society leads to the difficulty of framing ecosystems with well-defined stakeholders, where the outcomes of participation might not be visible to all stakeholders at various stages of a development (Brereton and Buur, 2008; Dalsgaard, 2010). Asad et al., (2017) state that temporary engagements between communities and public

organizations remain transactional, in contrast to relational constituents of participatory civics.

In these new forms of distributed inquiry, designers are not only facilitators but also “networkers” who configure dynamic engagement of various actors across various stages of transition efforts (Ceschin, 2014; p. 16). This leads the objective of participatory design to shifts towards imagining processes that can accommodate participation as a continuum. Bjorgvinsson, Ehn and Hilgren (2010) frame this as ‘infrastructuring’ , creation of new socio-material assemblies that bring different groups together in activities that help them surface their diverse understandings of issues, explore frictions and collaborative approaches.

While design has been developing methods, tools and processes to stage collective inquiry beyond institutional constraints, there is still a lack of understanding of how these relate to broader landscape of collective inquiry, which involves entangled activities of understanding, problematizing, negotiating, and instituting change in non-linear ways, where making sense of change is inseparable from putting it into action. In her account of participatory prototyping in the policy field, Kimbell and Bailey (2017) point out the "yet unresolved questions about how a processual, materialized and local understanding of problems and solutions intersects with formal democratic structures and processes". They state that these formal structures involve "other kinds of analysis and experimentation, such as randomized control trials which rely on different logics and routes to legitimacy" towards delivering policies at scale (p. 222).

Moore and Elliott (2016) raise a similar conflict between the need for capturing the situated and tacit knowledge of the public and collecting large amounts of data in

large-scale public planning projects. They state that the use of traditional participatory design methods such as design charrettes or workshops do not necessarily enable the transfer of knowledge to the final decision-making, while on the other hand, large-scale data collection fails to capture the tacit knowledge of participants resulting in misrepresentation and decontextualized information.

This work builds on contemporary PD principles of pluralism and contestation in making sense of systemic issues and proposes narratives as a mechanism for enacting these principles. I acknowledge the critiques about the gap between alternative participatory spaces and formal decision-making and call for tending to the entanglement of these assemblies with broader mechanisms of collective inquiry. Central to this approach is the commitment to account for inherent tensions of diverse ways of knowing, between prevailing structures of knowledge production and decision-making, and alternative assemblies staged by design

Critical and Participatory Futures. My approach to design interventions is informed by critical and participatory futures practice, with an emphasis on the entangled practices of knowledge production and visioning. Ramos (2017) identifies five modalities in Future studies: (1) *predictive* and (2) *systemic* futures that aim to forecast macro social and economic trends at increasing complexity, (3) *critical* futures that scrutinize discursive construction of specific future visions and their political implications, (4) *participatory* futures that aim to foster collaborative and generative exploration of futures within groups and lastly, (5) *action-oriented* modality driven by a shared commitment to

materializing desired futures. While these modalities build up on one another, they differ in their conceptual grounding and orientation.

In an earlier taxonomy of Futures research, Sohail Inayatullah (1999, p.1) distinguishes Critical Futures from predictive and interpretive approaches by its focus on “creating distance from current categories” that underpin how societies’ make sense of the past, present and future worlds. Rather than accepting concepts such as “population” as fixed variables, Critical Futures treats them as social and linguistic constructs. Through this critical distance, it aims to disrupt visions that derive their legitimacy from the perceived inevitability of these notions and open up alternative ways of thinking. This research is focused on this discursive construction of futures in collective sensemaking practices and its implications on stabilizing visions of systems change, rather than predictive capacity.

In recent years, a broader field of practice has been evolving at the intersection of design and critical futures practice. In approaches such as speculative design (Dunne and Raby, 2013), design fiction (Bleecker, 2009) designers materialize aspects of futures that are obscured by dominant narratives as a way to foster critical reflection and debate on future possibilities and present issues. However, their capacity to engage a broader public in this capacity has been scrutinized; for instance, Candy (2018) points out the reservation of futures thinking to a small number of experts, and Malpass remarks positioning of works in spaces like academia and galleries, limiting public participation (Mitrović and Auger, 2019). Moreover, as Forlano and Halpern remark (2023; p. 22:8), speculative design and design fiction have been critiqued for being “overwhelmingly white, male, able-bodied, wealthy and Western fantasies of utopia and dystopia”.

In the face of this challenge, participatory futures aim to engage broader public in critical reflection about futures (Mitrović et al., 2009) and elicit alternative futures from diverse groups. These engagements invite communities to materialize their aspirations, identities, and perspectives and open up possibilities obscured by institutional narratives (Baumann et al., 2017). For instance, Speed et al. (2019) co-created speculative prototypes to explore alternative social orders, allowing participants to physically engage with and perform futures outside the boundaries of their everyday experiences. Candy and Dunagan (2017) similarly emphasize the potential of experiential scenarios to invite people to inhabit and interact with future possibilities. Experiential futures create material-discursive environments where people can actively perform these futures, embodying alternative roles, mindsets, and relationships, thus moving beyond static representations of alternative futures.

However, the influence of participatory futuring practices on mobilizing change remains under-explored. An exception to this would be the participatory information systems in the public sector, which become vehicles for receiving input from the public; neglecting the inherent political aspect of participation (Puri et al, 2009). Examples of participatory futuring mostly involve materializing collective visions by convening individuals in self-contained co-design engagements. The engagement of participatory futuring with larger production of visions is a largely conceptual one, in the form of countering the dominant narratives. Yet, in order to translate these visions into practice (without being absorbed back into the dominant orders), participatory futures need to engage with the existing infrastructures of collective sense-making.

Several design and futures scholars have been expanding design futures with insights from Science and Technology Studies to envision critical engagement with futures in ways that account for the social and material mechanisms that sustain collective visions of desirable futures. Wong et al. (2020) invite designers to expand speculative engagement with futures beyond specific use contexts and account for the broader social, material and political environments that sustain specific future visions. Their approach of infrastructural speculations, bridges key ideas from Science and Technology Studies such as historic continuity of collective visions, institutional stabilization of social and material orders and the lived work of maintaining them with speculative actions. Similarly, Kozubaev et al. (2020) argue that design futures can achieve broader impact by attending to the representations and performances through which futures are framed, granted credibility, and materialized. Drawing on STS scholarship, they highlight how specific temporal representations, material interactions, institutional arrangements, and linguistic schemas shape how people envision and navigate futures, proposing this as a means to foster a more reflexive design futures practice.

The STS-informed sensitivity to how future visions get shaped, maintained and negotiated in the making of publics can help expand the potential impact of participatory futures practice. Accounting for the institutional structures, social networks, and infrastructures that span beyond the context of specific interventions is key for influencing not only debates but the very making of futures. Rather than focusing on future visions as self-contained representations, this work considers the embeddedness of future visions in various activities of knowledge production in contexts of movement

building, policy-making and scientific research, and explores what shifts in these contexts can sustain alternative visions.

1.5 Methodology

This research is driven by the following question: How might design engage narratives as a sensemaking mechanism to foster pluralistic, inclusive, and critical collective inquiry in systems change? To explore this research question, I conducted a series of studies using a qualitative approach. My aim was to understand how the social and material context of sensemaking activities influence co-production of knowledge and imaginaries in collective inquiry for systems change. The goal is to propose an approach for designing these contexts to foster pluralistic, inclusive, and generative ways of making sense of systems change.

1.5.1 Context of Study: Food Systems

This research is situated within the food system domain, conducted primarily in Chicago, and framed within the broader context of the United States. The food system offers a compelling context due to food's centrality to humans' social and economic lives.

What Are Food Systems? Unpacking the term *food system* itself reveals a landscape of diverse meanings and contestation. Early conversations on the food system can be found in the 1930s in conversations on diets and supply chains (Brock, 2023). However, the concept gained broader public attention with a series of reports by the National Commission on Food Marketing in 1966 (Clancy, 2022), which raised concerns over monopolistic practices, market power concentration, and the implications for farmers and

consumers. In the seminal 1972 report *The Limits to Growth*, Donella Meadows and colleagues identified food production as one of five critical systems requiring significant transformation to address pollution, resource depletion, and other implications of unbound growth that push planetary limits. This interest in a systemic approach was brought forth by the new range of societal and environmental issues that are entangled with the food system, such as climate change and nutrition-related public health issues, that demanded more holistic approaches combining the perspectives and know-how of different disciplines (Peters and Thilmany, 2022).

Although food systems are most commonly understood as supply chains, tackling the complex issues in the food system requires a more holistic view that accounts for its interconnectedness with natural and agricultural systems, infrastructures for the distribution of materials, data, and information, and other socio-economic systems such as education, culture, and finance (Clancy, 2022). The definition by Pinstrip-Andersen and Watson II (2011, p. 3), “the aggregate of all food-related activities and the environments (political, socioeconomic, and natural) within which these activities occur,” underscores the need for such a holistic perspective. As Samara Brock (2023) notes, due to the interpretative flexibility of the term, when diverse actors such as governments, corporations, grassroots organizations, and NGOs discuss the food system, their framing may reflect differing—and potentially incompatible—perspectives and priorities. The need to account for diverse perspectives, navigate conflicting agendas, and find alignment without reductive interpretations makes food systems a productive context for the aim of this research.

In *Food Studies: Matter, Meaning, Movement*, Szanto, Di Battista and Knezevic (2022) emphasize that making sense of food requires a pluralistic approach that can account for its fluidity and complex ties to diverse aspects of humans' collective lives. The authors advocate for understanding the *materiality* of food, its *meaning* and its *movement* in relation to one another, resisting compartmentalizing these facets: "Food, divided, ceases to be food" (p. 2). They elaborate by stating: "While positivism, science and causality are useful frameworks for some aspects of food studies, so are poetry, wonder, affect and un-knowing" (p. 1). Building on this perspective, this research explores how narratives can open space for diverse ways of knowing within collective sensemaking efforts to engage with the rich complexity of systemic issues.

Problem Domains: Wasted Food and Food Insecurity. The studies in this research focus on the systemic and interconnected issues of food waste and food insecurity. A 2014 report by the United States Department of Agriculture (USDA) estimated that roughly a third of all food in the country goes uneaten, excluding losses at the farm and distribution levels, amounting to \$161.6 billion in economic losses for the baseline year of 2010 (Buzby et al., 2014). Beyond its economic toll, wasted food carries a significant environmental cost. The Environmental Protection Agency (EPA) attributes 58% of methane emissions from landfills to wasted food (Krause et al., 2023) and estimates the annual greenhouse gas emissions from wasted food in the U.S. to be equivalent to those of more than 50 million gas-powered passenger cars. Furthermore, the freshwater and energy lost to wasted food each year could meet the needs of 50 million American households (Jaglo et al., 2021).

In 2015, the USDA and EPA launched a national initiative to address the significant costs of food waste and loss, setting a goal of reducing food waste by 50% by 2030 (USDA, 2015). Since then, the EPA, and many other institutions have worked to mobilize collaborative efforts across diverse stakeholders, sectors, and scales of operation to identify and implement pathways for reducing food waste. Despite notable progress, the EPA's 2023 report reveals that the percentage of food exiting the supply chain continues to increase, underscoring the ongoing need for concerted action (EPA, 2023). The first and third studies in this research are situated within collaborative efforts to address food waste, within a multi-stakeholder coalition in Chicago and a nationwide research network in the United States, respectively.

The second study focuses on the distinct, yet related issue of food insecurity in the Chicago context. Despite the staggering amount of food wasted in United States, 13.5 of the American population is estimated to be food insecure in 2023 (USDA ERS, 2024). According to USDA's definition, food insecurity is the "limited or uncertain availability of nutritionally adequate and safe foods or limited or uncertain ability to acquire acceptable foods in socially acceptable ways" (USDA ERS, 2024)¹. With an ever-increasing American population experiencing food insecurity, food assistance programs have been expanding nationwide, with food banks as their primary mechanism (Bacon and Baker, 2017). However, despite their established role in the food system, food banks remain a contested solution. The debates surrounding them reflect conflicting visions of what constitutes a good food system and reveal distinct imaginaries about the broader organization of social life and the use of shared resources.

¹ This definition is based on the 1990 report by the Life Sciences Research Office of the Federation of American Societies for Experimental Biology (Anderson, 1990).

In her seminal book *The Sweet Charity*, Janet Poppendieck (1999) examines how emergency food assistance has evolved from a temporary measure into an institutionalized feature of the American economy, under the guise of solving hunger. Poppendieck interprets this shift as a symptom of a broader transition from social policies that recognize food as a human right to localized interventions that isolate and segregate those requiring assistance. “Fighting hunger has become a national pastime,” she observes, highlighting how emergency food programs serve as a “moral safety valve,” easing the discomfort of witnessing hunger persist amidst abundance (pp. 24–26).

In *Feeding the Other*, Rebecca de Souza (2019, p. 18) critiques the narratives that stigmatize those experiencing hunger and food insecurity, which frame poverty as the result of individual failures and lack of hard work—rather than as the outcome of historical discrimination and systemic neglect rooted in gender, race, and socio-economic inequalities. She contends that this neoliberal stigma underpins the charitable food industry as a boundary between those who give and those who receive, sustained by what she calls “*discursive privilege*”: the power to shape stories about who the Other is and who “We” are. Stigma, De Souza argues, is inherently tied to the power to craft narratives of similarity and difference, of Us and Them, using these divisions to justify and perpetuate systems of oppression.

Despite the critiques about their legitimacy as a long-term solution to food insecurity, the growth of food banks did not slow down. In 2021, Atlanta Community Food Bank unveiled its new 345,000 square-foot facility (size of approximately five and a half football fields), earning the title of largest food bank in the world (Constanzo, 2022). The trend can be seen in food banks all over the US, alongside an ever-growing need for food

assistance. Daniel Warshawsky (2023) argues that, despite being viewed as an efficient and independently-run mechanism for reducing food waste, food banks have not been proven effective in significantly reducing food waste or addressing food insecurity. This can partly be attributed to a lack of data and the complexity of variables that make evaluating systemic impact challenging. However, he contends that food banks serve another important function: by offering a symptomatic solution to food waste and insecurity, they obscure the shortcomings of governments and corporations in fulfilling their promise of an efficient food system that meets the population's demands. As Warshawsky highlights, seeing the charitable food system as the solution to food waste sidelines their related yet distinct roots, the poverty, costs of housing, inequitable access to food and resources that sustain food insecurity, as well as the inefficiencies and costly redundancies of industrial food systems that prioritize profitability.

As these debates show, the difficulty of tackling these issues partly stems from the divergent perspectives about their framing, significance and ways to address them. This research approaches wasted food and food insecurity as domains for amplifying the emergent forms of social, economic, environmental organization; and potentially shifting the deeper mindsets that underpin inequitable and unsustainable systems.

Food Systems at the Crossroads of Futures. The controversies concerning the issues in the food system reflect the divergent worldviews, mindsets, and values. Thus, negotiating the future of food means negotiating the future of interconnected social, economic, and natural systems, shaped by collective aspirations and tensions. In the critical essay collection *Utopian Foodways*, Botelho et al. (2019) discuss how a meal, as a socially

ordering practice, reflects humanity's orientation toward a better world—transcending mere survival to foster symbiotic relationships that underpin the production, distribution, and sharing of food. At the same time, food's centrality to environmental, economic, and social orders makes it into a vessel for collective hopes and fears: nostalgia for past worlds, conflicting visions of a better life, fears of planetary loss, or a deeper yearning to transcend bodily needs and planetary dependence.

Stock et al. (2018) propose food utopias as tools for mediating a rich, contested and hopeful dialogue for sustainable and just food systems. These utopias are not bound to the literary domain; they materialize in the ongoing experiments with alternative ways of producing and sharing food, in mutual aid networks, ecovillages, and food worker movements that express ways of being governed by values other than limitless growth and efficiency. In their 2014 edited volume *Eat, Cook, Grow: Mixing Human-Computer Interactions with Human-Food Interactions*, Jaz Choi, Marcus Foth, and Greg Hearn explore the role of technology in shaping food futures within urban environments. Through emerging interactions around growing, distributing and sharing food in urban contexts, they capture how “playfulness” (Choi, 2010) manifests as voluntary and creative acts towards liberation in spaces of collaborative, contested and hopeful inquiry for systems change. Food, entangled with economy, politics, culture, and the built environment, emerges as fertile ground for re-imagining collective human lives and cultivating new forms of civic engagement required for systemic transformation. For instance, Smith and Schuler (2014, p. 82) illustrate how urban agriculture initiatives mobilized to address community needs exemplify “civic intelligence.” This concept refers to organizations and coalitions collaboratively building learning networks to tackle

systemic issues by experimenting with and iterating on new approaches. Similarly, Willis, Frosch, and Struppek (2014) highlight how these become sites where communities tactically use diverse technologies from social media to spatial imaging to reclaim to commons and organize use of shared resources. The studies in this research are positioned at the fault lines of these emergent forms of civic intelligence and the institutionalized knowledge systems, where distinct narratives of what makes a good food system compete for legitimacy.

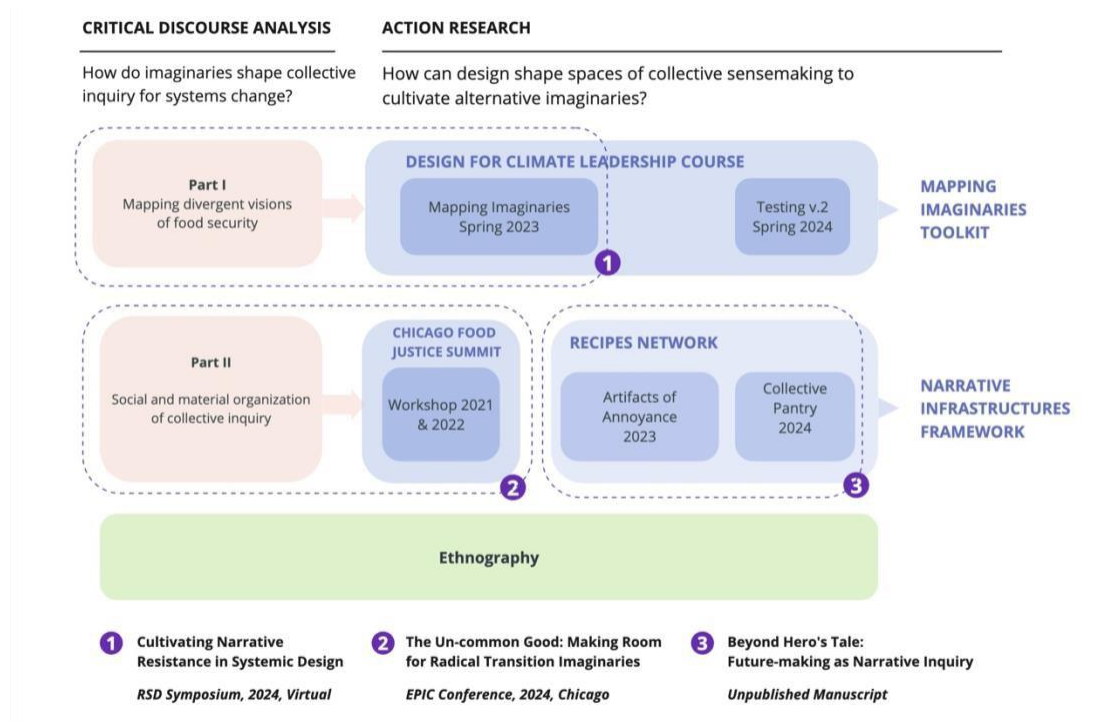


Figure 1.3. Research structure and contributions

1.5.1 Ethnography

This major stream of inquiry involved ethnographic observations of spaces where alternative food systems are envisioned, experimented with, and negotiated, as well as documenting my journey of learning to prevent food waste, from my kitchen to

participating in food rescue operations across the city. This dual approach not only deepened my engagement with the contextual complexities of food system transformation, but also allowed me to explore the fluid interaction between personal and collective ways of knowing, a thread that continued throughout the interventions.

I conducted online and in-person participant observations of local food organizations across Chicago, including conferences, summits, working group meetings, and volunteer activities. Initially, these ethnographic observations were limited to online engagements due to pandemic restrictions. The observations primarily took place within grassroots and non-profit circles, as well as local community gatherings and academic researcher communities, since I had limited access to business and government ecosystems where membership requires institutional representation. These observations also provided insight into informal and non-institutionalized spaces of discourse, where niche imaginaries were performed, primarily by community organizations. Throughout these encounters, I documented the ways in which groups expressed their imaginaries, ranging from place-based activations such as community potlucks and artistic interventions to artifacts such as zines, posters, and food itself.

These observations helped me navigate the contextual complexities of food system transformation in Chicago, revealing deep entanglements with racial, gender, and religious identities—including my own. In a city where food geographies remain marked by historic segregation and the oppression of Black, Brown, and other communities of color, understanding the historical struggles and lineage of resistance was essential. Most importantly, these encounters allowed me to build relationships with a diverse network of growers, advocates, and policymakers, helping me show up as an ally.

The ethnographic study also included my personal experiences with food and food waste. From January to June 2022, I documented my household efforts to prevent food waste and my involvement in food rescue efforts through volunteering with the Chicago Food Sovereignty Coalition. Reflecting on these experiences allowed me to explore the complexities and controversies of food waste through the lens of my own experiences and encounters, such as disparities in access to food retailers, cultural perceptions of food and waste, and deeper questions of accountability, such as the reliance on unpaid volunteer labor in food rescue. This reflection directly informed my intervention within the RECIPES Network, where I used storytelling to invite situated and subjective perspectives of network members to foster critical discussions around food waste.

1.5.2 Critical Discourse Analysis

This analysis frames collective inquiry as a negotiation of meaning and visions between competing socio-technical imaginaries, which stakeholders perform through their narratives of change. By analyzing public narratives from various stakeholder groups, the findings frame three distinct socio-technical imaginaries that shape collective efforts to address food insecurity in the Chicago context. Mapping these imaginaries in relation to one another, and surfacing the tensions between them, helped frame the systemic change in question as a constellation of diverse aspirations for the future, and associations with various social, economic and ecological concerns. This analysis led to the development of a methodological approach to incorporate diverse stakeholder visions in framing systemic issues, that was implemented in course context.

1.5.3 Action Research

Between 2019 and 2024, I participated in various collaborative efforts addressing systemic challenges within the food system. These efforts focused on two overlapping issue spaces: (1) food insecurity within the Chicago context, and (2) food waste, both within Chicago and the broader USA context.

Rather than validating a hypothesis, the prototyping was exploratory. In this phenomenon-driven design research, prototypes were used to stimulate new knowledge and engage with the entangled practices of knowledge production and visioning, rather than testing predefined hypotheses (Stappers et al., 2014; Wensveen and Matthews, 2014). By using the data from these interventions, researchers observed how narratives could foster groups' capacities to:

1. Engage with diverse imaginaries;
2. Contend with dominant framings of systemic issues and their underlying values and paradigms;
3. Translate friction into envisioning radical possibilities.

Intervention 1. Chicago Food Justice Summit: Seeds of Futures. The first intervention took place at the Chicago Food Justice Summit, where I staged a participatory futuring activity titled ‘Seeds of Futures’. Participants from diverse sectors of the food system used activity cards to articulate their vision for a better food system and the current ‘seeds’ they were planting to achieve it. This asynchronous booth allowed participants to display their contributions while engaging with others' submissions.

The collected visions were synthesized into key themes, reflecting the food system's significance for diverse aspirations around community, health, identity, culture, and ecologies. These themes, illustrated and published in the post-summit zine, highlighted the 'seeds' shared by participants. The analysis explored how the activity fostered diverse imaginaries by grounding anticipatory sensemaking in subjective and situated accounts of 'futures in the making,' emphasizing food's complex role in shaping ways of being. It also considered the activity's embeddedness in the food justice movement, focusing on the alignment of vocabularies, metaphors, and the role of researchers as co-creators of narratives.

Intervention 2. Design for Climate Leadership Course: Imaginaries Toolkit

Development. The second intervention involved developing a methodological approach to engage with diverse imaginaries in framing systemic issues and envisioning desirable futures. Derived from the learnings of critical discourse analysis, this approach translates the analytical framework into a practical toolkit for practitioners. The methodology uses stakeholders' narratives of change to identify their imaginaries and explore the systems change through the perspectives of these imaginaries. The tool provides a visual scaffolding, and step-by-step approach for analyzing, comparing and contrasting these narratives to uncover diverse interpretations of systems change, identify the tensions amongst these, and leverage these tensions to challenge dominant framings and hold space for marginalized views.

Initially, the toolkit was prototyped through self-application in the domain of mobility, serving as a preliminary evaluation of its efficacy. It was then integrated into

the Design for Climate Leadership course at the Institute of Design, led by Prof. Weslynn Ashton and Maura Shea. In Spring 2023 and Spring 2024, the course collaborated with the City of Chicago on food waste prevention projects. At the onset of the project, the students used the toolkit to analyze stakeholder narratives, map their imaginaries, and explore tensions, reintroducing these tensions as critical considerations in subsequent discussions.

The artifacts created by the students were analyzed to evaluate how the toolkit facilitated engagement with diverse imaginaries, shifted the framing of systemic issues, and contributed to more authentic stakeholder alignment by holding space for differences. Over two years, the toolkit was refined and will soon be published as an open-access tool for designers and systems change practitioners.

Intervention 3. Multiscale RECIPES Network. The third intervention involved a series of collaborative sensemaking engagements staged within a research network tackling the issue of food waste and loss, and experimented with storytelling as a way to foster pluralistic, critical and creative exploration of waste-free food futures. From 2022 to 2024, I participated in the Multiscale RECIPES (Resilient, Equitable, and Circular Innovation with Partnership and Education Synergies) Network, an NSF-funded, multi-institutional research initiative aimed at addressing food loss and waste. This opportunity was made possible by my advisor, Prof. Ashton, who had been an active member of the network since its initiation in 2021.

RECIPES takes a convergence approach, fostering transdisciplinary collaborations between diverse disciplines to understand complex drivers and impacts of

food waste and loss, and identify solutions that can foster sustainable, equitable and resilient food systems. Design occupies a central role in the network's convergence framework, positioned as a boundary-crossing discipline that fosters collaboration between diverse disciplines, and supports iterative co-creation of solutions rooted in human-centered and life-centered approaches. The network's structure, composed of multidisciplinary clusters, aims to foster fluid participation and knowledge sharing across teams, aiming to creating spaces for both in-depth collaboration and serendipitous encounters.

In 2023 and 2024, I collaborated with fellow designers and researchers on two interconnected projects that employed creative storytelling to examine food waste through the diverse perspectives and lived experiences of researchers. These projects utilized a blend of physical and virtual platforms, complemented by workshops, to create the conditions for cultivating alternative story systems on food loss and waste (FLW). The goal of these interventions was to disrupt the technoscientific imaginary of food waste prevention that primarily emphasizes efficiency and optimization of food and data flows and re-anchoring the exploration of a waste-free food future in its deeper ties to cultural, social, ecological values and mindsets.

The first intervention, *'Artifacts of Annoyance'* took place in 2023 and involved curating a small exhibition space that showcased artifacts exemplifying personal frustrations with the current food system - reflecting tensions between the practices that are normalized in the status quo and the aspirations for a different food system. The interactive exhibition was complemented by a workshop that invited participants to share their own annoyances with the current food system and use these shared frustrations as

gateway to explore the deeper values and mindsets that guide their imaginaries. The workshop culminated in the creation of speculative artifacts, through which participants imagined a future food system aligned with their values, which were added to the exhibition space.

The insights from this intervention informed the development of another project in 2024, titled '*The Collective Pantry*'. The project evolved from a self-guided storytelling activity staged in the in-person convening of the network. The activity invited network members to share their personal experiences of preventing food waste, and use these stories to explore *ingredients* for creating a waste free food future. By using personal accounts of food waste, the activity sought to uncover drivers of food waste prevention that tend to get excluded in scientific knowledge production, such as cultural perceptions of food or social practices of sharing food with neighbors. The stories collected through the activity were displayed in an interactive web platform, which included a virtual *pantry* of *ingredients*. This initiative culminated in a virtual workshop at the network-wide meeting titled 'Pantry Cook-Off', inviting researchers to explore how these ingredients could inform their approaches to creating waste-free food systems, in a playful 'cooking contest' format.

Combined, the two interventions leveraged storytelling as a sensemaking mechanism that engages situated knowledges of participants, in a space where ways of knowing other than scientific, empirical ones are given little legitimacy. The scientific portrayal of the food system often emphasizes the visibility and coherence of resource flows and causal relationships, aiming to present a clear, unified picture that can support strategic sensemaking for systems change. However, this portrayal and its promise of

singular reality to be acted on, risks overshadowing marginal approaches and flatten controversies, inadvertently reinforcing the transactional mindsets that contribute to the very issues it seeks to address. By engaging the diverse, subjective and lived perspectives of members, the interventions surfaced the messy but rich reality of diverse food ways, making room for a relational view of food system transformation that emphasizes the shifts in culture, mindsets, hierarchies of values needed for radical change.

1.6 Overview of Chapters

The findings from the studies were synthesized into three articles, making up the Chapters Two, Three and Four of this dissertation. The articles build upon the insights gathered through interventions focusing on three discursive levels of collaborative sensemaking and engage with different theoretical perspectives to explore intentional engagement with sociotechnical imaginaries in design practice (Figure 3). Together, they propose a broader understanding of how narratives function as a sensemaking mechanism across various scales of collective inquiry, considering the interactivity between wider discursive environments where imaginaries are stabilized and negotiated, and participatory collectives where alternative imaginaries can be cultivated.

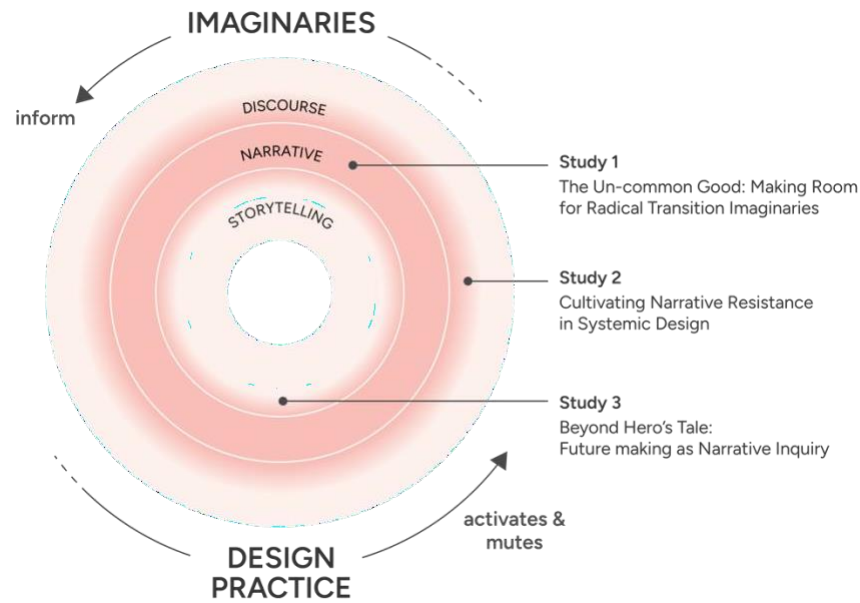


Figure 1.4. Positioning of Studies in Exploring Engagement with Imaginaries Across Three Discursive Levels

1.7 Scope of Research

The questions that drive this research and the approaches that inform it are part of broader debates, spanning across various fields of study and practice. For this reason, it is important to clarify the boundaries of this exploration by explicitly addressing what this research is not about.

While situated in the domain of food systems, this research does not seek to define the issues or prescribe solutions. My engagement with the extensive scholarship on food systems is at an introductory level, aiming to illustrate the significance of issues and capture key debates. I observe the current debates around the issues of wasted food and food insecurity primarily within the Chicago context, while also considering the broader implications within the United States. Although the insights from these studies

may inspire new ways of thinking and working for food system transformation, this research remains exploratory rather than prescriptive.

The second boundary concerns the scope and nature of narratives, discourse, and storytelling in this research. I adopt a design perspective to understand how narratives shape specific imaginaries of systems change within organizational settings—not within broader public discourse, which is far more expansive. While I draw from various interdisciplinary fields to understand how narratives are collectively constructed, circulated, and contested, my focus is on the role of designed social and material environments in enabling or constraining the articulation of different narratives of change. Narratives anchor my engagement with broader organizational discourse and storytelling practices, examining how the social and material contexts of each, by design, can enable or disrupt institutional narratives.

CHAPTER 2

THE UN-COMMON GOOD: MAKING ROOM FOR RADICAL TRANSITION IMAGINARIES

2.1 Introduction

In 2023, humanity experienced the hottest temperatures on record in its existence. We heard numerous calls for global cooperation, pointing to the planetary crisis that threatens the future of all species, including our own. Calls aim to radically transform our lifestyles, neighborhoods and cities, alongside the global flow of resources towards a ‘common good’. Yet, humanity does not agree on what that transformation ought to look like. Despite the apparent universal desirability of the United Nations’ Sustainable Development Goals or comparable overarching transition objectives, social groups often have disparate visions for what achieving these goals should look like, as can be observed in their competing narratives. This diversity of interpretations of common good is often regarded as a barrier to collective action, where we aim to create a shared vision that can guide collective efforts (Kania and Kramer 2013).

However, we argue that, rather than a barrier, this diversity is a fundamental and necessary condition for forming inclusive transition coalitions, and a creative resource for envisioning alternative futures beyond the dominant norms and mindsets that govern our world. In fact, the concept of common good, with its significant influence in Western political thought is a vague and contested one (Jaede 2017). Yet this very vagueness and the constant work of defining what common good means is a fundamental condition of democratic collective action (Mouffé

2000; Mansbridge and Boot 2020). Rather than rushing to align, we invite designers and transition practitioners to engage with this contested nature of common good through diverse narratives of stakeholders, and contend with the different value systems and mindsets that social groups uphold in imagining better worlds.

We propose a practical approach for designers and transition practitioners to use narratives as part of a collaborative sensemaking process for shaping inclusive and imaginative transition initiatives. We emphasize looking at narratives as a way to account for diverse visions of common good and surface the tensions amongst these, as a way to destabilize dominant ideals of common good in favor of radical possibilities. We illustrate how this approach can be used through an example of food system transition in Chicago and, more specifically, drawing on the Design for Leadership course at the Institute of Design (ID) at Illinois Institute of Technology. In this course, faculty and students facilitated a collaborative sensemaking process in the context of an emerging citywide initiative for food waste prevention and management in Chicago. We introduced a set of tools and frameworks to identify and draw out the tensions across stakeholders' narratives in the sensemaking process, in tandem with other design tools and methods.

In this paper, we share our experience using this approach based on the outputs of the course, as well as our lived experience, to examine the following questions:

1. How can practitioners, who are working towards transformative change, use narratives for a pluralistic framing of transitions that accounts for diverse and conflicting visions of common good that social groups uphold?
2. How might engaging with divergent ideas of common good promote more inclusive and creative imagination of transition pathways?

2.2 Theoretical Background

Seeing Transitions Through Narratives. Over the past decade, we have been observing a “narrative turn” (Goodson and Gill 2011) in sustainability scholarship, acknowledging the power of narratives, and storytelling, to stabilize and destabilize public opinion, as well as inspiring new ways of thinking (Bien and Sassen, 2020; Moezzi, Janda and Rotmann, 2019; Riedy 2020). “Narratives of change” (Dobroć, Bögel and Upham 2023; Wittmayer et al. 2019) or narrative change (ORS Impact 2021) illustrate how some interventions break the patterns of thoughts and beliefs surrounding an issue and build legitimacy around alternative pathways and imaginaries.

In the context of sustainability transitions, narratives are interpreted as meaning systems by which various stories are woven together to make sense of situations and events (Dobroć, Bögel, and Upham, 2023). As organizations and communities participate in the collective work of transformative change, they produce diverse narratives to frame their understanding of issues, portray visions of desirable futures, and advocate for specific pathways within the practical endeavor of the organization. They do so by purposefully making certain actors and relations visible, at the expense of others, and consequently legitimizing certain pathways as more desirable and feasible over others. Narratives take tangible forms, spanning from policy documents and impact assessments that guide large-scale initiatives to social media postings, artistic creations, and community spaces that offer glimpses into a world reimagined.

Narratives do more than represent reality, they actively construct it. And thus, they are a powerful force in how we make sense of the world around us, and envision its

alternatives. Jasanoff and Kim (2015) conceptualized this as SocioTechnical Imaginaries, “collectively held, institutionally stabilized, and publicly performed visions of desirable futures, animated by shared understandings of forms of social life and social order attainable through, and supportive of, advances in science and technology” (153). We draw from Science and Technology Studies (STS) and critical future scholarship to consider narratives as a mechanism through which our collective visions of the common good are constructed and performed as driving forces of these imaginaries.

Transition scholars utilize narratives to understand how actors form coalitions around specific imaginaries to contest the politics of transition policies, agendas and interventions (Friedrich et al. 2022, Moezzi, Janda and Rotmann 2017), and surface the controversies they involve (Baeris and Katzenbach 2021). While early research on imaginaries focused on public policy discourse and national agendas of science and technology production, a number of researchers have turned their attention beyond expert narratives to highlight local and translocal networks as spaces of resistance where alternative imaginaries are produced and performed (Chateau, DevineWright and Wills 2021; Dawson and Buchanan 2005). Tidwell and Tidwell (2018) state that when our collective notions of the “good life” are shaped by experts, “we are no longer looking at the ‘good life,’ but rather the ‘good’ life as perceived by a group whose perspective we are privileging above others.” (105).

We align with this view and argue that surfacing and nurturing alternative imaginaries in transition efforts is a fundamental capacity to move beyond the dominant mindsets that intentionally or unintentionally determine the rules and norms of our societies today. Seeing the world, and transitions through narratives offers a way to

embrace the multitude of lived realities that shape our understandings of the world and its possibilities – as opposed to a purely positivist mindset that seeks to seek a universal representation of reality, or bends to the “overarching politics of the real” (Inayatullah 1990). This is a call for alternative, inclusive, and pluralistic epistemologies that foreground situated understandings of the world (Haraway 1988; Paxling 2019; Suchman 2008) as opposed to a search for comprehensive and cohesive representations that the dominant view of transitions favor. As Paxling (2019) states, without changing how we produce knowledge, we risk reverting back to the normative categories that organize our understanding. Narratives, when seen as fragmented, situated and nonlinear accounts of lived and dreamed realities, can provide the flexibility for new and contested meanings to emerge.

Design as a Narrative Making Practice. Despite the growing interest in narratives, we observe a gap between their use for critical analysis of diverse claims and visions associated with transitions, and their generative application to encourage thinking and imagining beyond the dominant ideologies. In this context, the field of design occupies a significant role. The work of design is inseparable from the narrative construction of reality, and our collective imaginaries; both by shaping material arrangements that organize humanity’s collective life (Speed et al. 2019), as well as shaping the environments, processes, interfaces and tools through which we collectively learn, think and imagine.

In this sense, designers set the potential material conditions of how we make sense of the world (Krippendorff 2005; Krippendorff 2020; Tharp & Tharp, 2019), and

manifest it in social, material and verbal narrative forms. Joachim Halse (2013) proposes that unlike ethnography that is traditionally interested in the present and the past, design has the potential of tangibly articulating futures we would like to see while contesting the present. With or without intent to do so, design has the influence of stabilizing and destabilizing existing imaginaries (Forlano 2019) and enabling alternative ones (Southern et al. 2014). While this capacity of design has the potential to move our thinking beyond the dominant imaginaries, it can equally solidify the dominant ways of thinking, further obscuring radical possibilities. As Sohail Inayatullah (1998) states, “the discourse we use to understand is complicit in our framing of the issue” (820). We call for an explicit recognition of the narratives through which we make sense of the world and design its possibilities, given that we can’t address today’s challenges with the same mindsets that created them.

Reclaiming the Futures with Narratives. Transition narratives, which traditionally focus on pathways and drivers, get disconnected from the lived realities of communities who have been disproportionately impacted by the negative outcomes of policy and technology agendas or rendered invisible by these. In reaction to linear development visions that “empties the future” as a set of decisions to be made, Groves (2017) highlights community-based narratives that describe “lived futures” (34). Just like archeologists constructing a picture of past worlds from their fragments, narratives, in their fragmented nature, can allow construction of alternative worlds in ways that lure imagination away from mental constraints (Baerten 2019). As opposed to dictating a unified trajectory of events, critical futures invites participants to explore multiple

alternative worlds and historic trajectories through storyworlds that are anchored in daily experiences (Candy and Dunagan 2017; Howell et al. 2021; Mazé 2019).

Critical futures practitioners do so by cocreating alternative narratives of futures with communities, and centering their values and imaginaries in envisioning our collective future, through, for instance, experiential and provocative engagements or collaborative storytelling. They use narratives as a way of engaging with futures “from the inside” (Burdick 2019), in situated, subjective and provocative ways (75). These interventions aim to help participants defamiliarize with the present, engage with alternative imaginaries and challenge the dominant technodeterministic narratives of what drives our futures. They equally highlight how mainstream visions of technological progress often overlook considerations related to gender, race, and ability; ignoring the distinct frictions that marginalized groups experience, and moreover, their agency in shaping these futures as creators and innovators (Bauman, Caldwell, Bar and Stokes 2018; Harrington, Klassen and Rankin, 2022; Tran O’Leary, Zewde, Mankoff and Rosner 2019). By drawing on this critique, we call for an increased attention to the identities, values and imaginaries of local communities; not as beneficiaries, but protagonists of transformative change.

With the practical concerns of transition practitioners in mind, we propose that accounting for the existing transition narratives, and their counternarratives can be a way to engage with the diverse imaginaries often eclipsed by the dominant transition narratives. We argue that it is precisely at the intersections of these diverse imaginaries that we can reveal the tensions arising from different interpretations of common good, to inform a more inclusive and pluralistic framing of transition visions.

2.3 Approach

In this study, we propose a practical approach for contesting ideas of common good through the vantage points of different stakeholder groups in collective articulation of transition goals. We position narratives of change (Wittmayer et al. 2019) as a key mechanism through which sociotechnical imaginaries are constructed and performed: solidifying specific interpretations of public matters and visions of desirable alternatives, as well as legitimizing certain pathways while marginalizing others. Unlike more objective and linear formulations of change processes, such as theories of change, seeing transitions through narratives can help us more authentically and critically engage with the messy and often controversial nature of transitions, and consider possibilities beyond the ideals, pathways and values that are reinforced by the dominant narratives.

Rather than seeking to build a singular vision for different stakeholders to align on and act upon, we prioritize manifesting and working with the dissonances between these diverse ideas of common good, with the intention of better problematizing the transition itself, as we envision collaborative pathways (Figure 1). Driven by these goals, we propose a practical way for working with narratives together with ethnographic design methods to support collaborative sensemaking with stakeholder groups, in building inclusive and pluralistic transition visions.

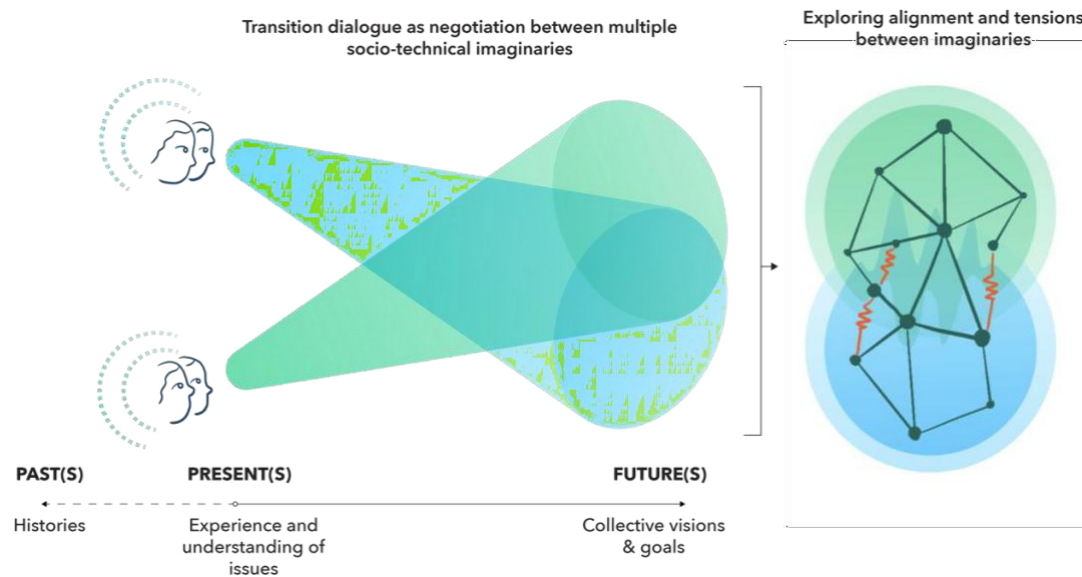


Figure 2.1 Exploring alignment and tensions between sociotechnical imaginaries of stakeholder groups.

Our methodology is grounded in Critical Discourse Analysis (CDA), an approach that has been widely adopted by energy and transition scholars in order to understand how distinct transition imaginaries get constructed and advocated for within policy, science and technology debates. CDA explores the social construction of the world (Gee 2014; 2017) and complex social phenomena through language as a system of representation. It differs from other discourse studies by its “constitutive problem oriented” (Wodak and Meyer 2018) perspective, as it seeks to understand how discourse “helps to sustain and reproduce the status quo” or transform it (Fairclough and Wodak 1997, 258). CDA pays attention to power relations of class, gender, race, ethnicity and other forms of social and political identities with the intent to “root out a particular kind of delusion” (Wodak and Meyer 2018, 7). Rather than a specific method, it is founded on this critical goal that can be accomplished through a combination of methods including text-based analysis as well as other ethnographic methods.

Drawing from CDA, we prototyped a set of tools to support making sense of and constructing transition visions with diverse stakeholder groups in an inclusive and pluralistic way. These tools aim to provide a scaffolding for:

1. Utilizing stakeholders' narratives to unpack diverse interpretations of common good that shape transition imaginaries,
2. Surfacing the differences in mindsets, values and worldviews beneath the overarching transition goals.
3. Contending with the tensions amongst these to foster inclusive and generative transition debates.

We prototyped this approach in the context of a graduate design course where we collaborated with an emerging transition initiative for citywide food waste management in Chicago, which involved a diverse group of stakeholders. We introduced a set of tools to uncover how different groups make sense of the issues, what values they are driven by, and what tensions they need to negotiate across their visions for the common good. Our learnings are based on the various materials that students created throughout the course project such as service maps, discussion notes, and online whiteboards. We also incorporated vignettes derived from our ethnographic observations and experience as project guides, collaborators and researchers within the Chicago food ecosystem.

Setting the stage: Making Wasted Food Matter. As the Institute of Design project team, we had been involved with national and local wasted food research for several years, and most recently through a National Science Foundation funded research network, Multiscale RECIPES. The workshop course emerged from our ongoing

conversations with organizational leads for the Chicago-based Wasted Food Action Alliance to align research and action across multiple organizations. We were asked to facilitate a collaborative sensemaking process for an emerging citywide initiative for fighting wasted food in Chicago. This effort was in support of the City of Chicago's participation in the Food Matters program of the Natural Resources Defense Council (NRDC). The program aims to support city administrations and local partners in leveraging their collective know how and efforts in implementing largescale food waste management policies and programs.

Chicago is a unique place for seeing how different food futures unfold next to each other, grounded in different ideas of what makes a good food system and the “common good.” The city aspires to become the “Silicon Valley of Food” with inclusive development of its food industry (Chicagoland Food and Beverage Network, 2021) and home to a growing number of food startups. Meanwhile, it has struggled to build food security for the 20% of its population without sustained access to nutrition, specifically in South and West side communities with a predominantly Black and Latinx population that have been historically disinvested and discriminated against. But this is not a tale of abundance and scarcity. In response to the lack of access to conventional food channels, there is a flourishing community-driven food system where communities are growing, circulating, sharing, and decomposing food in ways that build power, solidarity, and healing. In the absence of a public infrastructure or policy for tackling food waste, it is the community groups, public schools, smallscale composters, and urban farms that have been doing most of the heavy lifting for more circular food practices.

I take a break from the Chicago Food Justice Summit sessions on Zoom to join the ‘Future of Food’ webinar by Accenture that’s happening on the same day. It’s

a panel of corporate executives and food innovators. When asked about what he thinks about the future of food, one of the panelists tells the story of how he had recently been welcomed with his name as soon as he stepped into a high-end restaurant. *“They are caring enough to pronounce my name correctly”*, he emphasizes. I look back at my notes from the conversation I attended less than an hour ago. *“...radically reimagine our food systems for each other, for our families, for our communities and beyond. [...] our traditional food systems were never about access, it was about nourishing each other...”*. I feel torn between two realities and find myself thinking: *“Are we even living in the same world here?”* – Author’s personal reflection, February 2021

We indeed live in different worlds, where food has radically different meanings, told in different stories. Having been part of these debates in Chicago for several years, we knew that any conversation for initiating collective action would have to account for the diverse realities and imaginaries and consider the diverse meanings of the ‘common good’ for the groups who have been impacted by the existing systems in radically different ways. This included contending with our own role and assumptions as researchers who are part of an academic institution. Importantly, it involves contending with researchers’ legacy of knowledge extraction and exploitation of marginalized communities across the Chicago and US, where community members express weariness about their stories being taken and retold for someone else’s benefit (Chicago Beyond, 2019).

Course Structure. We organized the course as a platform to work with City-identified stakeholder organizations to map wasted food flows, identify current challenges for more effective management of these flows, and envision collaborative opportunities across the local food ecosystem. At the outset, our partners aimed to construct a collective vision statement and a strategic roadmap to guide the inception of a local initiative and rally

stakeholder endorsement through clear and tangible steps. While recognizing the need for actionable outcomes, we explicitly advocated for a less-linear approach that sets the stage for diverse positions to be made visible, with the aim of preventing superficial consensus around a singular vision.

Rather than prescribing a unified vision and specific pathways, the course sought to help stakeholders see the systemic issue of food waste from each other's vantage points both in an operational and critical sense. We adopted a pluralistic approach and aimed to reframe the transition as a contested space at the intersection of different imaginaries of stakeholders. We structured the course as a 14-week engagement where ten graduate design students worked with three key stakeholders representing the City, food waste rescue and recycling, and food service/retail (Figure 2).

Throughout the project, the students used a set of tools to analyze the narratives of stakeholders, with the goal of identifying the imaginaries that guide their efforts for creating a better food system and surfacing important shifts and tensions that need to be negotiated across these imaginaries towards a collaborative alignment. The students also used existing design tools such as service system mapping to illustrate the material flows of food and waste across food service, rescue and processing. By doing so, we combined critical reflection on transition goals with an operational lens, grounded in different perspectives of stakeholders.

The activities included two 'Chicago Food Matters Think Tank' events that convened a wider group of about 30 stakeholders from public policy, food service and retail, food scrap processors and local community organizations.

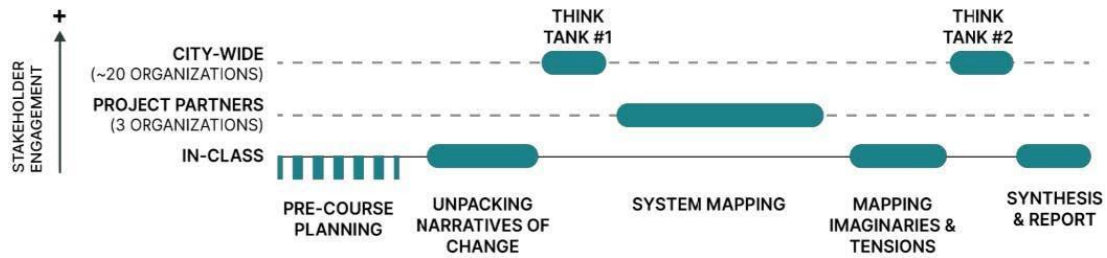


Figure 2.2. Timeline of course activities and stakeholder engagement.

Unpacking Narratives of Change. We grounded our course narrative in the proposal that there is no universal birds-eye (or God's eye) view of a system; rather, our understanding is partial, situated and subjective. It is partial and depends on where we are looking at it from. It is situated in our own context, worldview and intent. And it is socially constructed and subjective in that it is mutually shaped both by scientific knowledge as well as by our social and organizational realities and lived experience. We wanted to keep a distance from the idea of the all-seeing designer visualizing and taming a complex world. We initially formed four student teams, each working with a project partner to understand the issue from where they stand, making sense of these views, and constructing a patchy but shared picture.

We started the project by exploring how the project partners frame their vision for transformative change in their publicly available narratives (e.g., on their websites or social media), and how they position their role within this vision. We introduced a template to help unpack the meaning of overarching transition goals (such as economic development and community empowerment) through the narratives of the different stakeholder groups. We used a dialogic approach, where this template served as a shared scaffolding between the student teams, to compare how stakeholders interpret overarching goals of the transition efforts, by focusing on where they diverge and

converge. Since discourse can involve any material that joins the construction of meaning (Krippendorf 2005; Wodak and Meyer 2015), teams used a range of online sources such as websites, reports, social media pages of stakeholders to capture and unpack how each organization articulates its vision and strategy with textual and visual materials.



Figure 2.3. Analysis of a food rescue organization's narrative of change, student work using template developed by authors

Doing this initial analysis exposed the team to the array of considerations based on which stakeholders relate wasted food to different aspects of a longer-term transition of the food system, and its wider implications on society. Wasted food is often articulated as an environmental concern, and an operational challenge to divert edible food and inedible scrap out of landfills. Yet, as the analysis showed, it also involves the emergence of a green technology sector and its potential to create jobs; or from a community organization perspective, it concerns access to food as a human right.

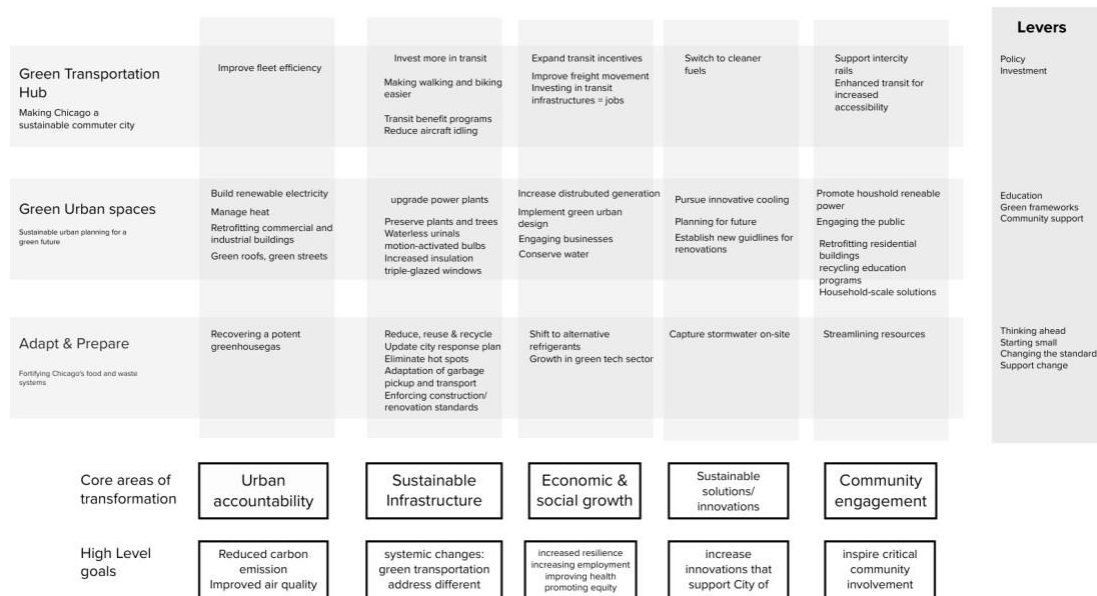


Figure 2.4. Analysis of local government's narrative of change, student work using template developed by authors.

But more significantly, comparing the meaning of these goals across different narratives helped us start surfacing the divergent aspirations of organizations that might be obscured by this seemingly shared transition vocabulary (e.g. economic development, sustainable food production etc.). One such term was 'community'. We found that, city government perspective that centers urban sustainability in understanding wasted food, the term 'community' signifies engagement of the public and promoting social equity through expanded food access and employment. From a food industry perspective, it involves "sharing harvest and knowledge with neighbors". And from a community organization perspective, it means empowering communities through education, spaces and organizing to foster sustainable lifestyles and community-based food production such as "equipping farmers with skills in high-tech growing" (Green Era n.d.). Not only do these imply different arguments for why 'community' matters in addressing wasted food, but they also suggest

different levels of agency and power for the community.

We used the online narratives of four stakeholders to have an initial understanding of what matters to them in making wasted food matter, and building a dose of skepticism towards the terms through which articulate these matters, prior to our first in-person engagement with stakeholders.

Identifying Imaginaries of Wasted-food Transition in Chicago. We used the overlaps and differences between the visions of stakeholders to define three imaginaries for wasted food transition in Chicago. These imaginaries offer distinct ways of seeing the issue, but they are not necessarily exhaustive or mutually exclusive in their considerations. Just as an organization can be a member of multiple social groups, an organization's vision might intersect with multiple imaginaries. In defining these imaginaries, we focused on what matters organizations' center in their vision, who they consider as the protagonists of change and what strategies they prioritize.

- **Sustainable Cities:** Centers sustainable and green urban infrastructures that are developed through citizen engagement and publicprivate partnerships to incentivize adoption of sustainable practices of food production and recycling and promote innovative solutions by an emerging green industry.
- **Strong and Sustainable Food Industry:** Centers positive environmental and social impacts of food businesses through commitment to local and sustainable food production that can create jobs for the residents and promote socially conscious consumption habits.

- **Community led Food Justice:** Centers radical transformation of food systems with innovative and community led approaches to fight racism, build economic justice and social equity by advancing hyperlocal, circular and culturally appropriate food flows.

These three imaginaries offered us an initial critical foundation and openness to diverse ways in which stakeholders might interpret current challenges and give precedence to alternatives, prior to our first contact with the wider stakeholder group. The Food Matters Think Tank event gathered 30 stakeholders at the Auburn Gresham Lifestyle Hub, a living testament to a communitydriven revitalization initiative striving for a variety of healthier, sustainable, and equitable solutions for the local community. The workshop activity aimed to kickstart the project dialogue by identifying the priorities and barriers for each of policy, food rescue, recycling and community organizations stakeholder groups at the event.

A portion of the workshop discussions revolved around practical considerations for scaling implementation of existing food rescue and recycling efforts, highlighting issues like the absence of clear regulations or adequate physical infrastructure. The conversations also brought forth tensions between the prevailing mindsets and norms that characterize the current state of the food system, and the more radical visions for the future, advocated particularly by community organizations – tensions that seeded our subsequent debates. One of the workshop posters featured the phrase ‘rethink, reduce, rescue, recycle’, emphasizing the need to reconsider the system while envisioning actionable solutions as framed in the usual 3R slogan of food waste prevention.

A discernible tension concerned the need for a more collaborative environment,

a priority that all stakeholders seemed to agree on – and challenged the possibility of authentic collaboration in the present competitive landscape of the food production and rescue environment. This landscape, with certain entities seeking profits from surplus food redistribution along with structural power disparities between mutual aid groups, major food banks and corporate donors, inadvertently fosters a competitive atmosphere rather than a collaborative one. The debate also highlighted the mindsets of ‘food scarcity’ and ‘charitable donation’ that perpetuate the systemic inequities, as opposed to ideas of wasted ‘abundance’ and ‘food as human right’ that are fundamental to the food sovereignty ethos of mutual aid networks.

System Mapping. In the next phase of the project, student teams collaborated with project partners to visualize their current operations in the domains of city-run organic waste management, food service & catering, and wasted food recycling. We initially developed three system maps from the operational perspectives of each project partner, which were later integrated into comprehensive visuals that depict the local service systems for food waste prevention, rescue, and recycling. These integrated maps showcased avenues for multistakeholder engagement across all tiers, and highlighted present issues.

However, combining an operations lens with a narrative one encouraged the teams to look beyond the issues at the surface level, and reframe challenges considering the norms and mindsets that are constitutive of the larger problem. One such issue was the lack consciousness and knowledge of the public about the importance of food waste prevention and recycling. This is a prevalent issue in wasted food discourse, pushing

consumer education campaigns to the top of food waste prevention strategies as in ReFED's (2023). Yet, leaning onto the debate of mindsets scarcity and abundance raised in the first Think Tank event, the team challenged the notion of abundance in food service. They reframed the issue as a cultural one that "equates quality and sense of abundance with excess food" as opposed to a sense of abundance rooted in sharing as framed in community led food justice imaginary.

Mapping Imaginaries & Tensions. While narratives of change of organizations often emphasize positive outcomes for common good, focusing only on what is deemed desirable tends to blur the lines between different positionalities that are in tension with one another. After all, both agriCorp giants and their grassroots opponents claim to build a world where none goes hungry. Imaginaries concern what their proponents are up against with respect to the status quo, as much as what they for. We prototyped an imaginaries mapping tool (Figure 2.5) to compare and contrast how different imaginaries problematize the status quo and what propose future alternative, in order to identify contested shifts that need to be considered and negotiated. The tool helps explore these shifts across various aspects of a systemic transition (e.g. cultural, political, data & technology etc.) based on eight capitals framework (Nogueira, Ashton & Teixeira 2019). We mapped each imaginaries' claims and aspirations using colorcoded stickies and used the framework as a conversational aid to surface these imaginaries' claims and counterclaims in relation to one another.

“On most days, I could sense the hesitation in the class, as this meant questioning and recreating the brief, while questioning our own belief systems and the narrative of [the] field of design itself. It often was a constant act of fostering debate while constantly hitting the breaks to try and see our own narrative and questioning the common ideals such as economic development, job creation, access that are the social impact vocabulary of many design projects. We leaned on our personal experiences, things that we were weirded about the world we live in, and the American food culture in particular, with its infinite grocery stores, perfect vegetables and giant portions”. - Author’s personal reflection, March 2023.

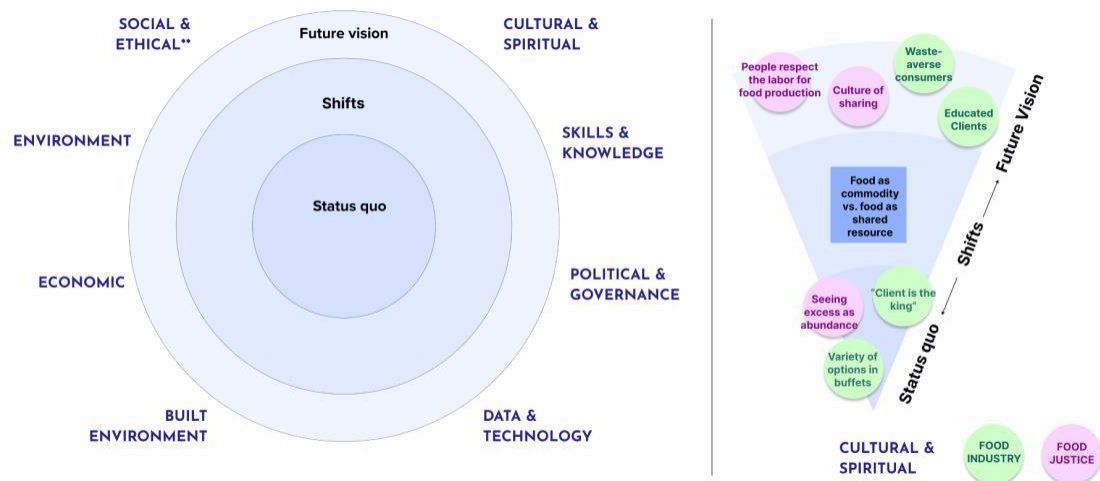


Figure 2.5. Imaginaries mapping tool (left), example use to compare food industry and food justice perspectives on cultural & spiritual dimension of wasted food (right).

Figure 2.6 demonstrates how we used the tool to debate the prevention of wasted food as a cultural transition and question the norms and mindsets that perpetuate wasteful practices of production and consumption. Through a comparative analysis of how food culture is presented in the Strong and Sustainable Food Industry's imaginary versus the Community-led Food Justice imaginary, we reframed the deeper cultural transformation as a matter of our society's relationship with food – a contrast of seeing "food as a

commodity vs. food as a shared resource." Each one of these shifts points to potential areas of generative tension where stakeholders' visions, values, and loyalties might diverge from one another and from the dominant narrative. Yet, we preferred to phrase these as critical shifts to emphasize their role of suggesting radical pathways rather than insurmountable conflicts. We formulated eight such dichotomous statements between the status quo and a critical systemic shift (Table 2.1). We then turned these into a deck of cards to support the ideation session with the stakeholders at the second Food Matters Think Tank event.

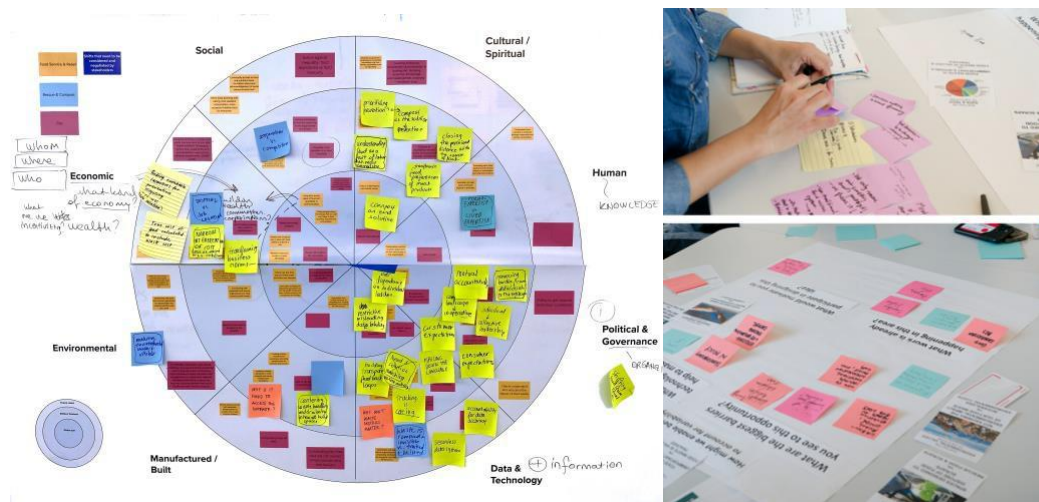


Figure 2.6. Imaginaries mapping tool used in class to identify critical shifts (left), and deck of critical shifts cards used during the Food Matters Think Tank event (right)

This event gathered over 30 participants from comparable stakeholder groups for a shareout and ideation session based on opportunity areas for prevention, rescue and recycling of wasted food, and the tensions between how different ideas of common good manifest themselves in these areas. The students facilitated discussions in eight breakout groups, with an opportunity area assigned to each group. As participants discussed barriers, emerging solutions and needs related to each opportunity area, we prompted

them to use the deck of cards to explore the pivotal shifts that they consider significant in this endeavor. Rather than prescribing specific paths, the cards were aimed to entice conversations on longer-term impacts that need to be considered, to build legitimacy around niche interventions, and challenge norms and mindsets that our current system is founded on. We encouraged participants to openly articulate the frictions they encounter, instead of presenting well-rounded solutions.

Table 2.1 Critical shifts proposed in the deck of cards

Financial	Economic	Cultural Values	Governance
Food waste organizations working cooperatively vs. competitively	Food scraps as waste vs. as a source of value (jobs, energy, soil)	Food as a commodity vs. food as a human right & shared resource	Responsibility of food waste mgmt is on the individual vs. on the systems
Data & Technology	Built Environment	Human Knowledge	Environment
Food waste is unimportant & invisible vs. food waste is valued & tracked	Spaces designed to dispose of food scraps vs. rescue food & scraps	Valuing expertise of executives or academics vs. the expertise of the people doing the work	Not knowing the environmental impact of our food waste vs. internalized environmental costs

Emergent Pathways. Following the event, the students synthesized the discussions into a list of opportunity areas and recommendations to inform development of a strategic pathway by our partners. The debates sparked at the event, challenged some of our previously held assumptions regarding the desirability of certain opportunity areas that we proposed and thus, led to new approaches. In addition to debates on cultural and economic paradigms that we previously discussed, two more emerged, concerning governance of collaborative efforts, and the role of data and technology.

Governance was a big question as the current landscape of food rescue and recycling in Chicago is fragmented across many grassroots efforts, without a city- wide infrastructure, or standardized process. Driven by a mindset of efficiency and seamless coordination, our opportunity areas proposed to “institutionalize and operationalize food waste prevention mindsets”. However, this received push-back from the group, on the grounds that ‘institutionalizing’ mindset perpetuated the asymmetries in power between mutual aid groups and the institutions that control access to decision-making and resources. The dominant imaginary assumed that efforts need to be institutionalized to succeed, overlooking the significance of decentralized governance from a food justice standpoint. This prompted a shift towards exploring networked yet decentralized solutions that amplify existing community-led infrastructures.

In the dominant discourse of wasted food prevention, data infrastructures are considered pivotal for scaling up food rescue and valorization, often portrayed softwares. However, from a community organizing perspective, seeing the system mainly through data can render grassroots and community based infrastructures invisible, and undermine their efficiency in responding to community needs. For instance, nearly twenty Love Fridges across Chicago neighborhoods serve as selforganized dropoff points for the mutual aid groups, while logging every delivery in pantries can be a hindrance in adhoc food rescue. This tension shifted our goal framing from more and better quality data to flexibility and inclusivity of data driven approaches, without shifting agency away from local groups who operate in a more relational mode.

We integrated these conversations into what we aimed to be a pluralistic synthesis, meaning, we sought to explicitly challenge the assumption of certain pathways

being inherently desirable for all, as in the case of data-driven approaches. We highlighted the ongoing grassroots and community-based efforts as legitimate pathways for addressing wasted food in ways that can help shift the dominant economic and social paradigms that our wasteful food system operates from. Perhaps one of the most challenging tasks was translating a pluralistic approach into a project report in a way that is easily accessible and usable by the project partners, providing clarity without flattening the ongoing debates.

2.4 Discussion

We presented a practical approach for using narratives to account for and contend with the diverse imaginaries of stakeholder groups in framing transition goals for collaborative action. We built on the recent work in transition studies, that advances narratives as a powerful mechanism to meaningfully engage with the controversial and conflictual nature of transitions, where under the seeming universality of high-level transition goals such as sustainability or equity, stakeholders portray different futures for the collective good. We introduced a set of tools in development that uses narratives for unpacking the diverse and conflictual meanings of transition goals, and building upon these divergent perspectives to challenge the dominant mindsets and norms through which transitions get framed. By explicitly engaging with narratives as tangible materials and abstract meaning systems that we design through, we were able to:

- Broaden our understanding of the transition in question, by considering it as a constellation of public matters (Latour and Weibel 2005) and aspirations extending beyond the dominant claims around the issue.

- Account for the diverse and potentially conflicting interpretations of high level transition goals that characterize distinct imaginaries of stakeholder groups.
- Surface the tensions between the goals and pathways that different groups advocate for, and engage with these in a generative way. This enabled us to challenge our assumptions concerning the fitness and desirability of specific approaches, as well as the economic and social norms through which we frame the issues and potential pathways.

As opposed to framing transitions as a set of strategic actions orchestrated around an assumed ideal of common good, narratives offer us a way to engage with the plurality of lived and dreamed realities that shape diverse meanings of collective good. This fundamental diversity of understandings of what that common good constitutes, is a critical and creative resource to cultivate the radical shifts that our world needs. Clearly articulating the points where stakeholders' imaginaries diverged from each other and framing these differences as specific tensions provided a foundation for cultivating pluralism by actively resisting the tendency to reconcile these differences. As Mouffé (2000) contends, pluralism acknowledges and values differences while challenging the quest for unanimity and homogeneity, which often proves to be illusory and exclusionary.

Surfacing and sustaining this divergence early on was foremost, a tactical act, as it reduced the risk of a superficial alignment that might otherwise conceal tensions stemming from stakeholders' hidden agendas. However, this required a balancing act of making room for the differences without stifling ongoing dialogue. It was equally a political act, as it helped us identify the norms and values that are reinforced by the dominant narratives and the regimes that they sustain, which promote particular notions

of common good. The challenge entailed tending to these differences without striving for immediate reconciliation, instead, allowing them to persist as the wedges that sustain open rifts, against the homogenization of perspectives within the prevailing narrative. And finally, it was a generative one, as it invited a reappraisal of the radical futures that are in the making, beyond the conventional blueprints of tackling wasted food. The "uncommon good" we envision encompasses these departures from the illusory and oppressive homogeneity of commonality, embracing the richness of diverse perspectives and paving the way for transformative and inclusive shifts in our collective path forward.

2.5 Limitations and Recommendations

This paper offered a methodological approach for using narratives to surface and work with the diverse and sometimes conflicting imaginaries of stakeholder groups with the aim of framing more inclusive and pluralistic transition visions. We would like to emphasize that the tools we prototyped here are not plug-and-play meaning generators but aids for deep reflection and questioning to help practitioners navigate the intricate landscape of values and mindsets that underpin multistakeholder discourse, including their own.

First of all, narratives are a slippery and elusive material to work with. Meaningfully using narratives relies a lot on the group's ability to look beyond the buzzwords, identify the nuances in the meaning and recognize the historic and political ties of specific concepts. Therefore, we suggest a certain familiarity with the specific context, as well as the larger discourse that surrounds the issue. For instance, the seemingly interchangeable concepts of 'food waste' and 'wasted food' for an unfamiliar

ear, have a significant difference in that the latter emphasizes the act of wasting a valuable resource, as opposed to normalizing it as another waste stream to be handled.

Organizational discourses tie the microworld of an organization and its reasoning to the grand narratives that make up the largescale reasoning of the world (Boje 2019). The external-facing narratives of organizations, such as reports, webpages, social media postings etc. which were the unit of analysis in this study, are often a highly regulated type of narrative, the facade of a messier web of stories, interactions and artifacts through which an organization makes sense of and articulates its vision. Thus it is important to account for the specific intentions of how an organization may choose to display or hide specific ideas from this facade, or explicitly counter the dominant narratives. Which is why the methodology presented here would be most effective when combined with other methods of contextual research, that can capture many meanings that might get concealed within the chosen and imposed limits of how an organization tells its story.

Lastly, we would like to remind readers that narratives are not limited to textual material, but encompass a wide array of digital and physical artifacts through which we construct meaning. Although its use was limited within this project, in previous iterations, we found visual ethnography, both digitally and onsite, to be a helpful complement to the analysis of textual material. Small-scale organizations and local communities, who may have limited online presence. express themselves through more informal forms of communication such as zines, murals, physical posters, placemaking, or oral storytelling.

CHAPTER 3

CULTIVATING NARRATIVE RESISTANCE IN SYSTEMIC DESIGN

3.1 Introduction

Many systemic issues that our societies grapple with today involve battles over competing narratives that advocate for specific problem framings, visions of desirable alternatives, and pathways for shifting resources to achieve those visions. The challenge of reaching alignment on the most pressing issues impacting our societies often stems from the distinct mindsets, values, and worldviews that undergird these narratives. The challenges facing our food systems are a perfect reflection of these dynamics, both on a global scale and within the United States, where our work is embedded.

In 2021, delegates from over 150 countries convened for the UN Food Systems Summit (UNFSS), envisioning a world where everyone has "access to sufficient quantities of affordable and safe food products." The goal was to rally governments, companies, and international NGOs to transform food systems in line with the Sustainable Development Goals. Despite this noble aim, hundreds of organizations boycotted the event, forming People's Autonomous Response to the UNFSS. The coalition declared that the summit's corporate-centric approach reinforced the extraction and oppression perpetuated by agricultural conglomerates and governments under the guise of 'feeding the masses.' Instead, the coalition advocates for a food system transformation grounded in food sovereignty, advocating for the people's right to define their own food systems, grounded in notions of solidarity and environmental stewardship

(Declaration of Nyéléni, 2007). This controversy persists, as the second summit in 2023 faced similar backlash.

The seeming intractability of food system transformation is not due to a lack of solutions, on the contrary, both platforms present a plethora of proposals, each charting a different course toward a better food future. These pathways, and their notions of ‘better’ are grounded in divergent views on the role of food in shaping how people relate to one another, to land and other species. The complexity of the UNFSS controversy lies in the irreconcilability of these narratives, the worldviews that sustain them, and the agendas that derive their legitimacy from them. Such narrative conflicts are not unique to food systems; they echo across many sustainability transition efforts today and are often blamed for stagnating collective impact.

A shared narrative is often seen as a powerhouse for driving collective impact at scale. Yet we argue that such friction is not only inevitable but essential. In areas ranging from energy to urban development, dominant transition narratives that push for broad alignment often perpetuate the same mindset of continuous growth and progress that has driven our planet to its limits and failed to deliver the promised well-being for many. The troubled times we are in call for a radical reappraisal of the values and mindsets that guide our collective lives with other beings of the Earth. Acts of narrative resistance, such as the People’s Autonomous Response to UNFSS, invite people to counter the mindsets that contributed to today’s planetary crises, holdings’ the complexities of transitions as they relate to our ways of being with each other, our planet and other species.

Narratives are not just competing proposals with distinct rationales; they are living story systems that guide how we collectively articulate systemic issues. How we

narrate systems change impacts what stories we can tell, and thus how we perceive and make sense of the world. Thus, we turn our attention to the infrastructures that facilitate collective articulation of public matters and organize the narrative work of systems change. We refer to the systems of artifacts and protocols that form the ordinary and ubiquitous grounds for organizational communication —policy papers, bullet lists, maps, and diagrams. While these infrastructures create a shared language that allows organizations and groups to communicate, they also inscribe specific ways of telling stories, setting the conditions for which worlds can be narrated into existence and which are denied legitimacy.

Drawing from Feminist Science and Technology Studies, we critically examine the narrative legacies of these infrastructures:

How does the design of these narrative infrastructures enable specific ways of knowing, seeing, and dreaming, potentially at the expense of others, in the collective articulation of systemic challenges?

We explore how narrative resistance manifests through the tactical adoption of existing *narrative infrastructures* and the construction of alternative ones. We present two studies within the context of food system transformation discourse in Chicago:

- An ethnographic investigation into the discursive landscape of food system transformation in Chicago, examining how narrative resistance emerges in the socio-material structures through which various collectives and organizations address the systemic issue of food insecurity.
- A participatory storytelling activity at Chicago's annual food justice summit, where we explored alternative framings of food system transformation, inspired

by examples of narrative resistance in food sovereignty movement.

We distill our learnings from narrative resistance in food system discourse to reflect on the narrative agency of systemic design in relation to dominant infrastructures, and directions for cultivating narrative resistance towards radical possibilities.

3.2 Theoretical Background

Narrative Resistance in Systemic Transitions. In recent years, there has been a growing recognition of the power of narratives and storytelling in fields such as energy, climate research, and political ecology (Moezzi, Janda, and Rotmann; 2017, Vigliano Relva and Jung, 2021). Transition scholars emphasize alternative transition narratives' capacity to contest dominant framings and their underlying value commitments by surfacing diverse interpretations and proposals beyond official public agendas (Burke, 2018; Longhurst and Chilvers, 2019). This diversity is crucial not only for the alternative proposals it generates but also for the capacity of various groups to self-identify and form coalitions by countering dominant narratives with their own.

Historically, narrative interventions have served as a form of resistance for activists across social movements, who use the emotional power of stories to humanize social issues and counter dominant narratives (Polletta 1998; Polletta et al., 2011). McKenzie-Mohr and Lafrance (2017) describe 'narrative resistance' as the process by which individuals reclaim their stories to oppose harmful narrative constructions of their identities by oppressive systems, which can then be mobilized through communities to amplify its liberatory potential. A notable example is ACT UP's resistance against the dehumanizing official narratives that framed AIDS-related deaths as a consequence of

‘lifestyle choices’, as well as the public rhetoric that naturalized these as tragedy. The activists tactically used language and public performance to call out the hypocrisy of pitying for the dying, when the death tolls were result of intentional, systemic, institutional discrimination (Lance 2020, Tilsen 2023). Similarly, Latin civil rights movements have been using various forms of art, chants, subversive and symbolic vocabularies to foster solidarity, make the struggles visible and articulate visions for alternative visions (Argueta, 2023).

There is a growing recognition that narrative resistance needs to be fostered as a networked capacity to make a lasting impact on policies, supported by networks of organizations that adopt, disseminate, and amplify these narratives. An emerging body of public and philanthropic organizations is investing in mobilizing narrative change to alter the deep-seated mental models and paradigms that sustain harmful systems (Davidson, 2016), by changing “the underlying system of stories that help people make sense of the world” (Davidson, 2022). As design increasingly positions itself as possessing the capacity for making sense of complex systemic issues and injustices (Dorst, 2015; Meharry and Carey, 2021), it becomes crucial to evaluate design’s narrative agency and—or in some cases, its subservience—to the broader story systems it shapes and is shaped by, especially in framing and contesting the problem space (Constanza-Chock, 2020; Shaw and Nickpour 2022; Sungu et al., 2023).

One direction for narrative resistance in systemic design is shifting these story systems in support of alternative framings of systemic issues towards radical and values-driven possibilities. In shaping the social and material environments through which people make sense of the world, design sets the conditions for the narrative construction

of reality (Krippendorff, 2006; Tharp & Tharp, 2017, Jones & Bowes, 2017). As Feminist scholar Donna Haraway famously wrote: “It matters what matters we use to think other matters with; it matters what stories we tell to tell other stories with.” It equally matters “where those stories came from, from whom, and under what conditions.” (DiSalvo, 2022). We argue that, in order to have meaningful narrative agency in influencing the frames through which we make sense of systems change, designers need to be attuned to narratives not only as static frames but as evolving systems of meaning, created, maintained and disrupted through distributed storytelling practices that are structurally supported and stabilized. Thus, the narrative praxis of design, needs to be understood within the larger web of institutions and practices that sustain the story systems that guide the collective articulation of systemic issues.

Resistance in a World of Bullet Points. To better situate design’s sensemaking role within the wider ecosystems we turn to Science and Technology Studies (STS). STS considers the collective formation of issues beyond a discursive lens, and pays attention to socio-material performances of deliberation and the ways of knowing enacted by these (Dewey, 1946; Latour, 2005; Marres, 2007; Chilvers and Kearnes, 2016). STS scholars emphasize the role of participatory collectives in enacting alternative forms of democratic participation and deliberation, outside of formal norms, protocols and technologies of participation, with the possibility of redefining objects of political concern (Chilvers and Kearnes, 2016). People’s Autonomous Response to UNFSS is an example of such participatory collectives. Participatory collectives are important spaces where alternative story systems are cultivated, creating resistance against dominant narratives.

The dominant structures that organize institutional discourse tend to privilege what's commonly considered as scientific evidence and rely on quantitative measures to represent issues, which overshadow alternative meanings and interpretations, leading to a narrow framing of issues (Dawson and Buchanan, 2005, Moezzi et al. 2017; Puri et al., 2009). Moreover, the mental models and worldviews that undergird Western scientific knowledge systems, such as western societies' dualistic representation of nature and humans' place in it can lead to a (Vigliano Relva and Jung, 2021, Boehnert et al. 2023). When formal spaces, structures, and languages confine issues to a narrow set of narratives, legitimizing certain forms of storytelling over others, alternative platforms for collaborative sensemaking can create space for narrative resistance other stories to be told and for a broader range of possibilities and considerations to be explored.

As Joanna Boehnert (2018, p. 64) states “The things we make are extensions of our ways of knowing.” The narrative grounds of transition debates, with its familiar slide decks, bullet points, statistics, arrows and circles, offer little help for conveying notions such as care, empathy, or inter-species relationality. Therefore, in order to cultivate alternative ways of being, we need to hold space for other ways of seeing and knowing. This idea is rooted in the work of feminist STS scholars who confront the purely positivist ideal of scientific knowledge as the objective unifying truth by foreground situated, partial, local and pluralistic epistemologies that are attuned to the relationship of humans to other species and beings (Haraway, 1988; Suchman, 2008). These partial views are not parts of a jigsaw which can align to present a complete vision of the whole, but they are always connected with imperfect seams and co-exist in tension.

Several scholars and practitioners are experimenting with centering stories and storytelling as an inherently situated and subjective ways of making sense of the world, resisting the reductionist storytelling and rationalization of diverse lived worlds by institutions in power. They highlight that, stories, by including individual subjectivities, can contribute to a more empathetic understanding of systems (Ollove & Lteif 2017; Talgorn and Hendriks 2021), and foster the pluralism and dissent needed for more diverse civics (DiSalvo, 2022). While these works demonstrate how design can mobilize storytelling as a form of resistance, the issue of how these localized interventions can contribute to narrative change at scale remains underexplored. What happens to the alternative publics when they intersect with formal structures and processes (Kimbell and Bailey, 2017)? What happens to the cracks they open, once they are removed from the site? (Sungu and Tekogul, 2023). The aftermath of interventions is rarely spoken of, but we note a few accounts of participatory civics where researchers looked back.

Di Salvo (2022) recounts a participatory civic intervention in Atlanta where researchers engaged residents in creating speculative stories of smart cities to counter institutional, techno-scientific narrative of smart cities with frictions from everyday life. The project produced 'Fictions of Smart Atlanta,' a booklet of stories shared electronically. The capacity of these stories to infiltrate into the territory of formal government depends on storytellers who keep them alive: "It is an endeavor of telling stories that dwell in the complications of contemporary civics, as residents, government, and industry mingle together, sometimes in uncomfortable ways." (p.67). As Von Busch and Palmås (2015, 2023) illustrate, the translation that happens through the mundane artifacts and protocols of deliberation, such as roundtable discussions, slide presentations,

technical papers and press releases, lead to a systemic filtering out of dissensus and marginal voices, ‘betraying’ the interests of participants. Yet it is precisely this material grounds that design must navigate in order to cultivate resistance in a world of bullet points, rather than temporarily counteract the dominant narratives.

Cultivating Narrative Resistance as a Networked Capacity. We approach narrative resistance as the cultivation of diverse story systems that can shift collective articulation of issues beyond the dominant narratives by activating diverse ways of seeing and knowing and mobilizing coalitions. We lean on two complementary concepts to illustrate how these living story systems are organized by socio-material structures and practices in collective articulation of systems change: narrative infrastructures and narrative ecologies.

The collective articulation of systems change occurs across various, interconnected arenas of public debate, from think tanks and government bodies to social media and street art. Chilvers and Keanes (2016, p. 36) describe this as a “rhizomatic articulation of matters of concern as a network.” Gabriel’s (2017) metaphor of ‘narrative ecologies’ illustrates the dynamic evolution of master and counter-narratives within these complex and distributed deliberations. He identifies ecosystems like “narrative jungles,” where numerous narratives compete for dominance, and “narrative monocultures,” where a single narrative prevails. In the context of systems change, this can be seen as the “narrative landscapes” (McLean, 2016) that organizations navigate, made up of overlapping ecosystems.

‘Narrative infrastructures’ was introduced by STS scholars Deuten and Rip (2000, p.76) to describe how a complex "mosaic of stories" within organizations becomes coherent storylines through the collective adoption of certain narrative elements. Schulschenk (2018) likens this to an evolving network of rails that guide and constrain the flow of new narratives. In systems change, this process involves diverse organizations and groups weaving their knowledge and perspectives into plausible narratives of change. The social and material structures that organize these story systems—such as artifacts, networks, platforms, protocols, norms, and languages—are significant as they stabilize storytelling traditions while also holding the potential to disrupt them. ‘Narrative infrastructure’ has been emerging in philanthropy discourse (Robinson, 2020; Tracy Van Slyke, 2022; Choi and Kalra, 2023) to emphasize the need for creating and investing in narrative infrastructure to mobilize narrative change at scale.

These two metaphors—narrative ecosystems and narrative infrastructures—help us frame narrative resistance as a capacity to be relationally cultivated, and structurally supported. Narrative ecosystems highlight the diversity and competition among story systems cultivated by various coalitions, while narrative infrastructures focus on the built environments that shape these ecosystems, fostering certain narratives while constraining others, ultimately reinforcing or challenging existing boundaries.

3.3 Methodology

This research explores social and material organization of systems change narratives. Our goal is to identify directions for the narrative praxis in systemic design, in ways that can cultivate the narrative resistance in collective articulation of systems change, to foster radical possibilities. We wonder: How can intentional design of collaborative inquiry can help diversify narrative landscapes, and make room for wider ways of seeing and knowing?

Our work is embedded in the food system transformation efforts in Chicago, where we have engaged as researchers, advocates, and neighbors. Chicago is a city where one can witness the diverse, and at times conflicting futures in the making, through the lens of food. Despite a thriving food economy, one in five Chicagoans remains food insecure (Greater Chicago Food Depository, n.d.) part of a broader issue of inadequate nutrition across the U.S. This disparity is most pronounced in predominantly Black, Brown, and Latinx neighborhoods of South and West Chicago, where residents often travel a mile or more to access a grocery store due to historic redlining and ongoing disinvestment. In response, communities and activists are building community-driven food systems to not only feed people but also foster food sovereignty by restoring relationships among people, land, and ecological networks. Amid these disparities, efforts to improve the food system reflect a struggle over the meaning of food and its role in the collective well-being of people, other species, water, and land.

3.3.1 Narrative Landscape of Food Insecurity in Chicago.

From a larger list of over 200, we sampled 16 organizations actively engaged in combating food insecurity, representing public, private, non-profit, and community sectors. We compiled a corpus of textual and visual materials from these organizations' public resources, including websites, social media, and reports. Additionally, we gathered field notes and snapshots from participant observations at various grassroots organizing sites, both physical and online, such as community gardens, events, and local gatherings, to capture narratives from collectives with limited online presence.

We applied an analytical framework by Chilvers and Longhurst (2019, p.4), which defines four dimensions for studying the distributed and collective production of alternative transition visions. We adapted these dimensions for our study:

- **Knowings:** Ways of knowing that are acknowledged and legitimized (e.g. Scientific, experiential, ancestral)
- **Meanings:** Ways of stabilizing and countering meaning (e.g. Vocabularies, taxonomies, metrics)
- **Organizings:** Governance of collectives and discourse (e.g. networks, membership)
- **Doings:** Material arrangements and practices (e.g. artifacts, spaces, events)

Three overlapping narrative territories emerged from this analysis: (1) public policy, (2) food industry and (3) grassroots organizing, characterized by shared ways of seeing and framings food-system transformation, facilitated by the distinct socio-material

environments through which collectives exchange knowledge and advocate for pathways (Table 3.1).

1. *Seeing like a government*, where diverse data sources are combined with the perspectives of stakeholders to generate an accessible core framing of the issue at scale, and value commitments to anchor goal-oriented coordination of efforts and resource allocation. Frames food insecurity as a systemic public health equity issue with disproportionate geographical and racial impacts, requiring policy-backed and coordinated interventions that can establish food as a human right and a resource for building community wealth.
2. *Seeing like a business*, where the data and expert perspectives interpret patterns to frame opportunity areas, often with a clear problem and solution coupling, and depoliticized language. Positions food and beverage industry as an engine for building a healthier and environmentally sustainable food system, with the capacity to address hunger and unemployment through charitable donations and hiring from “disadvantaged communities”.
3. *Seeing like a community/movement*, which emphasizes relational learning alongside evidence-based documentation of interventions to advocate for alternative pathways. This perspective frames food insecurity as a symptom of systemic injustices and exploitation of people and resources, where healing the relationship with food can be a path for building justice, liberation, community sovereignty and healing.

Table 3.1. Narrative territories of food insecurity discourse in Chicago

	Public Policy	Food Industry	Grassroots Organizations
ORGANIZINGS	Multi-sector coordination between public agencies, nonprofits, NGOs, and academia organized by task forces and collaboratives around shared program frameworks	Private sector coalitions, organized by formal (and paid) membership to networks, partnerships with local nonprofits	Place-based membership to neighborhood coalitions, spiritual communities, membership to race and gender-affirmative communities, volunteer and working groups
KNOWINGS	Birds-eye systems view towards actionable policy recommendations informed by databases and GIS, baseline assessments, impact evaluation, stakeholder inputs, and academic research	Knowledge of industry experts, success stories, private-access databases trend analyses and industry reports that emphasize statistical data	Community-engaged research, experiential knowledge, expertise of elders, embodied knowledge (herb-tasting, community cooking), spiritual and ancestral knowledge, affective knowledge emphasizing love and care
MEANINGS	Program-based vocabulary and metrics and frameworks, official issue commitments	Shared industry vocabulary: (“good food movement”) charity language, standardized visual vocabulary with stock images and icons.	Movement vocabulary (“food apartheid”, “food sovereignty”), Value commitments (care, rest, healing), artistic expression
DOINGS	Reporting and data assessments, workshops, public engagement, maps and databases, programming, architectural drawings and renders	Corporate responsibility and charitable giving, Sustainability actions and impact reporting, webinars and office hours	Placemaking participatory planning with communities and grassroots organizations, townhalls, proposals and reports

Translation Across Narrative Territories. These territories can be mapped/charted on two dimensions that characterize what the organization of collective discourse: degree formality of governance of collectives and discourse (horizontal), and the scale at which

collectives convene (vertical), from neighborhood to national level. Figure 3.1 shows the various components of narrative infrastructures that characterize these ecosystems.

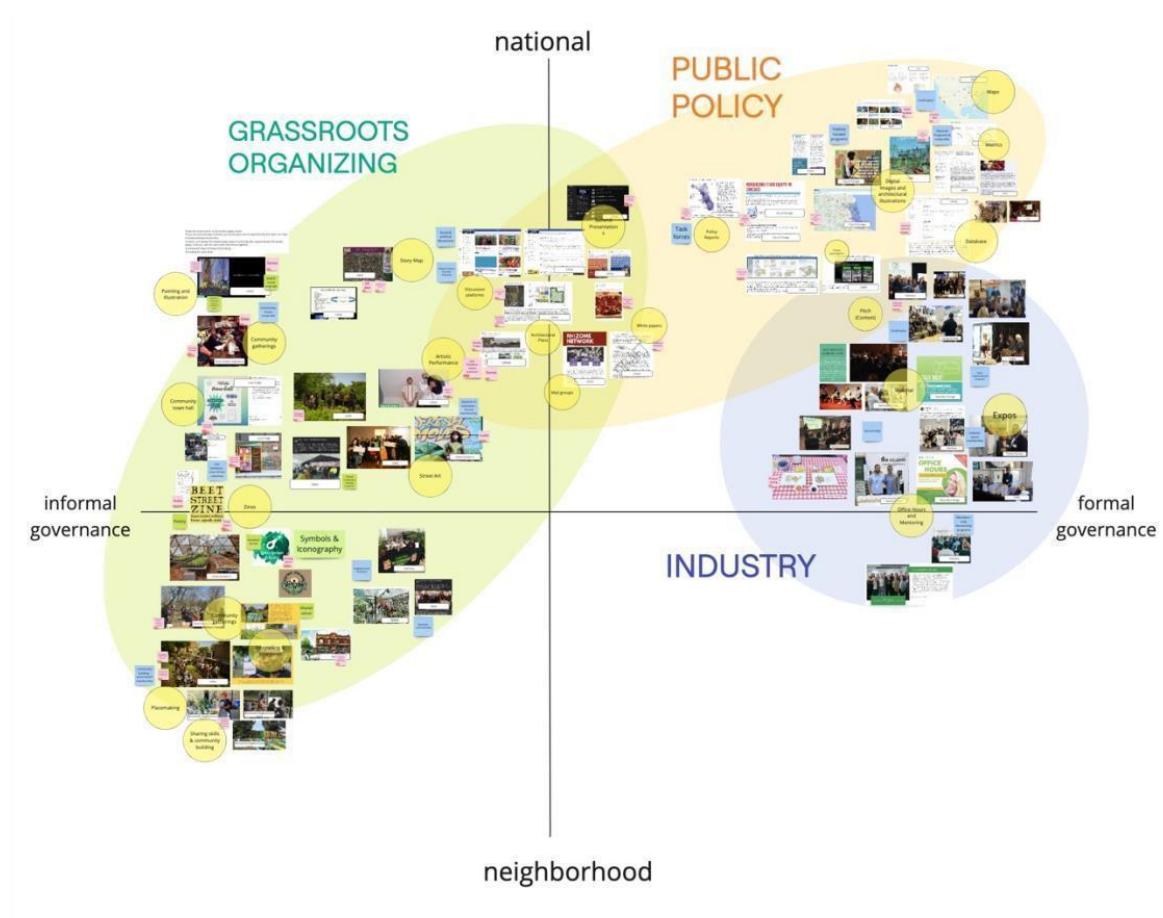


Figure 3.1. Narrative territories of food insecurity discourse in Chicago

Across this landscape, we observe a continuum from local and community-based spaces to large-scale, expert-driven ones, where narrative infrastructures increasingly prioritize strategic meaning-making over relational approaches. This shift becomes apparent in how meanings are translated across different narrative infrastructures, particularly in overlapping territories where various collectives deliberate and negotiate systems change. Often, this translation involves a gradual process of abstraction,

standardization, quantification, and rationalization, adapting narratives to align with the dominant framing. As these efforts scale and become more institutionalized, the framing of transformative change tends to become more formulaic. As a result, only pathways and visions that conform to dominant categories of knowledge and meaning are likely to be expressed and legitimized.

The ability of grassroots organizations to advocate for their alternative food futures and negotiate pathways depends on their ability to claim legitimacy within these frames while also countering them. Narrative resistance in the food sovereignty movement manifests in multiple ways, such as the tactical appropriation, creation and use of spaces, artifacts, language, in ways that invite and activate wider ways of knowing and alternative modes of organizing. We present a few examples that demonstrate narrative resistance, by situating, embodying, complicating food and its relation to diverse identities, places, histories; as opposed to strategic narratives that abstract, simplify and standardize.

Mapping as Resistance. Maps are frequently used in food insecurity discourse to show the connection between food insecurity, often tied to the absence of grocery stores, and historic racial inequities in neighborhoods. Chicago Urban Agriculture Mapping Project challenges this narrative by showing the growing landscape of urban growing as a viable path for communities to feed themselves and more importantly, giving the communities a tool to advocate for investment (Sheikh et al., 2021) The initiative involved researchers visiting 260 community gardens and documenting their food production and their stories (Rosing, Helpland & Delorenzo, 2022). Unlike conventional maps that focus on mapping

food sources, the map includes stories that show the broader purpose community gardens fulfill in neighborhoods as places” where children can grow their own sunflowers, sugar snap peas, and bean teepees.”

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The Tale of two Cities is another map , created by Inner City Muslim Action Network (n.d.), that juxtaposes the satellite views of Lakeview, a predominantly white, north-side Neighborhood with Englewood, a predominantly Black and Brown, south-side neighborhood, often referred to as a ‘food desert’. The map highlights the 111 vacant lots and 240 abandoned buildings in Englewood, against the fully occupied streets of Lakeview, to underscore the historic segregation and discriminatory policies that left Englewood residents deserted of many necessities, including jobs and housing. In showing the overlaps between lack of access to healthy food and historic injustices, the

map argues for an understanding of ‘food apartheid’, in a pushback against the term ‘food desert’ which connotes a natural condition.

Vocabularies of Resistance. A significant source of narrative resistance lies in the verbal and visual vocabularies of the movement. These vocabularies echo ancestral, spiritual, and affective ways of knowing food, countering the standardized and often neutral language of institutional discourse, with its stock images, icons, architectural renderings with silhouettes of people —portraying a world that could be anywhere but is nowhere, inhabited by anyone and nobody. A visible example is the representation of food itself, abstracted into mere sustenance and detached from cultural significance, as seen in the ubiquitous healthy cafeteria plate filled with apples, carrots, peas, and greens, or charitable donations of protein-enriched hot chocolate powders. In contrast, the celebration of food through sharing recipes in zines and community cooking events reclaims cultural ties to food and restores connections to ancestry and land with dishes like okra stew or medicinal tea made from sacred herbs.

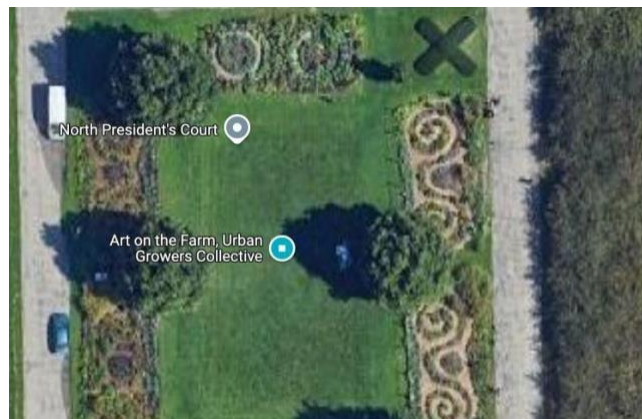


Figure 3.2. Satellite view of Sankofa landscape art in ‘Art on the Farm’ by Urban Growers Collective

Ancestral, spiritual and ecological ways of knowing are also echoed in the myths, symbols, and metaphors that permeate the vocabulary of the food sovereignty movement. For instance, the "Three Sisters"—an indigenous tradition of planting corn, beans, and squash together—serves as a metaphor that embodies the values of collaboration, collective resilience, and reciprocal care that the movement upholds. Another powerful symbol is Sankofa, from the Akan proverb meaning “It is allowed to go back and get what you forgot.” Erika Allen (2023), a movement steward, explains that Sankofa grounds her approach to circular food systems by “re-embracing indigenous heritage through the earth [leading] into new kind of paradigms”. The Sankofa symbol is featured in Urban Grower Collective’s ‘Art on the Farm’ garden as a land art installation by Erika Allen and at the Bronzeville Neighborhood Farm, which showcases Adinkra symbols to anchor the values and identity of West African heritage (Figure 3.2).

3.3.2 Seeds of Futures

Inspired by the narrative resistance in the food sovereignty movement, we present a participatory storytelling activity called "Seeds of Futures." This initiative emerged from the first author’s volunteer role on the leadership team of the 18th Annual Chicago Food Justice Summit, working with fellow creatives Alexandria Knapik and Isabel Galic. The activity aimed to explore diverse visions of a better food future by surfacing the 'futures in the making' embedded in the practices of growers, advocates, and educators. Using the metaphor of seeds, we sought fragmented and personal accounts of aspired futures, while emphasizing participants' agency in cultivating radical possibilities. Participants were

invited to describe their vision in a self-guided activity by answering three questions on an activity card (Figure 3.3):

- What future do you aspire to?
Complete this sentence: "I tend to a future where..
- What seeds do you plant for this future?
- What does this seed need to be nurtured with?

The responses of the participants highlighted various ideas about what makes a good food system and connections between food and the wider constellation of matters that shape our collective lives, around the overarching vision of “everyone having access to fresh and nutritious food”, a core commitment of the food justice community. The contributions emphasized fostering caring ways of being with each other, the land, other species, as well as oneself, where food was positioned as means to heal these relations, rather than the end. Some of the themes that emerged are:

- **Healthy and Nourished Communities:** Envisioning a future where “access to healthy living is not a luxury but a human right,” allowing people to come together to flourish”, and “fulfill their passions without mental and ecological pressures”, in a peaceful and accepting world”, and “not competing with each other”
- **Healed Connection with Land:** Imagining land that “feeds and houses everyone that needs it,” where people “feel connected to nature and earth again” and “steward native ecosystems to care for the community.”
- **Prevention of wasted food:** Appreciating food “as a product of many people’s labor [that] cannot be wasted”, with excess food “feeding our ecological communities



Figure 3.3 ‘Seeds of Futures’ pop-up at Summit venue (left) and activity card submitted by a participant (right)

Among the seeds of these futures, participants shared practices that promote community care, including healing and listening circles, growing, cooking, and sharing food, and fostering self-reliant communities with decentralized resources. Some of these were exemplified in everyday technologies such as public kitchens or the “little free apothecary”, a sharing box installed in the urban garden of a local ‘food as medicine’ advocate, where they put herbal remedies to be picked up by neighbors. Some participants also mentioned policy and investment support for local food infrastructure, with one contribution stating “...looking backward to our past will be a benefit to our future... a fusion of new tech/ industry/ government influenced by our vast global histories”.

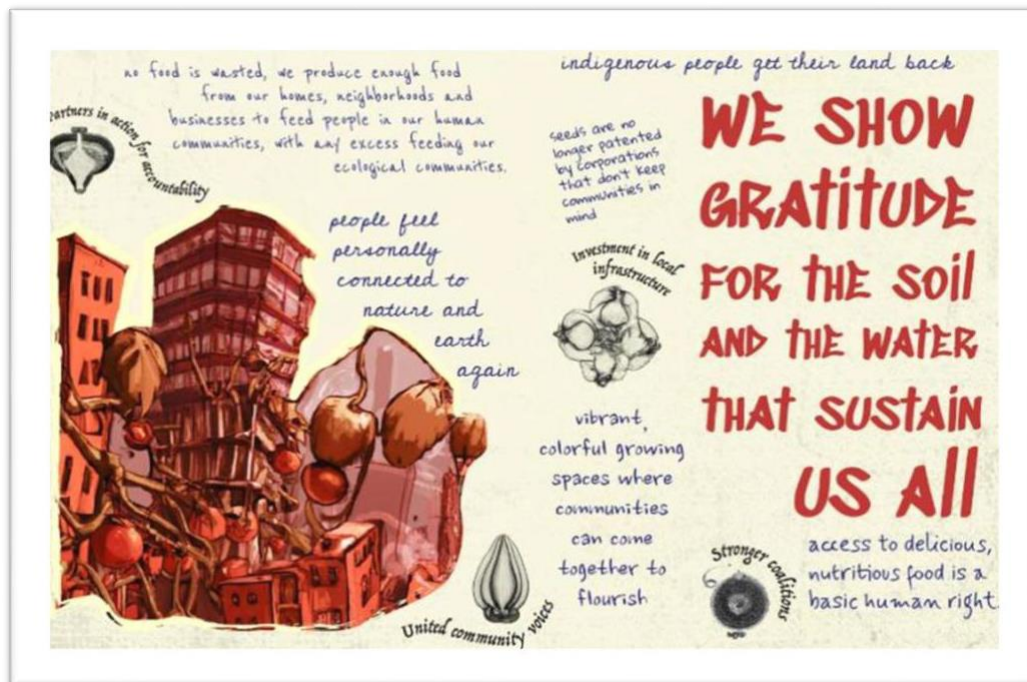


Figure 3.4. Zine page illustrating a theme from participants' contributions.

After gathering responses from people in a forum format, we created a set of collages that illustrated three themes throughout the responses, these were published as part of the Summit zine (Figure 4). In creating these spreads, the initial idea was to show the diverse meanings that shape understandings of a good food future. In this process of gathering fragments together in new stories, the researcher became a co-author, rather than merely curating and translating contributions. We worked with the format and language of zine, in its poetic celebration of food justice and people who steward it. Due to constraints of time and capacity, we used DALL-E, an AI image generation tool, to create illustrations based on participant input. The resulting artifact was made accessible in a digital format and shared with summit participants, as well as a limited number of

physical copies. The dissemination of this artifact was mostly limited to attendants of Chicago Food Justice Summit.

3.4 Discussion

We presented two studies that approach collective articulation of systems change from a narrative lens, specifically looking at the material environments that organize how groups and organizations manifest and disseminate narratives of systems change. We explored the food insecurity discourse in Chicago through the notions of narrative ecosystems and narrative infrastructures, tracing how narratives and counter-narratives are mobilized by collectives who advocate for competing framings of food insecurity, and the material organization of these narrative performances. Our analysis showed three distinct but overlapping narrative territories that reflect what the groups consider to be the ethos of systems change, grounded in shared ways of seeing, knowing, and being. These territories are supported by infrastructures that inscribe specific narrative performances, combining artifacts, spaces, vocabularies, and protocols that provide a scaffolding for the flux of stories, information, and meaning that are constantly woven into the narrative fabric.

Narrative landscapes can be seen as a blend of living story systems and the socio-material environments that shape them. This perspective broadens the understanding of narratives and narrative resistance in design practice. Instead of viewing narratives as static frames and stories as mere data points, it highlights the dynamic, distributed story systems that guide the collective articulation of issues and the institutional and material practices that sustain them. We envision narrative resistance in systemic design both as

an intentional engagement with the existing story systems that inform the design process and as the tactical creation of narrative infrastructures that foster diverse story systems, inclusive of broader ways of seeing, knowing, and being.

The examples of narrative resistance of food sovereignty advocates show that narrative resistance builds impact when it is a collective act. Just as these interventions build upon shared vocabularies, meanings, and networks; in describing future-making as seeding and cultivating alternative possibilities, we aligned with the vocabulary of food sovereignty movement. This alignment wasn't merely a linguistic, it marked our grounding in the ecological, relational, and caring worldview that is central to the movement. This approach was reflected in the open format of the activity, the intentional slowness of pinning cards on threads, and the collage format of the zine spreads that welcomed fragmented, composite expressions. It was relationally supported as we showed up as allies in the wider ecosystem. We recognize that the artifact would likely have little impact outside the network it was created in—it's hard to imagine it circulating on the desks of CEOs—but that was never the aim. The goal was to activate the implicit stories within the community, understanding that it would need to be presented differently to reach beyond this circle.

The key implication is that systemic design practice is always embedded within narrative ecosystems, where the sensemaking structures it proposes intersect with broader infrastructures that organize the collective articulation of public matters. We argue that meaningful and impactful narrative resistance requires intentional engagement with these infrastructures, their ways of knowing, meaning systems, networks, and material practices, whether that involves countering, aligning with, or transforming them. As

Barcham (2018, p.2) states, it's a matter of creating "joint horizons that do not involve the subjugation of one 'way of being' to another".

We propose the following critical compass for cultivating narrative resistance in systemic design:

- **Knowings:** How can the narrative infrastructures we design activate ways of knowing that are suppressed in dominant knowledge systems?
- **Meanings:** How can we intentionally use language—whether textual, visual, or performative—to genuinely create space for alternative meaning and value systems without translating them to institutional vocabularies?
- **Organizings:** How can we intentionally position our narrative practices within existing networks and channels to foster story systems that transcend boundaries and enhance coalitions' ability to drive narrative change?
- **Doings:** How can we creatively disrupt the narrative traditions embedded in our existing tools and shape new sensemaking spaces that can cultivating more diverse story systems?

These categories are not meant to be definitive or exhaustive; they are intended to be flexible and adaptable. As narrative inquiry and activation continue to evolve within design practice, this framework should be seen as a starting point—a guide that encourages ongoing reflection and adaptation.

3.5 Conclusion

This paper contributes to systemic design praxis by framing narrative resistance as a collective, networked, and distributed activity, enabling designers to build on the forms of resistance within their networks as they navigate and expand the boundaries of diverse narrative territories, blending relational and strategic ways of knowing. However, integrating this approach into systemic design practice is not without its tensions.

Narratives are complex and operate on multiple levels, from our inner worlds to the broader social and material contexts in which we operate. Thus, our approach must be multi-layered. We need to *look within* to examine our own worldviews, internalized narratives, and biases regarding the legitimacy of different ways of knowing and *look beneath* to identify the knowledge legacies embedded in the systemic design tools we use. Additionally, we should *look around* to understand the narratives that surround our work and find ways to connect with and amplify existing acts of resistance and *look back* to account for the reach and consequences of the narratives we cultivate.

This critical reflexive practice is fundamental for systemic designers to better navigate the complexities of narrative resistance and foster more inclusive and transformative discourse through their work. Our reflection surfaced some tensions concerning the implicit preconceptions about the role of designers in collaborative inquiry. These tensions relate to the balance between seeking alignment and holding space for differences, as discussed in our previous work (Sungu et al., 2023). This manifests in framing the designer's role as that of a 'translator': efforts to create a shared language between different groups can risk flattening the nuances of diverse vocabularies into dominant terms. At a deeper level, such translation risks reducing fundamentally

different worldviews to mere 'perspectives.' It also challenges the notion of designers as external facilitators or neutral intermediaries— 'storytelling' or 'stewarding story systems' is not an objective position. In curating, weaving, and materializing stories, designers become co-authors. It equally challenges assumptions of 'neutrality', engaging in narrative resistance as a collective act means designers must position themselves as allies, while collaborating with proponents of competing narratives. Additionally, this approach challenges the idea of the context-independent designer. It requires deep engagement to truly understand the narratives at play and their significance within a specific context, a process that can take years, as our own experience has shown.

We hope that these frictions and tensions will inspire further exploration in the field of systemic design. Embracing these complexities, rather than avoiding them, can lead to more nuanced and authentic engagement with diverse narratives as seeds of radical possibilities.

CHAPTER 4

BEYOND HERO'S TALE:

NARRATIVE INQUIRY IN FUTURE MAKING

7:30 PM, Thursday. My partner and I are packing surplus food donations in the parking lot of the grocery store. This is a regular pick up of our local food rescue network.

8 PM. Our fingertips sting from handling packets of frozen salmon filets at the community food hub. Half of them are already defrosted, and the best-by date is today. I check the guideline on the fridge. We can't risk storing them for donation.

9 PM. I'm parked on a dark street, handing a 2-pound salmon filet to a friend. Two more friends are scheduled to pick up their share from my apartment.

10 PM. I'm kneading ground salmon into homemade cat food. It's the first time I've touched meat in years. My partner, a lifetime vegetarian, can't stand the smell.

11.30 PM. I'm standing in the alley, staring at the pile of the remaining salmon I had to dump in the trash. The discount label reads "50% off"—that's the price of life, I guess. I feel defeated, tired, and angry. My mind races with ways this could have gone differently.

On paper, this fits the recipe for rescuing food from going to waste at stores. A grocery store willing to donate surplus food, a local hub for redistribution, and volunteers who can move it. On paper, cold and smelly fish filets become a symbolic arrow traveling between two points in space, carried by invisible forces. This neat circuit does not speak to the feeling of accountability for the sacrifice of life, the relationships, and the culinary creativity it took to make up for the shortcomings of a wasteful food system. It does not speak to the deeper cultural and economic paradigms that reduce food to a mere commodity. When some issues never get raised, some future possibilities remain unavailable.

Novelist Amitav Ghosh (2017) calls out humanity's imaginative failure in the face of the climate crisis as *The Great Derangement*. Ghosh observes this in contemporary literary fiction's inability to conceive the unexpected leaps and contradictions of the climate crisis while relying on modernist narrative traditions that order life in a gradual unfolding of events. Similar narrative traditions are embedded in the institutional languages and processes with which societies navigate contemporary social and ecological crises. The circles, arrows, icons, and numbers that populate policy papers, reports, and slide decks tame complexity at a cost. When institutional narratives smooth out the tensions, contradictions, and fragments that don't fit in, the much-needed twist in humanity's storyline becomes unimaginable.

Humanity's imaginative failure partly stems from this institutional pursuit of tackling the current crises with the same mindsets that have contributed to them. Despite an ever-increasing capacity to collectively make sense of these crises, the efforts remain governed by the technoscientific ethos of rationalization and efficiency. In order to conceive our collective lives beyond these paradigms, we need to shift how we make sense of issues and articulate change.

4.1 Future Making as Narrative Inquiry

To tackle this, we turn to a fundamental yet overlooked sensemaking mechanism: narratives. From ancient cave paintings to policy papers, humans give meaning to their lived and imagined worlds by telling stories (Bruner, 1991). Seen through this narrative lens, collective knowledge production is not a neutral act of capturing and interpreting information. It involves the cultivation of networked and distributed story systems

through which to stabilize shared ways of seeing the world and envisioning its possibilities.

Alongside future-oriented representations that foreground certain possibilities and issue commitments (Whyte et al., 2022), the stories about the present and past play an equally important role by setting the frame, focus, and protagonists of future worlds. The institutional story system of regulations, projects, data, and studies that frame food rescue as a logistical challenge, establish surplus food as the permanent backdrop of other possibilities. When dominant story systems narrow down the framing of issues and conceal possibilities, cultivating alternative story systems may help broaden the imaginative capacity of future making practices.

On the one hand, design, as a fundamentally discursive practice (Krippendorff, 2005), is well-versed for this task. The field of design has evolved a rich repertoire of using stories to critically and creatively explore complex issue spaces beyond what can be articulated in the conventional narrative grounds of institutions. These examples help foster the emergence of perspectives and new directions (Pendleton-Jullian and Brown, 2022), facilitate the flow of ideas between stakeholders (Price et al., 2018), or engage intuition and affect to foreground intangible aspects of systems (Talgorn and Hendriks, 2021). Yet, shifting story systems requires an understanding of stories as institutionally sustained, dynamic meaning systems beyond specific project contexts. We see this as part of a broader need for designers to engage with institutions and understand their role alongside them in their efforts to foster systems change (Lee, 2024).

On the other hand, fields like transition (Tidwell and Tidwell, 2018), organizational change (Hansen, 2019), and policy studies (Jones and McBeth, 2010) have built substantial knowledge of how narratives shape institutional and societal responses to change. However, their insights often remain at a descriptive level and pay little attention to material contexts through which narratives are performed. To bridge this gap, we draw on the concept of *narrative infrastructures* (Sungu et al., 2024). Narrative infrastructures describe the social and material contexts through which story systems are cultivated, maintained, and filtered in collective knowledge production. These infrastructures encompass the artifacts, vocabularies, meaning systems, networks, spaces, and platforms that are deeply embedded in everyday sensemaking practices. They play a critical role in future making by stabilizing the underlying system of stories through which societies envision their collective lives.

When dominant story systems narrow down the framing of issues and conceal possibilities, cultivating more diverse story systems emerges as a space of resistance to foster radical shifts. How can we then, shape narrative infrastructures in ways that can disrupt reductive framing of issues and filtering of possibility spaces? How might more diverse story systems expand the imaginative and critical capacity of future making practices?

This paper presents findings from a two-year action research project in which we deployed narrative infrastructures within a multidisciplinary research network addressing the systemic issue of wasted food. Drawing from Feminist Science and Technology Studies (STS) and critical futures, we focused on centering situated and subjective perspectives in sensemaking activities. We explore how these alternative narrative

infrastructures can help disrupt the dominant technoscientific framing of food waste and foster more radical directions. We also reflect on the ongoing work of cultivating and stewarding these story systems, drawing attention to the tensions and contradictions that arise when diverse ways of knowing come into contact.

4.2 Theoretical Background

Knowledge systems, with their embedded languages, norms, frameworks, and material practices, shape how societies make sense of the world and envision its possibilities. As Marie Puig de la Bellacasa notes, “ways of studying and representing things have world-making effects” (2017, p. 17). This influence becomes particularly visible at the science-policy interface, where the norms and structures that govern knowledge production influence how issues get framed and which strategies are deemed credible. For instance, the dominant framing of biodiversity as an economic asset to be managed privileges market-based solutions and marginalizes relational, ecological models such as environmental stewardship.

Beneath these framings lie the cultural, social, and economic paradigms that guide how people make sense of the world and envision their collective futures. As Sheila Jasanoff (2004) reminds us, “the ways in which we know and represent the world (both nature and society) are inseparable from the ways in which we choose to live in it”. Fundamental assumptions within Western knowledge systems, such as the human-nature dichotomy, limit the recognition of human interdependence with other species, sustaining worldviews that contribute to current ecological crises (Boehnert, 2018). As design joins

the pursuit of new horizons at the science-policy intersection, it must confront the systemic narrowing of possibilities embedded within dominant knowledge practices.

Narrative infrastructures provide a lens for approaching this as a design problem, emphasizing the dynamic interplay between social and material contexts of knowledge production and the narratives they legitimize. The artifacts, networks, and meaning systems that make up narrative infrastructures shape what stories are given legitimacy, and which ones are silenced. To disrupt the systemic filtering of possibilities in knowledge production, designers must recognize how their practice is embedded within these broader infrastructures, as they create alternative ones.

Narrative Conventions in Western Scientific Knowledge. The contemporary call on science to mediate national and global policy challenges is marked by a modernist portrayal of science: as a detached, objective, and unifying force capable of transcending political interests and conflicts that hinder cooperative action (Rungius and Flink, 2020). While this view of scientific knowledge has been extensively critiqued (Sismondo, 2010), it continues to shape the narrative conventions in scientific and policy discourse, as well as broader institutional contexts. These conventions are embedded in the social and material contexts of knowledge production, shaping what stories can be told. We turn to Feminist Science and Technology Studies to discuss three such conventions that characterize formal articulation of systems change: (1) detached gaze, (2) cohesion and reconciliation, and (3) progressive timeline.

Detached Gaze: Western scientific narratives often depict the world through a distanced, objective lens. Donna Haraway (1991) critiques this as the “view from nowhere,” which bases its claims of universality on being freed from body, context, intention, and equally, accountability. Instead, she argues for situated knowledges, acknowledging that perspectives are always embodied, partial and located in specific contexts. Situatedness of knowledge should not be seen as rejection of objectivity for relativity. Quite the opposite, it advocates for stronger objectivity by accounting for the assumptions, interests and values that shape knowledge production (Harding, 2008).

Cohesion and Reconciliation: Another defining feature of Western scientific narratives is the drive for coherence and reconciliation. According to Haraway (2016), this inclination toward shared, harmonious representations of reality, comes at the cost of stripping the world of its inherent contradictions and heterogeneity. Instead, she advocates for a language that does not resolve contradictions but rather embraces them, keeping threads open for new connections and insights to emerge. Lucy Suchman (2002) emphasizes the importance of lived, relational work of knowledge production for holding space for contradictions and controversies and avoiding premature closure of debates. In this view, story systems are not assembled as pieces in a puzzle but are continuously woven together, where open threads matter as much as the connections.

Progressive timeline: The third narrative convention is the progressive, forward-facing orientation prevalent in science and technology discourses. De la Bellacasa highlights the ethos of constant, urgent progress that fuels this perspective, driven equally by fears of

impending crises and the promise of innovation as a solution (2017, p. 174). Unlike the “forward march of progress” typical of late modern Western societies (Wenzel, 2020), the contemporary orientation resembles anxious pacing between future horizons to find the best way forward, yet equally resistant to looking backward. Christopher Groves emphasizes that future horizons are not “free-floating cultural structures”, but inscribed into the material environments, tools, devices, and infrastructures through which people know and experience the world (2017). As such, temporal orientation is embedded in and reproduced by the narrative infrastructures.

Holding Space for the Stories Untold. Ursula K. Le Guin’s “Carrier Bag Theory of Fiction” (2019) provides a compelling metaphor for the narrative infrastructures we envision. Le Guin frames science fiction as a carrier bag for the diverse worlds, dreams, and lives that are left out by the triumphant, linear narratives of progress often found in scientific and technological discourses. She traces the dominance of this heroic story back to early days of humanity: “It’s the story that hid my humanity from me, the story that mammoth hunters told about bashing, thrusting, raping, killing, about the Hero.” The crises of our time, Le Guin argues, point to a tragic end for this tale: “It sometimes seems that that story is approaching its end. [...] Hence it is with a certain feeling of urgency that I seek the nature, subject, words of the other story, the untold one, the life story” (2019, p. 33).

In her introduction to Le Guin’s essay, Haraway (2019, p. 10) emphasizes that this call extends beyond fiction to the broader story systems that shape how societies understand the world and its possibilities: “It matters what stories we tell other stories

with; it matters what concepts we think to think other concepts with.” She further poses, “the burning question is how to join in telling the needed stories” (Haraway, 2019; p. 18). The carrier bag metaphor serves as a guide for our vision of narrative infrastructures that embrace this mission—a container that welcomes complexity and difference without forcing them into a singular plot, holding space for stories to coexist without the demand for unity or resolution.

Critical and speculative futures practices offer many examples of narrative infrastructures that embrace contradiction, multiplicity, and situated perspectives. For instance, *Sankofa City* (Baumann et al., 2017) uses participatory multimedia storytelling to ground urban futures in the identities, dreams, and lived experiences of Leimert Park residents, a historic hub of African American culture in New Jersey. Another example, the *Feral Atlas* (Tsing et al., 2020), experiments with fragmented and non-linear storytelling through a digital platform. The project uses artistic cartography to invite visitors to explore hybrid ecologies emerging from the remnants of human infrastructures, conveyed through evocative audio-visual poems. Meanwhile, *Reimagining Work* (Forlano and Halpern, 2023) uses a board game format that invites participants to explore the intersection of technology and labor beyond the constraints of a progressive, linear future orientation. The game opens history as a speculative space, encouraging players to view the past as a resource for reimagining future possibilities.

There is much to be learned from this repertoire of narrative infrastructures that hold space for the diverse lived and dreamed worlds often erased from institutional narratives. By challenging the conventions that filter out varied perspectives, complexities, and contradictions, these interventions interrupt the continuity of dominant

storylines. Yet, despite their contributions, their influence on broader future-making practices remains underexplored. A key challenge lies in their positioning as temporary events within artistic or cultural spheres (Mazé and Wangel, 2017), which can limit their impact on institutional practices of future-making. Designers need to find ways to cultivate critical distance without fully distancing themselves from the structures they seek to disrupt.

4.3 Methodology

We conducted a two-year action research study where we designed and deployed narrative infrastructures within a national research network focused on the systemic issue of food loss and waste. The issue of food loss and waste exemplifies the challenge of driving transformative change while operating within primarily technoscientific frameworks. Dominant framing of food loss and waste focuses on shifting consumer behaviors and optimizing the flow of materials across food supply chains to prevent waste of resources. This framing downplays the need for transforming the culture and collective values for preventing food waste before it occurs and obscures the contradictions of such a shift with the current economic paradigms (Messner et al., 2020). Conceiving such a cultural transformation requires paying attention to the meaning of, and relations with waste in our social lives (Turner, 2019).

The network brings together researchers from 14 institutions, including two of the co-authors. It is organized around the principle of convergent science, aiming to foster collaboration across disciplines to understand systemic drivers of food loss and waste, and make science-backed recommendations for intervention. The collaboration is

primarily virtual and organized in clusters such as policy, data, education, and co-design, complemented by an annual in-person meeting.

In this context, we created a series of interventions that invited network members to explore ‘what makes a waste-free food future’ through their personal experiences, values, and frustrations. In a setting where scientific narrative conventions are the norm, we aimed to create space for open-ended and contested inquiry that foregrounds situated and subjective perspectives. Our goal was to problematize the issue of wasted food through a broader range of considerations, as a gateway to conceive possibilities beyond the dominant framing. We co-created each intervention with designers in the network, through several months of iteration and feedback from other network members to ensure alignment with the network’s goals of transdisciplinary engagement.

Intervention I: Artifacts of Annoyance. During the first in-person meeting of the network, we staged a small, interactive exhibit titled *Artifacts of Annoyance* (Figure 1). The approach centered personal frustrations—our “annoyances”—as an entry point to problematize everyday practices and structures that sustain a wasteful food system. As opposed to a purely positive articulation of impact, we shared what we are ‘up against’ to tap into deeper contradictions and value conflicts in the organization of food systems.

We curated a set of artifacts that represented current practices that frustrated us, with critical museum labels offering historical context (Figure 4.1.). For example, the dandelion, reclassified as a “weed” in the 20th century, critiqued manicured lawns as status symbols that glorify unproductive land. Another pedestal displayed canned beans, spam, and instant noodles, questioning the adequacy of food bank staples and critiquing

the rise of food banks from temporary solutions to a permanent fixture in a wasteful food system. By presenting these practices as historical inventions, we aimed to disrupt their perceived normalcy and inevitability, and open space for approaches rooted in different values and priorities.

To signal this transition from critique to possibility, the exhibit concluded with speculative artifacts such as a bag of 'Powdered Super Greens,' envisioning a future where grocery stores process surplus produce on-site, shifting the burden of responsibility from consumers and volunteers to the broader system. We invited participants to interact with the exhibit by sharing their thoughts with sticky notes.



Figure 4.1. Artifacts of Annoyance Exhibit

Following the exhibit, we held a short workshop to invite participants to explore their shared annoyances as a gateway to problematizing wasteful practices in the food system. Participants first toured the exhibit and wrote their annoyances on sticky notes. We formed groups based on these mutual frustrations. The teams used worksheets to examine the systemic value commitments that sustain these annoying situations and

reimagine them in an alternative future where different values are prioritized. We emphasized 'valuing' as a dynamic process embedded in everyday choices and systemic tradeoffs rather than values as universal moral principles.

The teams used craft materials to illustrate the future they envisioned with speculative artifacts. To encourage reflection beyond technological fixes, we provided participants with a deck of cards that contained prompts for reimagining categories like culture, labor, and environment. We welcomed a variety of contributions, such as objects, news stories, or abstract pieces, without needing to align into a cohesive scenario. We replaced the artifacts on the exhibit with these creations and re-titled the exhibit as *Artifacts from Alternative Futures*. In this process, the exhibit served as a scaffolding for the flow of ideas, shifting stories from our own to those created by our colleagues.

Intervention II: The Collective Pantry. Our second intervention, *The Collective Pantry*, took place the following year and unfolded as a series of playful activations centered around the question: “What ingredients go into making a waste-free food future?” Our goal was to invite network members to think about the *ingredients* that get discarded in academia, such as culture, relationships, and mindsets.

Similar to the previous project, we began by drawing on our own stories. Before this study, the first author documented their year-long journey of learning to prevent food waste, both at home and by volunteering in local food rescue networks. By using this ethnographic data, we experimented with a storytelling format that framed each experience as a recipe for preventing food waste, some successful, and others failed attempts at it. Each recipe listed the ingredients that shaped this effort, such as food itself,

containers, and spaces but also relationships, mindsets, and feelings. These stories offered glimpses of waste-free food futures emerging in community-driven food systems, accounting for foodways beyond the industrial supply chains.

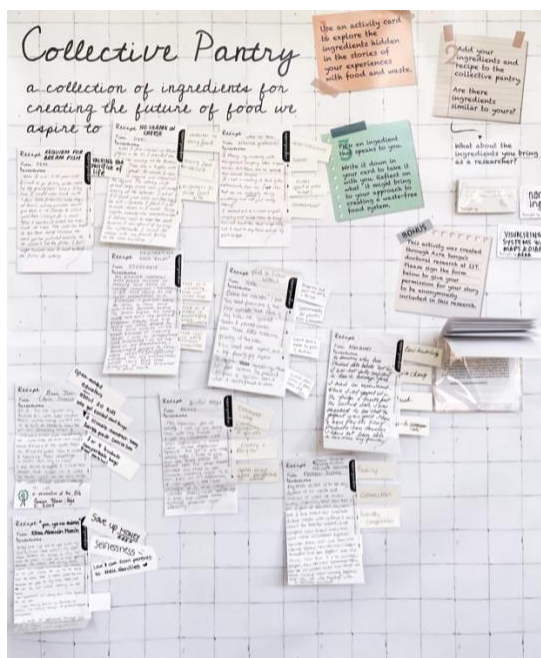


Figure 4.2. Documentation of Ethnographic Data as Recipes

By using this format as a template, we staged a self-guided storytelling activity (Figure 4.2) during the in-person convening of the network. We invited network members to share their own stories with food waste prevention, and identify the ingredients they bring from their personal journeys. We displayed two of our own stories on the activity board to help participants navigate the metaphor. The decision to follow a self-guided format was made due to time constraints of the convening.

After the convening, we created a website to display the stories (Figure 4.3). We designed it as a *virtual pantry*, listing the ingredients and their origin stories. We organized the ingredients into six broad categories —*Mindsets & Culture, People & Relationships, Play & Creativity, Knowledge, Objects & Spaces, and Access to Funds*—

with a wildcard category to signal the evolving nature of the pantry. In the meantime, we informally met with various network members to welcome their ideas for shaping the next phase of the intervention.

To bring these ingredients back into the conversation, we organized a virtual co-creation workshop with the whole network. Our goal was to explore how the diverse perspectives embedded in the ingredients could spark new ways of thinking about food waste and its prevention beyond the filters of academic discourse. We designed the workshop as a virtual cook-off. Participants individually selected ingredients from the pantry and worked in teams to envision how these ingredients might be combined to create impact toward a waste-free food future. These could take any form, such as interventions, situations, or approaches.



Figure 4.3. Storytelling Activity and Virtual Pantry

As participants selected ingredients, they shared what motivated their choice. We encouraged them to reference the origin stories of ingredients for inspiration. Participants could add "secret ingredients" of their own into the mix, such as their research expertise, skills, or contributions outside of their formal role. Our goal was to blur the lines

between personal and professional spheres and explore how these intersections influence collective efforts to reimagine a waste-free food system.

We piloted the workshop with the co-design cluster, so a few members participated twice. This was helpful in guiding the eleven breakout groups, as we couldn't assist each group individually. At the end of the session, we held a brief discussion where members reflected on their experience and how it relates to their work and networks' convergence goal. We also distributed a short survey to gather additional feedback and thoughts.

Data collection and analysis. We collected data through photographs of artifacts created by both researchers and participants, as well as written documentation of processes and observations (Table X). Following the first intervention, we conducted an initial analysis to inform the second intervention. This analysis focused on how participants framed the issue and the shifts in perspectives reflected in the artifacts. After the second intervention, we transcribed over 600 minutes of audio recordings from the workshop breakout groups using the AI transcription platform Rev, followed by manual revision for accuracy. Our analysis followed a grounded approach, where insights were derived inductively from the data. We began with open coding in MaxQDA, applying it to both transcripts and screenshots of virtual whiteboard spaces. In the second phase, we performed axial coding to identify recurring themes. Finally, we carried out selective coding to refine these themes and re-coded data from the first intervention. This allowed us to identify continuity between the two interventions and to capture persisting frictions.

Table 4.1. Data Collection

Activity	Setup	Time	Number of Participants	Data collection	Goal
Interactive exhibit	In-person	2 days	58	Photo-documentation of sticky notes added by the audience	Assess participant willingness to engage with the exhibit and responses shared by visitors
Workshop I	In-person	75 mins.	13	Photos of artifacts created by participants	Capture the key issues explored and the focus of proposed food waste prevention solutions
Self-guided storytelling activity	In-person	2 days	8	Written stories contributed by participants	Document personal reflections on food waste prevention and individual considerations
Pilot workshop	Virtual	50 mins.	10	Documentation of artifacts created by participants on virtual whiteboards	Evaluate platform engagement and gather feedback to refine the workshop format
Workshop II	Virtual	50 mins.	38	Video and audio recordings of breakout group discussions	Observe the process of deliberation and ideation; document the role of stories in the process; capture frictions encountered
Survey	Virtual	NA	10	Documentation of survey responses	Gather feedback and evaluate the workshop

4.4 Findings

The projects aimed to create a space where personal perspectives are legitimized and valued, cultivating alternative story systems through which the issue of food waste is problematized and pathways are re-configured. By combining physical and digital platforms with workshops, both interventions created adaptive scaffoldings for participants to engage with diverse stories and consider future possibilities beyond the filters of traditional academic discourse.

The findings demonstrate, first, how personal engagement and storytelling helped problematize food waste in relation to a broader range of considerations, known but often

left unspoken in scientific research. Secondly, we discuss how this personal engagement foregrounded relational considerations in prevention of wasted food, where participants brought in fragments of knowledges, insights, questions from various spheres of experience, considering their role as agents of change beyond their formal expertise. Finally, we explore the tensions that surfaced in this inquiry between the technical and relational approaches to future making, echoing a deeper dichotomy between structural and relational considerations in future making.

Problematizing Food Waste Through Personal Perspectives. The projects created platforms for participants to problematize food waste through their personal frustrations, aspirations, and curiosities, raising issues often left unspoken in academic research. This served as a gateway for deeper critiques of the cultural, political, and social underpinnings of a wasteful food system.

In the first intervention, the 'annoyances' activity provided a light-hearted and engaging entry point into this critical engagement. By sharing frustrations with everyday practices—such as over-purchasing, best-by labels, or buffets—participants unpacked the systemic value commitments that sustain these practices. A key theme was capitalism's role in sustaining wasteful practices, including cultural perceptions of excess as abundance and hustle culture that make it difficult to plan meals. This allowed participants to articulate critiques that don't often surface in scientific discourse and engage in a values-driven exploration of future alternatives and systemic tradeoffs.

This demonstration of how personal perspectives can disrupt and enrich systemic discussions set the stage for the next intervention. Building on this, we used the metaphor

of ‘ingredients’ to position personal experiences as valuable sources of knowledge, despite often getting discarded as ‘scraps’ in scholarly discourse. The pantry included items like ‘sentimentality for plants,’ ‘risk-taking,’ and ‘kids’ excitement’, that were drawn from personal stories of members, inciting considerations beyond what might seem immediately relevant to food waste.

As participants explored the virtual 'Collective Pantry' and selected their ingredients, they shared personal motivations, curiosities, and frustrations, often blended with stories. In this process, ingredients served as anchors for diverse concerns, with each story conjuring another in an open-ended process of deliberation. For instance, one participant shared what made her choose the ingredient ‘seeing food as sacred,’ through a powerful and intimate story:

"And I think what really kind of started me on this kick is when I was breastfeeding, and I had shipped a ton of breast milk when I was moving from [...] to [...]. The shipment got lost, and they asked me to estimate the value of my package loss when I went to claim it. And I was like, it's irreplaceable."

Through this story, she critiqued the commodification of food that reduces it to a monetary value, overlooking the labor, care, and resources behind its production. Another participant in the group connected her choice of ‘sentimentality for plants,’ to this conversation, questioning how perceiving plants as non-sentient and humans as separate from ecosystems reinforces the commodified view of food. This sparked an exploration on fostering a 'sense of gratitude' as a way to foster mindful consumption behaviors, “acknowledging whatever life allowed that food to feed your family and in those lives of people that you care about.” As the group explored this direction, they raised fundamental questions about how we assign value beyond financial terms and grappled with the challenge of embedding these alternative values into systemic interventions.

Unlike the first intervention, where participants departed from shared annoyances, selecting distinct ingredients anchored the conversation in diverse concerns. Participants responded to each other's stories, searching for intersections, and grappling with frictions as they explored a broad range of issues. While this problematization was essential to disrupt reductionist framings of food waste, we were interested in how these critiques could become productive in broadening future possibilities.

Weaving Possibilities Through Diverse Spheres of Knowledge. In the second part of each workshop, we explored how these problematizations could inspire new approaches to preventing food waste. In the first workshop, participants reimagined the annoying situation in alternative futures aligned with their values. Ideas ranged from a catering system that subverted norms of abundant food displays to multi-generational housing that preserved traditional knowledge of reusing leftovers. While this process foregrounded cultural, social, and political influences, whether these critiques and radical possibilities could inform ongoing practice wasn't clear. This raised the risk of alternative visions remaining as marginal *what-ifs*, disconnected from the realities they sought to challenge. We leaned onto this gap and shifted our focus from imagining distant futures to exploring how futures are actively shaped in the present. We used the metaphor of *ingredients* to emphasize *what goes into making futures*, and we invited participants to see themselves as active agents in the making of these futures.

This broader framing encouraged participants to consider non-technical ingredients they bring to the table, such as creativity, humor, or relationships with their community. One group explored how they could use their access to sustainability-

focused, large research grants to push their institutions to commit to concrete food-waste prevention strategies rather than “paying lip service to sustainability.” By highlighting current initiatives where this shift is already sprouting, they envisioned their agency within broader collective efforts.

Groups drew inspiration from various spheres of life, recalling childhood memories, family traditions as well as interesting encounters in their research work. One group explored how shared living and cooking spaces could foster collective value commitments to prevent wasted food. This vision challenged the dominant view of single-family households individually managing food and positioned food waste prevention within the broader question of “How do we live differently?”. Reflecting on their various encounters with intentional living communities, cooperatives, and mutual aid groups, they gathered a mosaic of possibilities from spaces that already embodied the shifts they envisioned.

While this framing encouraged participants to consider their diverse spheres of experience as valuable sources of insight, very few participants explored how their scholarly know-how might be combined with these insights. One participant noted the difficulty of combining their expertise with the culturally oriented considerations drawn from personal stories. This boundary between personal and professional spheres reflected the deeper dichotomy between structural and relational approaches that we explore in the next finding.

A Persisting Divide Between Relational and Structural Approaches. Both interventions aimed to create a scaffolding where multiple possibilities could coexist

without needing to reconcile into a single cohesive narrative, allowing participants to explore a range of perspectives. However, a clear dichotomy emerged between relational approaches—centered on cultural and mindset shifts—and those focused on more structural solutions, such as legal frameworks, policy changes, and technological innovations.

In the first intervention, some groups gravitated toward technological solutions, while others leaned into cultural transformations. In the second intervention, we leaned onto this tension by encouraging participants to explore unconventional combinations of *ingredients* in fostering a waste-free food future. The approach brought relational and social considerations into focus but also surfaced deeper divides between these and more structural approaches.

While many could envision how shifts in culture and mindsets might influence behavior within small communities, conceiving these shifts on a larger scale was more challenging. One participant voiced skepticism about whether fostering a 'sense of gratitude' for food could be effectively translated into a systemic intervention, saying: "I guess my inclination is always to try and go for a more systems approach rather than a small program that teaches just a few kids." This sentiment suggested that relational approaches might be relegated to small-scale efforts.

Another group grappled with this tension while exploring how *kids' excitement* might be combined with *food law awareness*. Reflecting on this unusual combination, they reframed food donation liability concerns through an emotional perspective, likening it to kids' reluctance to try new things unless it comes from someone they trust. When they struggled to translate this idea into a concrete approach, we suggested approaching it

as a fusion cuisine experiment. They came up with an approach that centers relationships of trust between organizations to address fears of liability. One participant reflected on the challenge of navigating this dichotomy:

"I was surprised at how hard it was for me and maybe [for] our group as a whole to reconcile very different ingredients. Two of our ingredients were very much about relationships and people, and then one was more structural and institutional. We'd create solutions and then think, wait, how does this work with the law part? Or we'd think about the law part and suddenly people weren't part of it anymore."

They reflected on the cultural norms around flavor combinations as a metaphor for the inherent challenge of convergence science: "It does [...] speak to what we do here as a network, and that uncomfortable space when things don't easily go together." These metaphors provided a different syntax that allowed participants to explore the implicit norms and boundaries in scientific knowledge production and reflect on their implications for future making.

4.5 Discussion

This research pursued a design approach to counter the systemic narrowing of futures in knowledge production. We expanded on the concept of narrative infrastructures as a mechanism by which this systemic filtering of possibilities can be observed and disrupted. The interventions tactically combined artifacts, spaces, metaphors, and interactions to foster alternative story systems. Through the sharing of stories, we grounded collective inquiry in situated and partial perspectives, favoring loose entanglements over tight connections.

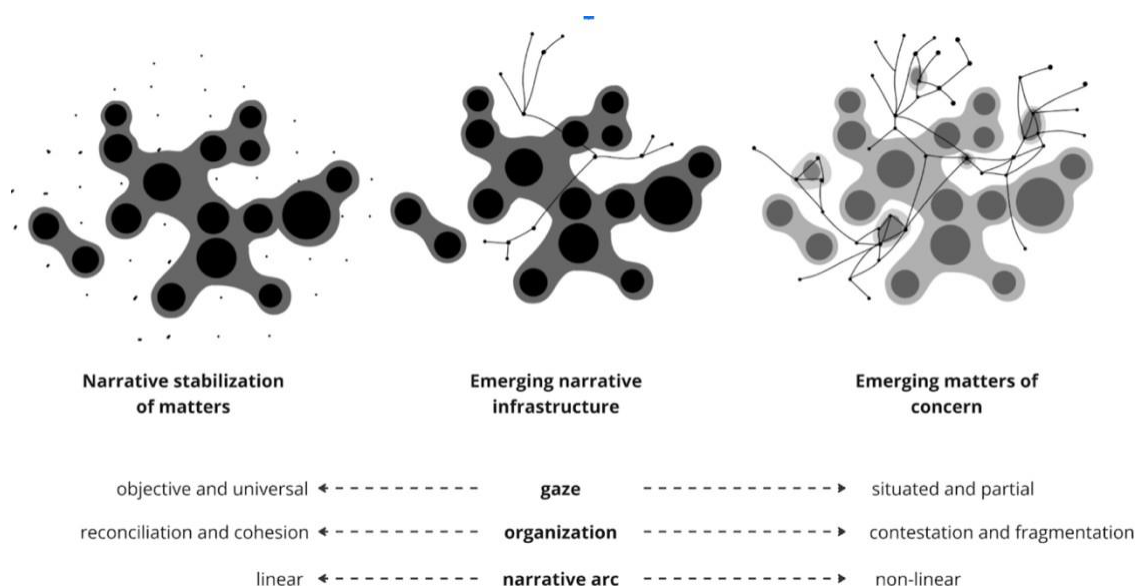


Figure 4.4. Re-configuration of Matters through Emerging System of Stories

As stories multiplied and branched into one another, new areas of concern started to emerge, such as the underlying cultural and political paradigms that are often overlooked in scientific discourse. The evolving story system became a mosaic of possibilities, with fragments gathered from researchers' diverse spheres of experience and knowledge. For instance, combining children's enthusiasm about food with companies' liability concerns led to novel approaches to that center relationships of trust and beyond awareness-raising. People could see themselves as protagonists in their stories, not only as researchers, but through their diverse roles in future making.

It is not enough to contrast dominant narrative conventions with alternative ones, but navigating the space between, which is no simple feat. As participants reflected on the frictions, they noticed invisible boundaries in knowledge work and their implications for future making. These included barriers between personal and professional identities, or the divide between technology and policy-driven approaches and those that center

cultural and social shifts. These had direct implications on how researchers perceived their agency, what they considered relevant, viable and compatible. Rendering these boundaries visible was a first step to the longer work of disrupting them, which echoes Suchman's emphasis on relational work of translating knowledge.

A core aspect of our approach was the cultivation of story systems, emphasizing slow, adaptive growth. As the stories grew and expanded, we tactically created new scaffolding around them. Our continued engagement with the community was essential as we co-created each stage through iterative feedback and trust-building, constantly aligning the interventions to those of the network. A few members adapted the interventions for use in educational settings, showing that the infrastructures could extend beyond their initial context. As such, the narrative infrastructures served as living carrier bags, stretching, tearing and morphing with the stories, woven and carried by many hands along the process.

4.6 Limitations and Future Directions

The insights gained from this research offer pathways for expanding design's role in future-making, particularly at the science-policy intersection. Key among these is the importance of embedding narrative infrastructures within ongoing practices, moving beyond standalone interventions to create lasting impact. This means creating friction from within, by surfacing and challenging the implicit boundaries of dominant knowledge systems. While our research centered on a single community of researchers, a compelling future direction is adapting narrative infrastructures across scales, from

smaller communities to larger organizational ecosystems, navigating the distinct cultural dynamics and structural conditions at play.

CHAPTER 5

CONCLUSION

At the heart of this research is the notion of collective inquiry in systems change—distributed and networked practices through which diverse collectives generate knowledge about and frame systemic issues, envision desirable futures, and advocate for different pathways of change. Yet, despite an ever-growing capacity for collectively making sense of systemic issues, the collective action needed to address the pressing social and ecological threats remains frustratingly stagnant. For many years, design, like other fields, has focused on the complexity, ambiguity, and unpredictability of systemic transitions as the core challenge. This focus on enhancing our capacity to make sense of complexity—through increasing volumes of data and ever-more sophisticated frameworks—misses a deeper issue.

Systemic issues often sit at the juncture of divergent visions of the common good, imaginaries. Tackling these issues, therefore, is not only a matter of navigating complexity but a battle over the meaning these issues hold in our collective lives. While the conflict between diverse interpretations of systemic problems is frequently seen as a barrier to collective action, this research considers it an essential resource—a way to foster the critical and creative mindsets necessary for transformative change. When dominant imaginaries reinforce the mindsets underpinning current crises, the potential for radical shifts depends on recognizing and disrupting these entrenched patterns, and making room for alternative imaginaries. Without accounting for the role of imaginaries

in sensemaking process, design risks becoming complicit in perpetuating dominant paradigms and sidelining radical possibilities growing at the margins.

This research centered narratives as a mechanism by which design can intentionally engage with imaginaries, accounting for how they shape and are shaped by sensemaking efforts across scales. In design practice, narratives have always played a central role, as design fundamentally is a practice of creating meaning and shaping discourses (Krippendorff, 2006). However, the field's understanding of narratives as dynamic mechanisms that operate across various scales is still nascent. Conversely, fields such as transition and organizational change studies have explored how narratives shape institutional responses to change. They show that narratives do more than convey information; they actively shape how groups understand problems and imagine possibilities. However, these insights often remain at an analytical level, lacking practical approaches for intervention. This research aimed to bridge this gap by framing narratives as living meaning systems sustained by social and material contexts of collective inquiry.

5.1 Summary of Insights

Adopting a narrative lens, I approached collective inquiry as a distributed practice in which organizations and diverse collectives curate stories, construct and disseminate narratives, and advocate for change. This lens offers a way to surface the multiplicity of interpretations, contradictions, and conflicts embedded in systemic issues. Through three studies, I explored how narratives can enable intentional engagement with imaginaries in design practice, accounting for the interconnectedness of various scales in collective articulation of systems change. Together, these three interconnected studies, published as

separate papers, shaped an approach that centers on narratives' capacity to both stabilize and disrupt imaginaries and the translation of this capacity into design practice.

The first study was situated within an emerging multi-stakeholder coalition, examining how narratives act as collectively stabilized framings of issues that anchor perspectives of stakeholders. This study proposed a practical approach for scanning and analyzing the narratives of change within a given issue space, unpacking the diverse, and at times conflicting, interpretations, goals, and value commitments of stakeholder groups. By unpacking systemic issues through diverse narratives of change, the study grounded the framing of systemic change as a constellation of matters of concern, and demonstrated how narratives can be used to make room for considerations and contradictions that are otherwise smoothed out in purely strategic framing of systems change.

The following two studies expanded the understanding of narratives, moving beyond their role as stabilized framings of systems change to approach them as living meaning systems within the collective articulation of systems change. These studies shifted focus to the social and material contexts through which groups, organizations, and larger ecosystems codify their learnings and shape interconnected story systems. This work informed the development of narrative infrastructure: a conceptual framework that illustrates the dynamic interplay between the social and material contexts of collective inquiry and the story systems that these contexts sustain and mobilize. This framework acts as an adaptive scaffolding, enabling a view of social and material environments as designed spaces that stabilize specific ways of seeing the world and imagining its possibilities.

Narrative infrastructure serves as an analytical tool to understand how social and material contexts, by design, perpetuate specific ways of seeing and making sense of systems change. The framework's four dimensions—knowings, organizings, meanings, and doings—invite designers to consider the diverse ways in which stakeholder groups construct, articulate, and disseminate stories of systems change, using platforms, artifacts, and shared meaning systems. By moving beyond formal, institutional contexts that often reduce systems change to logical equations, and instead engaging with diverse story systems, designers can become more attuned to the social, cultural, and political complexities often smoothed out in official narratives. To effectively navigate these diverse territories of meaning and build a networked resistance to dominant paradigms, designers must also consider how their practice is embedded within broader narrative landscapes.

The second use of narrative infrastructures is a critical one that reflects on the narrative traditions embedded in design's methodological repertoire. In staging spaces for sensemaking, designers determine which stories about systems change can be captured and told, and which ones can not. As designers increasingly engage with institutional efforts to address systemic issues—working alongside governments, corporations, and scientific communities—the shared language and representations they create risk subordinating alternative story systems to institutional ones. Narrative infrastructures can serve as a critical compass, enabling designers to question and disrupt the ways of knowing they privilege, the language and meaning systems they employ, and their positioning within broader networks. By examining how they enable or constrain diverse

stories of systems, designers can intentionally leverage their narrative agency to resist reductive and exclusionary framing of issues.

The third use of narrative infrastructures is a generative one. Narrative conventions within Western knowledge systems—such as objectivity, detached perspectives, cohesion, reconciliation, and a progressive view of change—systematically narrow down framings of issues and possible futures. The promise of universal representations can come at the cost of flattening complex topics, strip away important controversies, and detach issues from identities and local contexts. Designers can disrupt these limitations by creating space for situated perspectives that embrace fragmented viewpoints, tensions, and open-ended questions, while also drawing on the past as a valuable source of knowledge. By holding space for alternative story systems, designers invite a broader range of considerations, help problematize overlooked issues and paradigms, and open up new spaces for possibility.

Integrating these perspectives requires shifting understanding of sensemaking. Rather than a process of assembling information and drawing connections, it emphasizes the living story systems that shape and are shaped by sensemaking activities. These story systems influence which perspectives are given legitimacy, which matters are considered relevant, and which futures are deemed possible. This approach disrupts the view of systemic issues as puzzles or equations to be solved. Instead, it attends to the entanglement of concerns, aspirations, values, and worldviews that make up the deeper complexity of systems change. It also accounts for situatedness of sensemaking within existing story systems, rather than assuming it a blank slate for new meanings. Third, this approach considers the interconnectedness of various sites and structures of sensemaking.

How groups make sense of issues is inseparably linked to broader contexts of collective inquiry through discourse, as well as social and material contexts.

5.2 Contributions to the Field

This research contributes to design practice both methodologically and theoretically. By proposing the framework of narrative infrastructures as a compass, it aims to guide designers and change practitioners in leveraging their narrative agency as a capacity for resistance across. The framework emphasizes fostering diverse story systems by continually working with various contexts of collective inquiry, rather than temporarily countering dominant narratives with alternative ones. This may involve micro-disruptions within the artifacts, spaces, and social contexts of sensemaking—holding space for tensions, open questions, and marginal perspectives that don't fit into master narratives. It can also support the creation of entirely new narrative infrastructures, helping groups gather not only their knowledge but also their concerns, hopes, and frustrations, thereby shaping the narrative foundations of transformative collective inquiry.

The second contribution is a toolkit developed from the findings of the first phase of this research, designed to help designers navigate divergent visions of change in multi-stakeholder efforts. This toolkit distills insights from critical discourse analysis into a step-by-step process for scanning, analyzing, and contrasting stakeholder narratives of change. It creates space for diverse perspectives and, importantly, provides a way to productively engage with tensions. The toolkit is intended as a flexible, adaptive

scaffolding for critical observation and action, and does not claim to replace the depth of critical discourse analysis.

Lastly, this research expands the theoretical grounding of knowledge production in design by drawing on Science and Technology Studies (STS) and Feminist scholarship. The studies illustrate how the notions such objectivity, universality, progress manifest as narrative conventions in science and policy discourse, constraining what stories can be told about the world and its possibilities. These narrative conventions are embedded in artifacts, languages, and institutional structures, reinforcing dominant paradigms in practices of codifying and sharing knowledge. This research calls on designers to recognize the influence of dominant worldviews within the very tools and methods they use to make sense of systems change.

5.3 Implications

The transformative power of narratives has long been known in social justice movements, but recent initiatives in social and public sector highlight the systemic mobilization of knowledge and resources needed for narrative change. Foundations such as the Ford Foundation and Open Society have been investing in the knowledge, networks and funding needed for narrative change toward racial justice and health equity (Open Society Foundations, 2019; Martinez 2022). Initiatives like Pop-Culture Collaborative and Liberation Ventures are advancing the knowledge about the narrative landscape of racial justice and reparations work and strategic know-how for narrative change. Notably, Color of Change and Narrative Initiative have advocated for narrative infrastructures as the networked capacity needed to cultivate transformative story systems

(Swenson-Lengyel, 2019; Robinson, 2019). More recently, Centre for Public Impact, a global organization dedicated to re-imagining government, launched resources that explore the power of storytelling and story-listening in advancing community-led systems change.

Narrative-informed design practice can play an important role in these efforts. By accounting for the diverse narratives within a given issue space and the stakeholder commitments they anchor, designers can more productively engage with the tensions inherent in narrative change. The narrative infrastructure framework proposed in this research can help bridge the gap between design's expertise in shaping material environments and social justice movements' focus on networks and coalition building. To become authentic allies in transformative work, designers must work alongside advocates, experts, and creatives who have long been advancing the strategic know-how of narrative change.

During the final months of this research, I collaborated on such an initiative with Community Food Navigator, a local nonprofit in Chicago. The project amplifies the stories of local growers to highlight the various resources that support them in their journey. It exemplifies leveraging digital media and platforms to enhance the capacity of local coalitions to connect through stories and advance a vision of relational food system transformation.

A broader implication of this work is considering narrative change beyond messaging and communication, but as a capacity embedded in distributed sensemaking activities across various domains. For instance, intentional engagement with narratives can help policymakers account for diverse stakeholder perspectives and reframe policy

debates beyond techno-scientific visions, incorporating social, cultural, and environmental dimensions of systems change. Researchers and educators can adapt these tools and frameworks to foster collective learning spaces where people can bring their values and identities to explore complex environmental and social challenges more critically and creatively. In the private sector, this approach can help companies align their business practices with broader societal goals by grounding this alignment in a deep understanding of how diverse communities and partners experience systemic issues and envision change, rather than relying on superficial alignment on high-level goals such as sustainability or economic development.

Some of these possibilities have already materialized through instances where our collaborators applied the outputs of this work. One collaborator from a government agency shared how she continues to use the decks we created for a workshop (See Chapter 2) , which illustrate critical shifts to consider when framing food waste prevention. She described the decks as a tool for fostering dialogue across diverse perspectives, saying, “I look at the cards and think about how someone who is not me may be thinking about the issue. The cards give me language to use when I speak to others about the issue, or it helps put me in their shoes so I can communicate to them in a way they will understand.”

Another example involved a researcher from the RECIPES Network adapting the Collective Pantry workshop format for a course on wasted food for undergraduate students. This approach invited students to center their personal values in envisioning approaches to prevent food waste. The researcher reflected that it provided a “unique academic experience for students.” Meanwhile, students shared that centering their values

in envisioning solutions deepened their sense of purpose and agency, as articulated by students, while also presenting tensions in balancing trade-offs between competing values. It is important to note that these examples are offered as anecdotal reflections rather than empirical evidence.

5.4 Future Research

This research opens up several avenues for further exploration at the intersection of design, narratives, and systemic change. One promising area concerns the role of technology and algorithms in narrative change. Narrative infrastructure framework can be adapted to account for the agency of algorithms in relation to other social and material components and develop actionable recommendations for advocacy groups advancing narrative change. Future research could also consider how the features and affordances of diverse platforms influence the capacity of collectives to cultivate narrative change. Additionally, understanding the translation between physical and digital components of narrative infrastructures, as well as their continuity across different scales of organizing, presents a critical opportunity to investigate how narratives maintain coherence and adaptability in increasingly hybridized spaces.

Another emerging line of inquiry concerns design's role in creating spaces for engaging diverse ways of knowing on collective articulation of systems change. This research showed that, while giving value to lived experiences, emotions, intuition, imagination or ancestral ways of knowing can [...] a more holistic [...] of collective intelligence, this is fraught with tensions. Future research can explore how collective learning spaces can be designed in ways to foster diffusion between these diverse ways of

knowing on an ongoing basis, and how this might promote critical and creative inquiry on systemic issues. As distributed and transdisciplinary learning collectives gain influence in driving science and technology, engaging with ... This equally call for attention to : Emerging debates on human and machine intelligence often emphasize rationality, potentially sidelining emotional, social, and contextual dimensions of intelligence.

Another emerging line of inquiry concerns design's role in engaging diverse ways of knowing in the collective articulation of systems change. This research demonstrated that valuing lived experiences, emotions, intuition, imagination, and ancestral knowledge can contribute to a more holistic understanding of collective intelligence, yet this process is often fraught with tensions stemming from structural barriers and cultural norms embedded in knowledge systems. Future research could explore how collective learning spaces might be designed to foster ongoing diffusion between these diverse ways of knowing, and how this might promote critical and creative inquiry into systemic issues. As recent developments in AI transform sensemaking practices across all domains, resisting a reductionist view of intelligence as reason is essential to cultivate the caring attitude needed to shape radical possibilities.

5.5 Final Thoughts

This research departed from a simple yet powerful premise: we cannot tackle today's pressing crises with the same mindsets that created them. The social and ecological crises of our time demand a fundamental shift in how we perceive our place in the world and act within it. Although the outlook may sometimes seem dark, radical

possibilities continue to sprout at the margins, their seeds carried across generations. For these possibilities to take root amidst the rigid foundations of current paradigms, we need new approaches to amplify and sustain them in our collective consciousness. This calls on designers to re-evaluate their roles in systems change, to see themselves as stewards of alternative possibilities, grounding their impact in relationships, and tending to the soil in which these possibilities grow.

APPENDIX A
CRITICAL DISCOURSE ANALYSIS DATA

The contextual study explored how organizations negotiate diverse understandings of transformative change through discourse.

The aim of this study is to frame collective inquiry as negotiation of socio-technical imaginaries ordered by infrastructures. This exploration is driven by the following questions:

- What are the diverse aspirations, values and meanings associated with high-level goals of transformative change?
- What are the socio-material arrangements and infrastructures through which organizations collectively make sense of transformative change and negotiate their visions?
- What different ways of seeing, knowing and sensemaking are activated in these socio-material arrangements, and which ones are excluded?

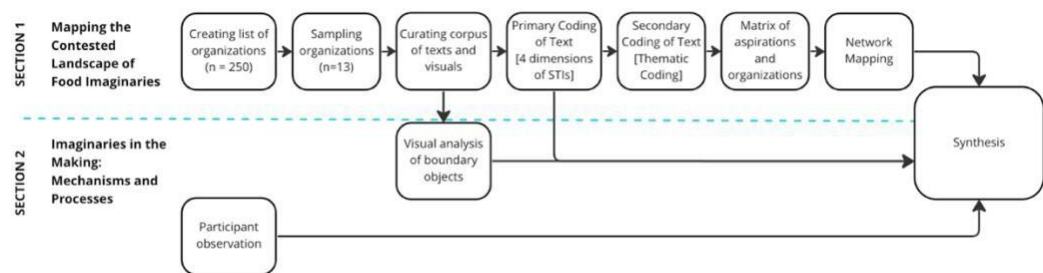


Figure A.1. Structure of the study

Sampling and Data Collection. I conducted critical discourse analysis on the issue of food insecurity, by selecting a corpus of online narratives of 13 organizations (see Appendix) from the Chicago food ecosystem. This sampling was based on an initial list of 250 organizations that I collected by web-crawling the participant and partner lists of

large-scale convenings such as the Chicago Food Justice Summit. These 13 organizations were selected with purposeful sampling to represent different types of stakeholders from the ecosystem to have diverse size, type, sector, to get to the different voices and their respective channels and platforms. The main criterion was organizations addressing food access, including mentions of 'food (in)security, food sovereignty, food justice' all of which reflect different social and political connotations.

The corpus consists of the visual and textual materials published by these organizations online, collected from a variety of channels depending on the channels most actively used by the organizations, such as web pages, online publications (policy documents, reports, media statements), and social media channels.

Primary Coding. The documents were coded using descriptive coding (Saldaña, 2009) that captured organization's positioning, issue framing, future vision and approach to transformative change (Figure A.2). I started the analysis with in-vivo coding, and clustered the codes into categories of the analytical framework proposed by Jason Chilvers and Noel Longhurst (2019, p.4), who compare socio-technical visions of energy futures through the co-production of 4 dimensions as “meanings, knowings, doings and modes of organizing”. Instead of centering STIs on the dominant narratives and national agendas, authors consider the niche imaginaries to be equally important for making sense of transitions and focus on how these visions are performed by adopting a relational co-productionist approach. Departing from the framework of Chilvers and Longhurst (2019), the analysis is informed by the four high-level dimensions below:

- **Knowings:** What are the ways of knowing and epistemic orders that inform these discourses? What are the socio-material conditions that organize the flow of knowledge and ideas? Whose knowledge joins the discourse and whose knowledge is excluded?
- **Organizings:** What are the forms of governance that organize these collectives and their discursive performances? Who is included and who is excluded? What are the spaces and platforms [that facilitate these?]
- **Meanings:** How do the collectives construct and stabilize the meanings, framings of issues and approaches that they coalesce around? What does the material construction of meaning look like?
- **Doings:** What are the practices and material productions that the collectives perform?

This framework is useful as a structural guide for making sense of the complexity of imaginaries. Unlike the analytical frameworks that focus on the visions, this framework foregrounds the collective performances of sensemaking that produce imaginaries. However, authors' framing of these imaginaries as competing rationale risks missing the deeper layers of meaning that the visions are rooted in, as well as excluding the perspectives that do not fit into logical formulations of transitions. While I use the framework as a structural guide for organizing findings, I expand their framing of dimensions to account for alternative ways of making sense of reality that are present in marginalized collectives, with an emphasis on social and material conditions of these practices.

Organization	MISSION	AIMS	STAKEHOLDER	GOVERNANCE	PRACTICES	KNOWLEDGE	MATTERS	LEVERAGES	FOOD	ISSUES	PATHWAYS	VALUES
Love Fridge	The Love Fridge is a Chicago-based initiative created to nourish our communities through mutual aid. Powered by kindness, generosity, and mutual respect, Love Fridge believes that feeding oneself is not a privilege, but a right.	Local justice, kindness, generosity, "Feeding oneself is a right", community identity, mutual aid	community, mutual aid groups, donors	community-driven, horizontal leadership		Food experience of community members, data collected through activities	Food access, reliability, community identity	Mutual aid networks, local donors and growing spaces	Domestic products, city, or packaged food items	Food insecurity	Food distribution through community refrigerators, collaborations with local artists	care, solidarity, generosity
City of Chicago	Advancing Food Equity in Chicago: to transform the food system by removing barriers to urban farming, supporting BPOC food entrepreneurs, and better connecting residents with nutrition programs and healthy, affordable food.	Every Chicagoan having access to healthy and affordable food, equitable food system, support local BPOC growers, promote BPOC vendors, improve urban agriculture, wealth creation	City Departments, partner agencies, City Council members, community partners, community food system experts, BPOC food business and entrepreneurs, local BPOC growers, people with disabilities	partnership, cross-sector working group, workshops, policy and program agenda, democratic deliberation of procurement efforts, encourage non-city institutions	review past food plans, identify priorities, share data, programmatic and policy priorities, policy and program agenda, procurement efforts, encourage non-city institutions	cross-sector working group, neighborhood-level perspective of community members	percentage of food insecurity by racial group, dollar spent on food purchase, dollar shared funding	Vacant land, food businesses, increased collaboration, nutrition programs, Chicago's Food Fund Policy, urban farms and community gardens, BPOC-owned and food businesses	N/A	Food and nutrition insecurity, community experiencing disparities across health outcomes, decreased access to food system engagement, stalled implementation of plans, lack of transparency in decision-making, lack of connection of small producers with large-scale purchasers, BPOC growers can't access resources	Eliminate barriers to food system engagement, and maintain nutrition programs and benefits, leverage City and institutional government to support local BPOC growers, producers and plans, lack of transparency in decision-making, lack of connection of small producers with large-scale purchasers, BPOC growers with access to capital	transparency, accountability, community, coordination
Good Food Purchasing Program	The Good Food Purchasing Program is a national program that helps public institutions purchase food that is transparent and equitable food system built on fair labor values, local economies, health, vibrant workforce, animal welfare, and environmental sustainability.	value-based procurement for large institutions that highlight local economies, health, vibrant workforce, animal welfare and environmental sustainability, increase transparency in the supply chain, equitable food system, advancing a variety of viable equitable food system, normalizing value-based procurement, protect biodiversity, robust carbon footprint, place-based research, worker rights, freedom of association, diversity, equity and inclusivity, small and mid-sized agricultural and food processing operations within the local area or region safe and healthy working conditions and fair compensation for all food chain workers, generous portions of vegetables, fruit, whole grains, and minimally processed foods, while reducing salt, added sugars, saturated fats, and red meat consumption and eliminating artificial additives	Local, national, public institutions, large businesses, distributors, producers, food businesses, suppliers, public officials, city, departments of public health, fair Trade community organizations, farmers markets, universities, healthcare facilities	align with local efforts, multi-sector coalitions towards ownership and long-term sustainability, local political champions, identify needs and resources, support, develop local strategies, share results, support diverse and small producers, data assessment, outreach to food producers, working with food service management on new food service management contracts, pathways for BPOC producers and food businesses to scale, contract food vendors, equip their development, working to train producers, partnering with design school, GIS mapping	flexible framework to create benchmarks of procurement, measure and communicate impact, measure to understand relationship between distributors, vendors and suppliers, list of needs and resources, support, develop local strategies, share results, support diverse and small producers, data assessment, outreach to food producers, working with food service management on new food service management contracts, pathways for BPOC producers and food businesses to scale, contract food vendors, equip their development, working to train producers, partnering with design school, GIS mapping	transparency, accountability, ownership, food spend, food spend reported, useful data in current practice, annual report, public meeting, assessment, feasibility study, long goals, full mapping of regional forms and food information, share information to local producers	transparency, accountability, ownership, food spend, food spend reported, useful data in current practice, annual report, public meeting, assessment, feasibility study, long goals, full mapping of regional forms and food information, share information to local producers	nutrition of food and national partners, multi-sector coalitions towards transparency and long-term relationships, support for policy by coalition, BPOC-owned businesses, women-owned businesses, equity and well-being by offering generous portions of vegetables, fruit, whole grains, and minimally processed foods, while reducing salt, added sugars, saturated fats, and red meat consumption and eliminating artificial additives	Barrier of flexibility in some institutions, lack of equitable access to good food, lack of certification, supplier capacity	Transparency, accountability, ownership, food spend, food spend reported, useful data in current practice, annual report, public meeting, assessment, feasibility study, long goals, full mapping of regional forms and food information, share information to local producers	Transparency, accountability, ownership, food spend, food spend reported, useful data in current practice, annual report, public meeting, assessment, feasibility study, long goals, full mapping of regional forms and food information, share information to local producers	transparency, accountability, ownership, food spend, food spend reported, useful data in current practice, annual report, public meeting, assessment, feasibility study, long goals, full mapping of regional forms and food information, share information to local producers
Naturally Chicago	To make Chicago an epicenter for fast-growing natural products companies	Little region's good food action, support access to food that is better for people and environment, "more diverse and more accessible food system", growth of good food action, innovative good food companies, stimulate economic growth, create jobs	entrepreneurs, investors, retailers, industry experts, women and minority entrepreneurs, leaders, established businesses, early-stage businesses, influencers, industry leaders, local food startups	connected to naturally, network, membership, members-only events, networking, reception, panel, sponsorship	Provide platform for continuing education and mentorship, help early-stage entrepreneurs expand, promote events, "jobs and opportunities for under-represented communities", business opportunities for women and minorities, wellness, business strategies, member-only events, parties, networking breakfast, panel, reception, sponsorship, flash events, office hours	wellness, customer trends, "natural presentation", "all consumer trends, new businesses, eating	growth of good food action, consumer trends, new businesses, eating	through leadership, member network, branding, marketing, data	Good food food that is better for people and for planet, packaged food, testing samples, natural and sustainable products, food products, local meat, plant-based beverages	Challenges of food businesses, lack of skills, lack of food movement growth, lack of innovation	Accelerate innovation through knowledge sharing	Industry leadership, expertise, "specialized in finding emerging trends, measure ROI while leading with equity and inclusion

Figure A.2. Screenshot of Dataset After Primary Coding

The coded dataset can be downloaded at [this link](https://docs.google.com/spreadsheets/d/1fQ_BwFQ8Xp6sKYXV21DrmYHxYSgWMb/edit?usp=sharing&ouid=113200726518678014617&rtopf=true&sd=true):

https://docs.google.com/spreadsheets/d/1fQ_BwFQ8Xp6sKYXV21DrmYHxYSgWMb/edit?usp=sharing&ouid=113200726518678014617&rtopf=true&sd=true

Secondary Coding. I carried out a secondary coding on the themes of mission and aims.

I listed the transformation goals articulated by organizations and clustered them by the high-level aspect of transformation that they are referring to. For instance, community development, community strength, community power are labeled as different articulations of the community strength theme. These tags refer to the high-level goals articulated across discourses which anchor the collaborative efforts of organizations. This analysis resulted in a matrix (Figure A.3.) that highlights the distribution of 13 goals of transformation across organizations. The aim of this analysis is to surface the elements of visions that anchor collective imaginaries in a shared vocabulary, rather than being an exhaustive analysis of what matters to each organization and what not, as articulated in their online narratives.

	Green City Market	City of Chicago	Good Food Purchasing Program	Family Farm	Naturally Chicago	Bigger Table	CFBN	IMAN	Urban Growers Collective	Hatchery Chicago	Greater Chicago Food Depository	Grocery Run Club	Love Fridge
	"Green City	advancing	transparen	"Our enduri	make chic	Address hu	"Build a Pl	IMAN mode	Rooted in g	"The Hatch	Food distrib	Grocery Ru	Our goal is
Food Access	x	x	x	x	x	x		x	x	x	x	x	x
Equity / Inclusion / Diversity		x	x	x	x	x		x	x		x	x	x
Economic Growth		x	x	x	x	x	x		x	x	x		
Strong Industry		x		x	x	x	x						
Employment				x	x	x	x	x	x	x	x		
Local Food Businesses		x		x			x	x		x			
Thriving Growers	x	x	x	x					x				
Health		x	x	x				x	x			x	x
Community Strength								x	x	x	x	x	x
Environmental Sustainability	x		x	x	x				x				
Valued Workforce			x	x					x	x			
Nourishing Spaces													x
Racial Healing								x	x				

Figure A.3. Matrix of Organizations and Their Transformation Goals

In order to understand how these visions relate to one another, I visualized the network of categories of transformation goals articulated by organizations by using

Kumu.io, an online network mapping tool. The resulting network organizes transformation goals by their number of connections to organizations, and closeness to other goals (Koon et al., 2022). The goals that occupy the center of the network are the ones with the highest influence, and in this case the most commonly articulated ones, while elements with fewer connections are pushed to the margins of the map.

Mapping the Contested Landscape of Food Imaginaries. In order to capture the visions articulated by organizations and their relations to each other, I visualized them as a network connected by thematic categories. Each category corresponds to the high-level articulation of these visions, often found in policy documents, strategic recommendations, mission statements etc. For instance; economic growth theme includes aspirations such as ‘developing local economy’, ‘sustainable economic growth’, ‘Chicago becoming a trade hub’. Hence it is important to note that the important nuances between these aspirations gain clarity through the juxtaposition of the larger discourses, which is the focus of this analysis.

Tracing the connections between these aspirations tells us different accounts of what transformative change should be about. Is it about building a stronger industry that can feed everyone? Or about empowering communities to be the sovereigns of their own food? Hence the center of the map can be seen as the epicenter of the encounters of these different future visions, rather than their mutual agreement on what makes futures.

Imaginaries of Food System Transformation in Chicago. The future visions and strategies within the corpus, can be consolidated into three socio-technical imaginaries. I identified these imaginaries by looking at the goals at the margins of the network. These goals center the narratives of organizations on different issues framings and provide competing interpretations of the shared goals of transformation. Socio-technical imaginaries consider deeper meanings associated with visions and their drivers, and not solely the visions. I name these STIs as *Community-led food justice*, *Accessible healthy food* and *Strong food industry*. These imaginaries do not seek to be exhaustive, and instead aim to articulate the different layers of meanings that shape the future visions. The sources of imaginaries identified here are limited to narratives that explicitly address food access.

Strong Industry. The visions voiced by industry stakeholders aim for a strong, networked industry that can drive "inclusive economic growth" (Bigger Table, 2021) for the region, create more jobs and contribute to "neighborhood revitalization" (CFBN, 2021). "Inclusiveness" through supporting minority entrepreneurs. This imaginary paints a Chicago that is "The Silicon Valley of Food & Beverage" (CFBN, 2021) where

innovative food businesses and networked companies that can address consumer demands and ‘give back to community’ by supplying food banks and agencies with "nutritious food" through donations. This imaginary does not propose a structural solution to food access but aligns with the dominant imaginaries that feeding masses is possible through large-scale production. For instance, a post on Bigger Table website showcases the "No Sugar Added Hot Cocoa Mix with added Protein and Fiber" created by the collaboration of member companies to be distributed to food banks, and a truckload of almond protein bars to be donated. This imaginary promotes exchange of data and knowhow within the network of members to promote the emergence of new business and products; claiming that it can create more jobs and contribute to the local economy and communities.

Accessible Healthy Food. This imaginary shares some of the social concerns of the Community-led food justice imaginary, while positioning companies and economic growth as key drivers of change similar to Strong industry imaginary. It centers on increasing the availability of “local food” and “good food” by supporting sustainable farming practices, livelihood of farmers and investing in healthy food businesses. The foundational idea is that access to “nutritious food” is a right and necessity for healthy communities. For instance, Green City Market states, “farmers' markets are a powerful instrument of a local food system, and we're dedicated to creating widely accessible marketplaces for local food in Chicago where farmers can make a living wage.” This imaginary emphasizes the potential of food as a vehicle for addressing the climate crisis, yet the environmental and social aspirations of the imaginary are not strongly tied.

Family Farm's website notes that "good food movement is not as inclusive or equitable as it has to be".

Community-led Food Justice. This imaginary centers racial justice and equity in food system transformation and advances food sovereignty rather than simply making food accessible. This core vision is articulated with nuances across discourses with goals such as addressing systemic racism, historic segregation, supporting black and indigenous farmers. The collectives that align with this imaginary are mostly community-organizations, mutual aid networks and nonprofit organizations. It advocates for land-based approaches that can promote healing from racial traumas in communities. Hence farms are not only places of food production but also create space for community members to connect with each other, with land and their past. Environmental aspirations are also framed as cultivating a different relation with land to replace the extractive practices. The return to ancestral roots and knowledges do not exclude technology or economic advancement, but aim to orient those towards generating community wealth through alternative economic models.

APPENDIX B

DESIGN FOR CLIMATE LEADERSHIP SPRING 2023 COURSE DATASET

This dataset consists of assignments submitted by students of Design for Climate Leadership course offered at the Institute of Design in Spring 2023. The assignments demonstrate use of the Imaginaries Analysis and Mapping tools developed as part of the research. High-resolution files for the image can be downloaded at this link:

[https://drive.google.com/file/d/1Om4mhHNoQGDrUcUKm2Y5DIwAY94LoTuQ/](https://drive.google.com/file/d/1Om4mhHNoQGDrUcUKm2Y5DIwAY94LoTuQ/view?usp=sharing)

[view?usp=sharing](https://drive.google.com/file/d/1Om4mhHNoQGDrUcUKm2Y5DIwAY94LoTuQ/view?usp=sharing)

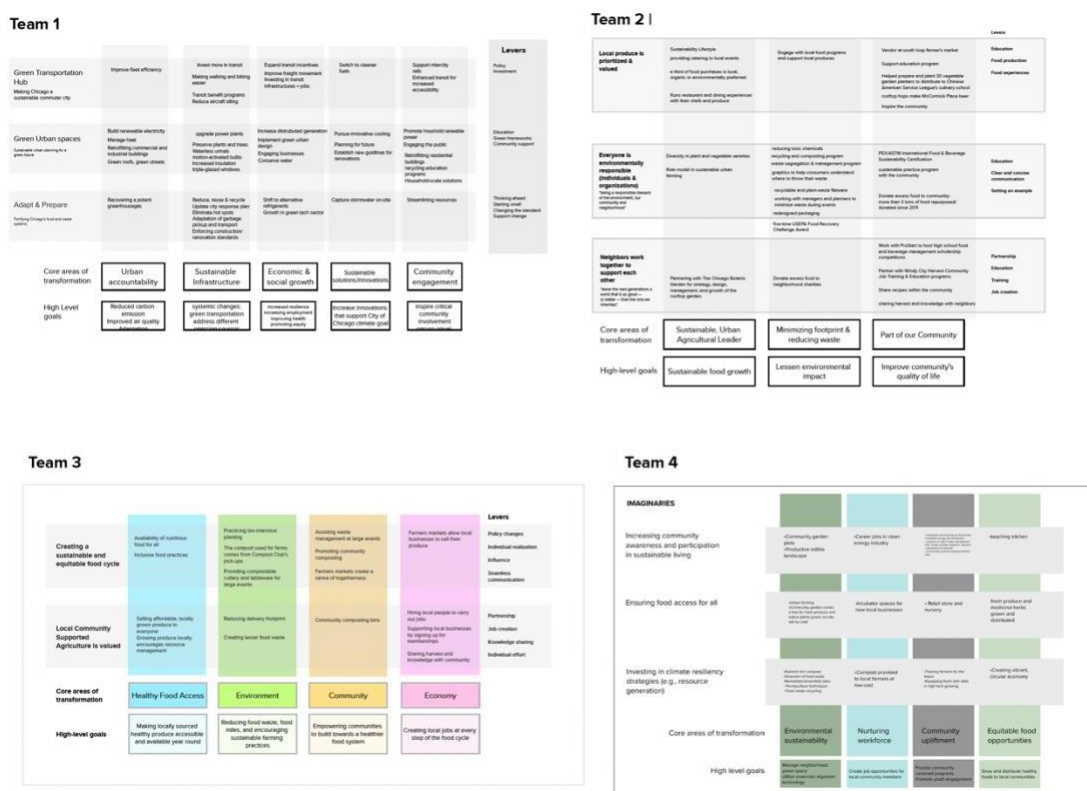


Figure B.1. Imaginaries Analysis Assignment Submissions

High-resolution files for the image can be downloaded at this link:

[https://drive.google.com/file/d/136gvbvA1dEysaWWipp_vZ-](https://drive.google.com/file/d/136gvbvA1dEysaWWipp_vZ-Pt9esOSNNv/view?usp=sharing)

[Pt9esOSNNv/view?usp=sharing](https://drive.google.com/file/d/136gvbvA1dEysaWWipp_vZ-Pt9esOSNNv/view?usp=sharing)

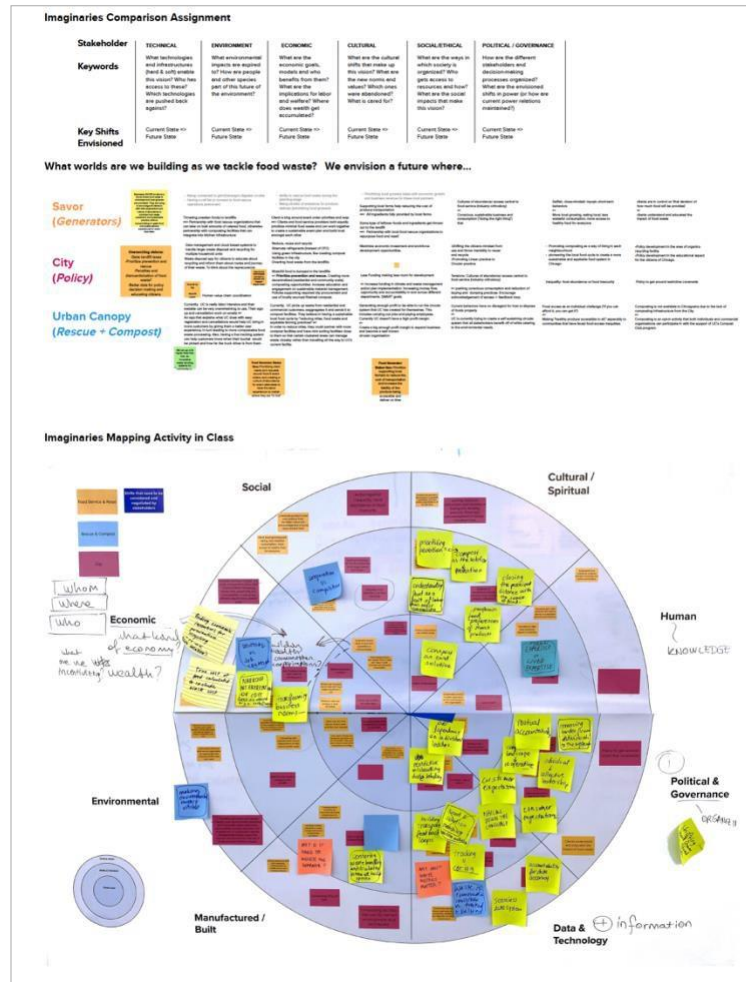


Figure B.2. Imagineries Mapping Assignment (top) and Class Activity (bottom)

APPENDIX C

RECIPES NETWORK WORKSHOPS DATASET

Artifacts of Annoyance Workshop (2023). The intervention consisted of two pieces:

(1) *Artifacts of Annoyance* exhibit, where we curated a set of artifacts exemplifying things that we find annoying in the food system, accompanied by a 'museum label' describing the object and its significance. The pedestals were placed in a corner of the conference room where the main sessions of the convening took place. Some of the artifacts included in the exhibit were food items commonly found in food banks such as canned produce and instant noodles, highlighting the historic growth of emergency food assistance into a permanent infrastructure for redirecting surplus food from corporations to people experiencing food insecurity. A dandelion and foraging book brought the politics of gardens into question and the erasure of edible landscapes with the rise of lawns as a status symbol. The final pedestal in the exhibit presented two speculative artifacts from our exploration, such as a bag of 'Powdered super greens' from a future where grocery stores are designed to prevent the waste of produce, dehydrating and repackaging these in a rooftop facility.

We left blank sticky notes and a marker pen on each pedestal, encouraging participants to react and share their thoughts.

(2) Workshop: We facilitated a 75-mins workshop where we used annoyances to facilitate a critical reflection on how the dominant values of the food system sustain wasteful and inequitable outcomes, and speculate on potential future alternatives. The workshop consisted of 4 activities:

1. **Wall of Grievances:** We asked participants to tour the exhibit and reflect on their own annoyances. We then collected and grouped these annoyances on the a board to form the groups for the next section.
2. **Teamwork:** In the next portion, the teams used the worksheets on the table to unpack their shared annoyance and speculate an alternative situation. We provided a deck of 'Re-imagine' cards on each table to encourage participants to think about different ways in which they can challenge the given situation and imagine alternatives.

Identify Roots of Annoyance

- Writing down the mutual annoyance and discussing what they find annoying about this
- Reflecting in what current systems are fueling this annoyance and what these systems value and care for

Identify Roots of Annoyance

- Discussing how the group's values can shape an alternative future
 - Reimagining the situation in an alternative future that reflects these values and 'bringing back' an artifact from this future to add to the gallery (a story, object, technology, news headline etc.)
3. **Shareout:** Each group briefly shared out their reflection and the artifacts that they prototyped. We added the artifacts that the groups created to the exhibit, replacing the existing artifacts in the exhibit.

High-resolution files for the images can be downloaded here:

https://drive.google.com/drive/folders/1Z8_63nO24cLFu13NSUAov1UvLWtOY4vI?usp=sharing

Artifacts of Annoyance Workshop - 2023

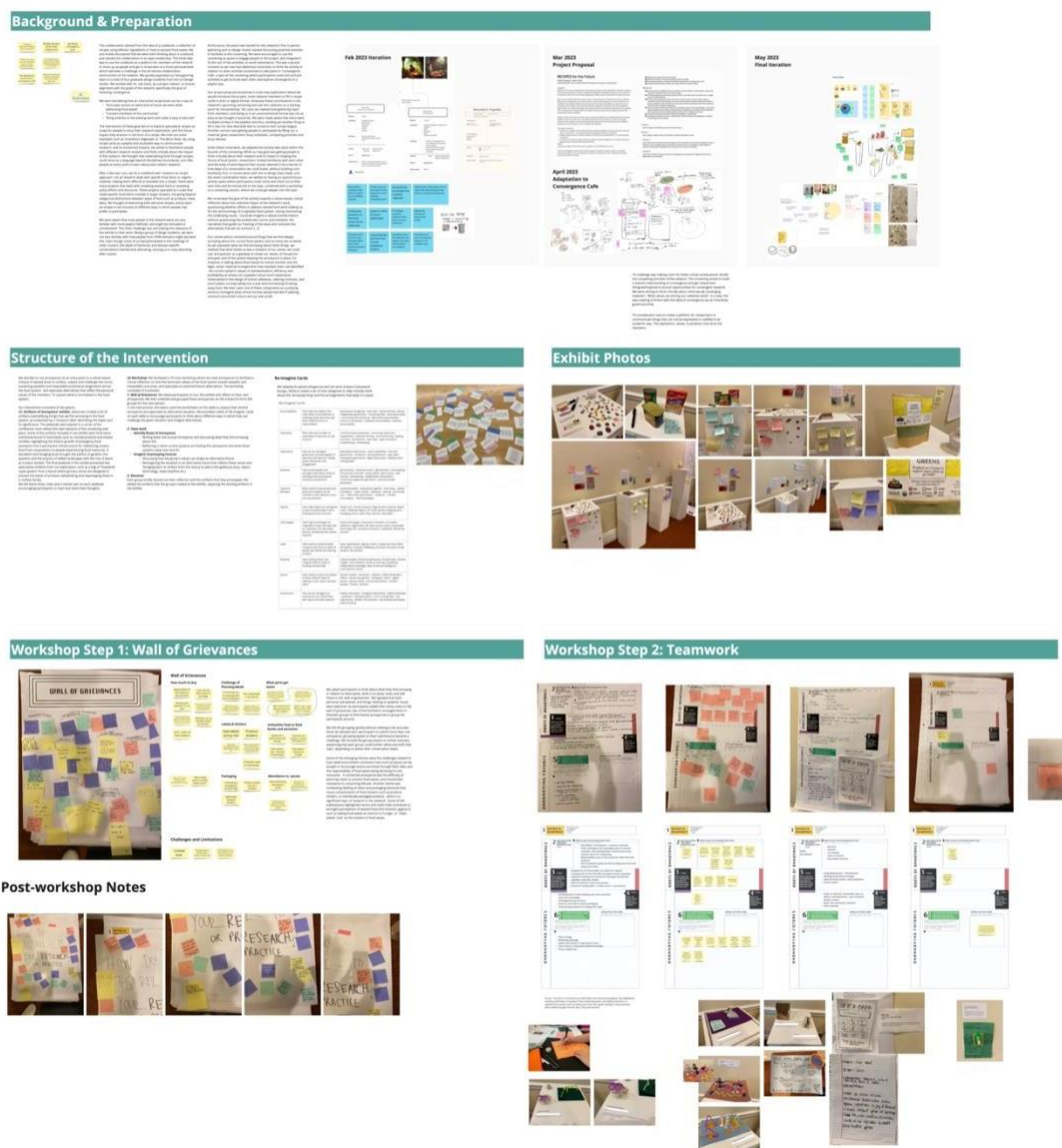


Figure C.1. Artifacts of Annoyance Process and Workshop Data

Collective Pantry Workshops (2024)

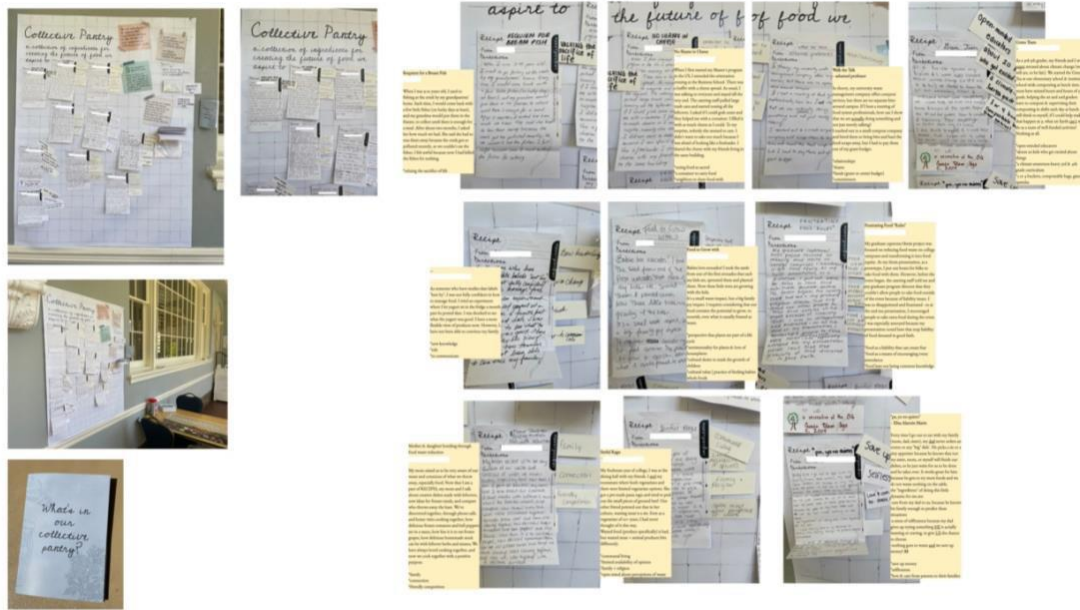


Figure C.2. Storytelling Activity Dataset

1 Pick & explore your Ingredients

15 minutes

1. Pick items from the ingredients list and place them on the table. Read the story of the ingredient to the group. Discuss and share your own story and your own.
2. Share your story about these ingredients with the group.

Ingredients from the Collective Pantry

Be in a group

- 1. Pick items from the ingredients list and place them on the table.
- 2. Read the story of the ingredient to the group.
- 3. Discuss and share your own story and your own.
- 4. Share your story about these ingredients with the group.

2 Get Cookin'!

25 minutes

Time to start cooking!

How can you combine these ingredients to create impact for a waste-free food future?

You are free to use what you create. It can be a situation, an intervention, an object, or anything that excites you.

For instance, you can imagine a way to combine these in your story to tell the world about, or imagine a future scenario.

Place your creation for the food future.

Check out the tip sheet below for inspiration.

Ingredients from the Collective Pantry

Be in a group

- 1. Pick items from the ingredients list and place them on the table.
- 2. Read the story of the ingredient to the group.
- 3. Discuss and share your own story and your own.
- 4. Share your story about these ingredients with the group.

Phrases that inspire

Be in a group

- 1. Pick items from the ingredients list and place them on the table.
- 2. Read the story of the ingredient to the group.
- 3. Discuss and share your own story and your own.
- 4. Share your story about these ingredients with the group.

Your creation

How did you combine your ingredients to influence a waste-free food future?

Share a brief description of what you created. This can be a situation, an intervention, an object, or anything that excites you.

What inspired you to create this?

What did you learn from this experience?

What did you learn from this experience?

What did you learn from this experience?

Tip Sheet

Food inspiration in stories

Consider flavor pairing

Consider...

Figure C.3. Virtual Co-Creation Workshop Breakout Group Example

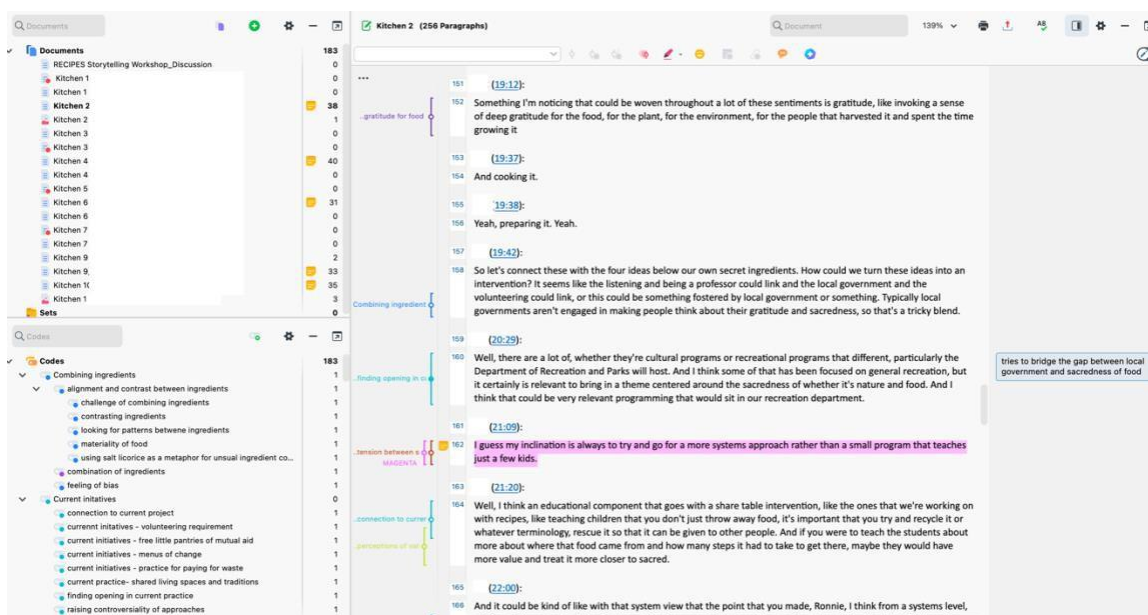


Figure C.4. Analysis of Workshop Data in MaxQDA

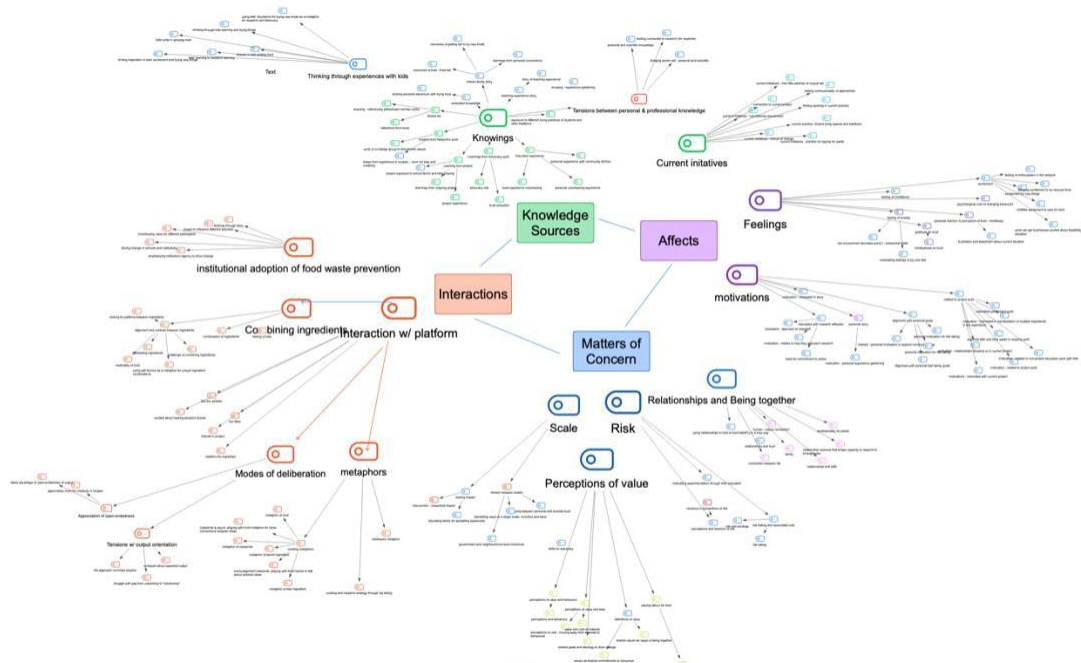


Figure C.5. Thematic Clustering of Codes in MaxQDA

APPENDIX D
POST-WORKSHOP SURVEY PROTOCOL

Post-Workshop Survey Protocol. This survey was conducted after the Collective Pantry Workshop to collect participant feedback and evaluate the workshop outcomes.

Collective Pantry, Post-Workshop Survey. Thank you for participating in the activity. This initiative is part of the research study titled 'Recipes for the Future of Food,' led by Professor Weslynn Ashton and conducted as part of the doctoral research of PhD Student Azra Sungu, at the Illinois Institute of Technology.

- This survey aims to learn about your experience and feedback about the activity for research purposes.
- This survey is anonymous. No one, including the researcher, will be able to associate your responses with your identity.
- Your participation is voluntary. You may choose not to take the survey, to stop responding at any time, or to skip any questions that you do not want to answer.
- Your completion of the survey serves as your voluntary agreement to allow the anonymous data gathered to be used in this study and in future research.

Any further questions about the research and your rights as a participant will be answered if you contact the research director (Prof. Weslynn Ashton, IIT Institute of Design, washton@iit.edu). Questions and complaints can be addressed to the Executive Office of the IIT Institutional Review Board at 312-567-7141.

1. How did this activity influence your thoughts on the issue of wasted food and potential approaches to addressing it? If it did not have an influence, please tell us

about it. If there were any ideas or moments that stood out to you, we'd love to hear them!

2. What are your thoughts on the relevance of this activity to the collective work of the RECIPES Network?
3. How would you rate your overall satisfaction with the activity?
(Very Unsatisfied / Unsatisfied / Neutral / Satisfied / Very Satisfied)
4. How would you rate the ease of navigating the steps involved in the activity?
(Very Difficult / Somewhat Difficult / Neither Easy Nor Difficult / Somewhat Easy / Very Easy)
5. We would love to hear a bit more about your impressions. What were the things you liked about this activity? What did you find challenging?

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